

BLOCKCHAIN

UNDERSTANDING THE TECHNOLOGY
OF BITCOIN AND CRYPTOCURRENCY



THIS BOOK INCLUDES-
CRYPTOCURRENCY INVESTING, BITCOIN, BLOCKCHAIN

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CRYPTOCURRENCY INVESTING

Unlocking the Lucrative World of Cryptocurrency

INTRODUCTION

Congratulations on downloading this book and thank you for doing so.

The following chapters will discuss all that you need to know about investing in cryptocurrencies. Cryptocurrencies are relatively new, with the first one being developed in 2009, but they have taken off like crazy. Bitcoin was once worth just \$1 for each coin, and now it is worth more than \$11,000 per coin in just a few short years. Many other digital currencies are seeing the same kind of increase in value and many people now prefer to work with these currencies rather than their traditional currencies.

Because the love for digital currencies has grown so much, now is the right time for investors to get into the market. The value of most digital currencies will continue to grow, and getting in at the right time and learning how to read the market, will ensure that you see a big return on your investment in no time. This guidebook will provide you with the information that you need to get started as an investor and see some good results.

Inside, we are going to touch on all the basics that you should

know to become an expert at investing in digital currencies. In addition to talking about some of the best strategies for trading in cryptocurrencies, we are going to talk a bit about what these currencies are, how to choose the best one, how to store the coins and keep them safe while you are investing, and even how to avoid scammers along the way. With the help of this guidebook, you will get all the tips that you need to effectively use your digital currencies without having a lot of questions along the way.

As a beginner investor, there are a lot of things you may be wondering about when it comes to getting started. Let this guidebook help you to answer some of those questions so you can get started on the right track in no time.

There are plenty of books on this subject on the market, thanks again for choosing this one! Every effort was made to ensure it is full of as much useful information as possible. Please enjoy!

CHAPTER 1: WHAT ARE CRYPTOCURRENCIES?

The idea of cryptocurrencies is relatively a new idea. Just a few years ago, it was an idea that no one thought would ever become a reality. Who would think that you could have a whole currency available online, one that you are not even able to get paper versions of, and that didn't have a central authority control them? Today, there are many cryptocurrencies available, with the most notable being options like Bitcoin and Ethereum.

While many people have heard about these various cryptocurrencies and some may even use cryptocurrencies on a daily basis, understanding how these currencies work is not that well-known. Most don't understand how these currencies started, how they work, and what technology is behind them to ensure that they will be trustworthy and work the way that people want them to. So, let's take a look at these digital currencies and what they are all about.

To start, a cryptocurrency is going to be a medium that you can use for exchange, just like what you will find with traditional currencies. The big difference that you will notice is that these

cryptocurrencies are designed to exchange information digitally through the blockchain. This means that you will be able to use these cryptocurrencies just like you can with your traditional currencies, but all of the transactions will be completed online, rather than with the paper currencies that you would use at the store.

There are quite a few currencies that are available for you to choose from. The first cryptocurrency was Bitcoin, and it was developed in 2009. Once Bitcoin started to gain popularity throughout the world, many more cryptocurrencies, which are now known as Altcoins, became available. Some work similar to Bitcoin, but there are a few that were designed for slightly different uses.

Cryptocurrency is going to happen completely online. You will not be able to print off the coins or convert them into paper money to use. This is because when the cryptocurrencies are used, either to send money or to finish another transaction, it is going to be converted into lines of code that have a monetary value. This is why cryptocurrencies are often going to be known as digital currencies.

In addition to being able to use these currencies online, another benefit of using cryptocurrencies is that there isn't a central authority that is controlling them. With the economic issues of 2008, many people throughout the world stopped trusting government control over their money and were excited that cryptocurrencies provided them a way to have a trustworthy

source of money without having any government interaction.

The reason that cryptocurrencies can be so effective without having to rely on the government is that of the innovative blockchain technology. We will discuss this technology more later on, but this technology has been able to change the face of many industries because it can process transactions instantly, is safe and secure, and the transactions that are placed on it can't be changed when they are done.

With many cryptocurrencies have been designed so that they decrease in production over time, which ensures that there will be a market cap with them. This is much different from traditional currencies because, with traditional currencies, it is possible for the government and other financial institutions to create more of the currency. This is how inflation occurs. On the other hand, the popular digital currency Bitcoin will never have over 21 million coins available to use. Not all of the coins are available in circulation right now, but the total amount will never get above this number.

While there are many different types of digital currencies that you can pick from, they are all going to be derived out of one of two protocols. These two protocols are known as proof-of-stake or proof-of-work. All of the digital currencies will be maintained by miners, who are basically people who have set up their computers and other machines who will participate in validating and processing all of the transactions that occur for a particular currency.

The history of cryptocurrency

Cryptocurrency came around at the perfect time for it to be successful. It would have had a lot more trouble gaining life and doing well if it had come about at any other time in history. Bitcoin was the first digital currency in existence, and it was first created in 2009. It is unknown who was the one to develop Bitcoin, but they go under the name of Satoshi Nakamoto.

Many people liked the idea of Bitcoin because they were tired of the economic failure that was occurring throughout the world at the time and they liked the security and freedom from government that they got when using these digital currencies. While it still took some time for Bitcoin to take off, it has reached big heights in recent times, with one Bitcoin being worth more than \$11,400 right now.

The first altcoin, known as Namecoin, started in April 2011. This was created to form a decentralized DNS, which made it more difficult to censor the internet. Then in October of 2011, Litecoin was released. Litecoin became the first digital currency that uses a scrypt as the hash function rather than relying on SHA-256 like Bitcoin did. This made it easier for the general public to mine these coins without having to get the specialized technology and machine that were required for Bitcoin.

As these major players started to gain more traction in recent years and more people throughout the world started to join the networks. There are literally hundreds of digital currencies that are available online now, and you can choose the one that is going to meet your needs the best.

The market capitalization of digital currencies

Right now, Bitcoin is considered one of the largest digital currencies in market acceptance, notoriety, volume, and capitalization. However, it is not considered the most valuable of the coins right now. But since it is one of the most well-known currencies and it was one of the first, it is often seen as the most important. After Bitcoin, Litecoin and then Ripple are seen as the biggest names in digital currencies.

Another option that you may have heard about, known as Dogecoin, is an interesting one when it comes to market capitalization. This digital currency is often third place in terms of trading volume, but the market cap is going to be lower. This one usually ranks as number six.

The security of digital currencies

Since there isn't a government agency or a financial institution that is running these digital currencies, many people who are considering joining the networks may be a little bit worried

about how secure their transactions would be on them. The security that comes with digital currencies is going to be two parts. The first part comes because there is some difficulty with finding hash set intersections, something that the miners will work on.

The second part is what is known as the 51 percent attack. This means that for any changes to be made in the network, for any of the transactions to be changed, the hacker needs to be able to gain control over 51 percent or more of the computers on the network. Since the users of these currencies are available all throughout the world, this is almost impossible to occur. This helps to keep the network as safe as possible.

In addition, these digital currencies are going to be less susceptible to being seized by the government, having issues with law enforcement, or having holds placed on the transactions like other things may. The transactions are instant and permanent, so you won't have to worry about missing out on payments or having to wait around. And with the help of the blockchain, you will be able to remain anonymous on these networks, making it harder for people to figure out what transactions you are completing.

These digital currencies are proving to have more security compared to traditional forms of payments and even compared to some of the big-name stores that we like to use. These big financial institutions and stores have run into a lot of trouble, with hackers always aiming to get on their databases and take

the information that is there. It is all too common to turn on the news and hear that information has been compromised yet again. This means a lot of time and hassle on the part of the customer, and it can seem frustrating and like the company isn't doing anything about it.

There are some better security measures in place with these digital currencies. The blockchain is really an amazing feat of technology that not only helps to get transactions done in no time, but it is designed to keep your information safe. If someone goes onto the blockchain and starts to mess around, anyone on the network will be able to notice because the blockchain becomes a mess. It is almost impossible for the hacker to get ahold of your information and steal your identity.

With that being said, you still have to take some precautions with your coins. While the blockchain may be pretty secure, there are still hackers who will want to get onto the network and take your coins right out of your wallet. You have to be the one to be vigilant about this issue; there really aren't any safety features in place to help with this. If your coins disappear because a hacker got into your account or because of a computer glitch, there is no one on these networks who will be there to help; the coins will just be gone. We will spend some time looking at the various backup options that you can use, such as hardware backup and cold storage, that will ensure your coins stay safe, even if something happens to them in your wallet.

Dealing with legality and taxes with cryptocurrencies

Another issue that you may need to deal with is whether these digital currencies are considered legal in your country. Most countries allow these digital currencies, with Vietnam and Iceland being the exception. In addition, China has already banned their financial institutions from working with Bitcoins, and Russia has said that it is illegal to purchase goods in the country with currencies that are not the Russian rubles.

While these digital currencies are allowed in your country, you may have to deal with the issues of taxes. This is seen in the United States where the IRS has already ruled that Bitcoin should be treated just like property when it comes to tax season. This means that any of the Bitcoin that you have can be subject to capital gains tax. This has put some restrictions on Bitcoin and the other digital currencies, and you have to be careful when reporting them on your tax return each year.

In the United States, you also have to be careful about other law enforcement. The United States has not prevented digital currencies yet, but they are making it more difficult to use. One of the main benefits of using digital currencies is that you can remain anonymous on these networks, making it harder for others to track your transactions. However, the United States government is setting up laws that would make it a requirement for the digital currency exchange sites to report information of all users, which would take away some of this anonymity that you are looking for.

It is important that you understand the laws for digital currencies in your country of residence. Technically anyone can get on a digital currency and use it, no matter where they are located throughout the world. However, if there are specific rules that prevent this in your country of origin, it is not a good idea to try and use the currency.

The best thing that you can do is take a look at some of the rules for your country of use before getting started. Most countries are fine with you using these currencies so that will not be an issue, but some do have limitations. In addition, you need to make sure that you understand the tax implications of using these coins, especially if you are making a profit from investing them. As long as you follow the rules in your own country though, you should be fine.

Services available with cryptocurrencies

There are actually quite a few services that you can use when you want to monitor various cryptocurrencies. This can be useful for the person who is interested in mining with digital currencies and who wants to fully do their research ahead of time. CoinMarketCap can be a good one to start with because this will allow you to see information about digital currencies including their market cap, the price, and the available volume and supply. Reddit is a good way to follow the trends that are going on, and CryptoCoinCharts will have a lot of information about the exchange rates of the different coins you want to work with.

Liteshack is another one to consider because it will allow you to view the hash rate for a variety of coins using the six different hashing algorithms. This site even provides you with a graph of these hash rates, which can help you to check on the trends that go with the coins you want to work with. This will make it easier to see if the coin is gaining in popularity or if they seem to be losing interest before you invest.

If you want to get into mining with coins (which can be very profitable if you learn how to make it work), then you need to use the website CoinWarz. This site will help you to look at the hash rate to determine which coin is going to provide you with the most profit. Mining can be difficult so figuring out where you will get the most money can help save you time and will make this endeavor more profitable.

There is a lot of potentials that comes with using cryptocurrencies, whether you want to use them for your own investing, you want to use them to make purchases, or you just want to learn a bit more about them. This is a new world that is still expanding, and there are a lot of great possibilities that are sure to come out of these digital currencies in the future.

CHAPTER 2: WHAT IS THE BLOCKCHAIN?

Before you can really use any of these cryptocurrencies, it is important that you get a good understanding of the technology that helps them to run well. Blockchain is the main technology that does all the work. It is there to record transactions and keep information safe on the network. And without this technology, it is unlikely that these digital currencies would ever become successful in our modern world.

Remember that one of the benefits of using digital currencies is that they do not rely on any government intervention. These currencies are found completely online, and there isn't going to be a central authority that gets to control how the currency works or how much it is worth. This is a great thing to many people, but you also have to wonder how the currency is going to remain safe and trustworthy. This is where the blockchain technology is going to come into play.

So, how does this blockchain technology work? Let's start from the very beginning. When you sign up for any kind of digital currency, whether it is Bitcoin or one of the other options, you

will become a part of the network, and you will receive your own link in the chain of that networks blockchain. As you use this network to send or receive money, all the information about that transaction is going to be added to the chain.

These chains can hold onto quite a bit of information, so it will take some time for you to fill it up. Some people can fill it up faster than others depending on how many transactions they can complete in a certain period of time. Once you have filled up that first chain with all of your transactions, it will then be placed into the permanent record for that network and safely stored on the system. The network will send you a new chain to fill up, and the process will continue for as long as you are on the network.

Depending on how many transactions you complete, you may have a short or long chain. The more transactions that you end up completing, the longer your part of the chain is going to be. But if you are only on the network occasionally and you do not send or receive a lot of money, you will not end up with a long chain.

When the chain is all filled up, it will join the permanent record, which is the main blockchain that the whole network uses. A good way to think about this is that the blockchain is like a bank ledger. When you complete a transaction, it is going to show up in the chain and be stored there. When the chain is all filled up, you will have a full chain that has all those transactions together, just like you do at the end of the month from your bank.

After some time, you are going to have your own blockchain that includes all of the transactions that you have ever done. You can think of these as your record from the bank. They will include all of the chains that you have created (or the monthly statements from your bank). You can take a look at these whenever you would like to check that the information is correct and to see that payments are taken care of.

Anyone on the network will be able to take a look on the blockchain to make sure all transactions are supposed to be there, but there have been some precautions taken to ensure that all information on the blockchain is as safe as possible. Your Bitcoin address, as well as the work that the miners do, will ensure that while anyone can look at the Bitcoin ledger, they will have a hard time figuring out who you are.

As mentioned, the blockchain is designed so that the miners can store information on it while protecting you. These miners are going to do the work of providing a unique code to the blockchain so that all the transactions are safe and secure from those who want to get ahold of the information. We will discuss the work of the miners a bit more later on, but there are some specific rules that need to happen when creating these codes to ensure that none of the blockchain is messed with and that all the transaction stay safe.

Anyone who is on the network will be able to look at the

network, but as long as the miners do their work the proper way, they can look all that they want, but they will not be able to see your personal information. This is nice because it is a good thing for people to be able to look at blockchain and make sure that everything is legitimate on the network, but you know that the information is safe from prying eyes.

You will also enjoy the security that this technology can bring to your coins. Remember how we mentioned earlier that there really isn't anyone who is in control of this network. You aren't able to call them up if someone takes your information and file a complaint like you can with a store or with a bank. However, the blockchain technology is going to ensure that your information stays as safe as possible, so you don't have to worry about it all.

This is going to work thanks to the unique code that is designed with the blockchain. Each of the characters that come in the code has to correspond with the parts of the code that were there before. This helps the different chains of the code to match up with each other. The miners are going to be responsible for this working, so if even one character is changed anywhere inside the code, everyone will be able to notice.

This is good news for those who are worried about how well the blockchain will work. You will be able to check in on the blockchain. All of the numbers should match up together, and there shouldn't be any issues. If someone tries to get onto the blockchain and tamper with one of the transactions though, it is going to mess up the blockchain. Each thing they try to mess

with will end up changing at least one of the characters in that chain. This will start messing up other parts of the blockchain as well. And when this mess starts, it will be easy for anyone to take notice and see, making it hard for a hacker or an unethical person to get away with what they want.

This technology has changed the way that digital currencies work and without the blockchain being around, it is unlikely that any of those digital currencies would be considered trustworthy and still around today. But while blockchain was designed and introduced at the same time and in conjunction with Bitcoin, there are quite a few other uses for it, and many other industries are starting to take notice.

One of the biggest industries that are trying to use blockchain to make things easier is the finance industry. Not only is blockchain able to keep personal information safe, but it is also able to speed up the transaction times. Think about how long it takes to get a transaction done at your regular bank or even with a credit card. You have to wait for your bank to verify the transaction and then you have to wait for the money to end up in the other account. It can easily take three or more days for this to occur. With the help of blockchain technology, you will be able to see these transactions occur immediately, which can provide a huge benefit to many financial institutions.

The financial industry is not the only place that can benefit from using this kind of technology. Many industries are working to develop their own form of blockchain to serve their customers

better. Whether they are looking to use this in the finance industry, insurance industry, for reservations or more, there are a ton of ways that blockchain technology can be used to improve customer satisfaction. While blockchain may have first been introduced as one of the technologies for Bitcoin and other digital currencies, there are a ton of other ways that it can be used in the future.

CHAPTER 3: HOW IS THE VALUE OF CRYPTOCURRENCIES DETERMINED?

You will find that when it comes to the value of digital currencies, the factors that determine it will be a little bit different compared to what you will find with traditional currencies. For most users of these digital currencies, this can be a really good thing. It means that the government stays out of the mix and that the users get to determine the value through supply and demand. There are some people who worry a bit about the value of these currencies though, worrying that they may become overinflated and that the value will drop really quickly, but right now, it seems to be following the market for the value.

When it comes to traditional currencies, it is up to the government or financial institutions who will decide how much the currency in that country is worth. They will release a certain amount of the currency and let it go into circulation to be used in trade. Over time, the government can decide to make more of the currency. They are not limited by anything for how much of the currency can be available.

In the past, our money was based on the gold standard. The government was able to make only enough money to match with the amount of gold that they owned to back it up. But today, money in our system is no longer on the gold standard, meaning that the government can print off as much as they would want. Each time that the government prints out some more of the currency and places it into the market, it results in inflation.

Now, when a government or financial institution is responsible and watches the market and is careful about the amount of money that they release into the market at a time, this is not a bad thing. A little bit of inflation can be good for the economy and can help it to grow and the people to be better taken care of. But there are plenty of examples where the government was not responsible. They saw that the economy was starting to fail, so they kept printing and releasing more money into the system.

This causes over-inflation and can make the currency pretty much useless. The people will decide to go back to trading rather than use the money because the currency may be worth something one day but with the bad inflation, it will become worthless by the next day. This can leave a country in turmoil and make it, so the country has to restart with their currency and build up the trust again.

Things are a little bit different when it comes to digital currencies. There isn't a central organization that is running the value of the currency, so the problems with releasing too much or relying on a government to make the right decisions are not

going to be an issue. After the worldwide economic collapse that happened in 2008 and beyond, many people were excited to find that there is a currency they can work with that is not reliant on someone else to determine the value.

When it comes to digital currencies, there are a few things that will determine the value of the coins. For the most part, the demand for the coins at a particular time will determine how much the coins are worth. If a lot of people would like to have the coins at one time, the value is going to go up through the roof. If there are not too many people who want to have these coins, then the value is going to go down.

The amount of coins that are in circulation will determine the price as well. Remember that most digital currencies have a limit on how many coins can be out on the market ever. For Bitcoin, there were only 21 million coins created, and no one can create more than that. Not all of them are in the market at one time though. The miners are going to do their work to protect the blockchain, and over time, the coins will be released.

If there are more people interested in purchasing the coins at one time, and the miners haven't been able to release more coins through creating hashes, then the value of the coins will raise up. If the interest has died down a little bit and some more coins are released by the miners, the value may go down a little bit.

This is a supply and demand issue, just like what we see in a free market model of the economy. And so far, it seems to be working well to help keep the market going. No one has to worry about an individual or an organization determining how much their coins are worth and they can rest assured that the value of the coins will be accepted no matter where they use them throughout the world.

CHAPTER 4: THE BENEFITS OF INVESTING IN CRYPTOCURRENCIES

When it comes to finding a good investment to work with, there are a few different options that you can choose from. Some people like to go into real estate because they like to get the different market trends and there is a lot of money that can be made. Some people like to start their own business and work at a job that they love each day. And still, others will spend their time with the stock market because of all the different options that are available with that one.

One option that you may be considering, especially with how much they are trending right now, is to invest in digital currencies. Many people have already decided to invest in Bitcoin and are seeing some good success with it, but there are many other digital currencies that you can choose to invest in. But why would you want to choose to invest your money with these digital currencies rather than with one of the other options listed above? Let's take a look at some of the different benefits of investing your money with cryptocurrencies and why it can provide you with a great return on investment.

They are growing quickly

It only takes a few days of watching the market to find out that digital currencies are growing like crazy. They are in high demand, and since they can be used by people all throughout the world, they are a great option that will continue to grow in popularity in the near future. If you jump in at the right time, you can see a huge return on investment in no time.

Let's take a look at the growth of Bitcoin. In February of 2017, the value of one Bitcoin was about \$2500. That was still pretty good, much better than the amount it started out with. But at the time of writing this guidebook, the value of one Bitcoin had jumped up to over \$11,400. If you had invested in Bitcoin just a year ago, you could easily see a huge return on your investment at that time just by leaving your money alone there.

The other digital currencies are seeing a big growth as well, although most of them are not as big as Bitcoin right now. What this means is that as an investor, it is a great time for you to get into the market and earn some money when you invest in digital currencies.

No government controls

One of the things that a lot of people like about working with digital currencies is that they do not need to worry about

government control. There is not one entity, either a company, a government, or someone else, who is in control over these digital currencies. This can pose a few problems, but overall, people like the ability to work with their money without having to worry about how the government is going to react to it.

With all of the issues that occurred with the economy during 2008 throughout the whole world, many people are having trouble trusting the government to take care of their money. They do not like the control that the government has over all parts of their finances, especially since the government was not very effective at doing it in the first place. With the help of these digital currencies, you will be able to work with your own finances, not having to worry about anyone being in control over your money at all.

Ability to be anonymous

When you do any transaction on these networks, you will be able to keep your identity hidden, something that is almost impossible when it comes to traditional financial institutions. With all the recent news about how hackers have been able to get on traditional databases and steal your information, it is nice to know that there is now a system available that will keep your information safe.

There are a few things that you can do to ensure that you can keep your identity hidden. When picking out an address to use,

make sure that it is unique and that you do not use your name or any other information that will make it easy to identify you to other people. In addition, many people choose to switch their addresses on occasion, which helps to hide their information even better. You are allowed to switch your address as often as you would like, so this is a great idea to hide your tracks a bit more.

Huge growth possibility

Right now, many cryptocurrencies are growing like crazy. People all around the world are hearing about these digital currencies, and they like the benefits that they present. They like that they can use these currencies, regardless of where they are located in the world, and this makes it easy for them to make a purchase. And with the security that comes from the blockchain, they can do these transactions in no time.

Because of all the benefits and the fact that the currency is available to anyone who has internet access, no matter where they are located throughout the world, there is a lot of growth opportunity. In just the last year Bitcoin has gone from a value of \$2500 to over \$11,000, and it is predicted to go way higher than that in the near future. There are many other digital currencies that are seeing huge increases in value as well, which makes it a great investment to go with.

While other forms of investing will require a lot of research

before you join the market and they can be really risky (or they will take a long time before you can earn any money), this is not the case with cryptocurrencies. You could easily make your money back within a year plus a lot more, and the risk is fairly low right now, as long as you pick out a good digital currency to work with.

Send money instantly

If you have ever sent money to others, whether you are trying to give them money while they are visiting another country, or you are sending money after finishing a transaction, you know that this process can be really slow. This is because both banks, your bank and the bank of the person who is getting the money, have to reconcile their records to include this money and verify the transaction. This could end up taking up to five days or more depending on the type and amount of the transaction.

No one wants to wait that long for the transaction to finish, but that is the way that the current banking system works out. However, it is slow and can really make things a pain. What if you want to send money instantly to someone? Luckily, with the help of the blockchain technology, you will be able to do this. With the blockchain, there is only one ledger that you need to worry about and this one works instantly. If you want to finish making a purchase or you want to send money to a friend in another country, you can get it done in just a few seconds rather than in a few days.

Saves money

The blockchain technology that is used with many of these cryptocurrencies can actually help you to save money. You will have to spend a little bit of money to exchange the coins from traditional currency to the digital currency and back again, but it is a lot less expensive compared to working with most banks and financial institutions. And since you can do the trading without a broker if you want, something that is really hard to accomplish with other forms of investing, you can end up saving a lot of money on your investment, putting more of that money back in your pocket.

Many options for investing

There are actually a few different options when it comes to investing in digital currencies. You are not stuck with one option that you don't like. And even if you try out an option and find it is not the best one for you, it is pretty easy for you to make some changes and try out one of the other options.

This guidebook will talk a bit about the different options that you can pick when it comes to investing in digital currencies. We will talk about the buy and hold strategy, the day trading strategy, starting your own business and accepting the digital currency as payment, and even how you can become a broker and help others to invest in these digital currencies. These are just a few of the options that you can go with when you are ready to put your

money to work while using digital currencies.

Don't require a lot of work

You will be surprised at how little work you will have to put in to get started with investing in these digital currencies. There are some strategies that you can use that will require a bit more work, such as in day trading where you need to watch all of the daily fluctuations of the market, but for the most part, you can place your money into the market and get results.

This makes it easier for beginner investors to see the results that they would like. You do not have to worry about watching out for a big crash because that is not likely to happen. While these digital currencies will probably level out at some point and remain steady, right now the demand for them is high, and the value just keeps going higher and higher. This makes it an easy investment choice, one without a lot of work in it, for those who are looking to make money with digital currencies.

Easy to access

The money that you can place into these digital currencies will be easy to access at any time. Even if you do the buy and hold strategy, which requires you to put your money into the market and then leave it there until the market grows and you make money, you can take out the money at any time.

There are some investments out there that will require you to put your money on the market and not take it out for a specified amount of time. If you do take the money out early, you will end up facing lots of fees and other penalties along the way. Even though it is considered your money, you are not able to use it the way that you would like.

However, when you invest in digital currencies, you will be able to instantly access your money at any time. Even though you are still investing it, there are no rules about when you can take that money out. If there is an emergency that comes up and you need the money for something else, it is just fine to take it out of the investment account. If you see that the market is going down and you want to get your money out before it all disappears, that is fine as well. You have no rules about when to take out the money when you invest in digital currencies, and this can help a lot of investors feel better about using them.

As you can see, there are many reasons why an investor, especially a beginner investor, will choose to go with digital currencies as their investment option of choice. It is hard to work in some of the other investment options, but digital currencies can actually be pretty easy. Take these reasons into consideration when you are ready to start out with a new investment.

CHAPTER 5: SETTING UP YOUR OWN CRYPTOCURRENCY ACCOUNT

Before you can decide that investing is a good option for you with digital currencies, you need to be able to find a way to get ahold of those currencies. You can't go to the bank and exchange out for those currencies, although this may be something that changes in the future. You will need to set up your own account and have it available, so you can reach the coins whenever you would like. Some people choose to work on mining to get the coins, but since this is a hard job and requires a lot of outside stuff to be successful, it is not the right option for everyone. The good news is, you can use an exchange site to get some of your own coins.

Most people choose to go onto an exchange site and exchange some of their current traditional currency for the digital currency of their choice. You will find with a little bit of research that there are a few exchange sites that will work and the one you will pick often depends on which currency you would like to invest in and which one has the best offers at the time. To help you get your site set up and ready to handle any and all transactions and investments that you want to do with digital currencies, let's take a look at how to find these exchange sites and how to get

started.

What are the best exchange sites?

The first step that you need to work with is finding an exchange site to switch your traditional currencies over to a digital currency. This is one of the easiest and fastest methods that you can use to get your Bitcoin because you will be able to get it all switched in no time. You will find that most of these exchange sites are set up like the traditional ones; you will exchange your traditional currency for digital currency just like you would exchange it for a currency from another country. This makes it easier to get the exchanges done without a lot of hassle. There are many exchanges to deal with, but some of the best choices that you can pick from include:

- Xapo: this is a good one that will also provide you with a few other services, such as a debit card and a wallet to hold your coins. You will be able to make deposits of traditional currency into the account before switching it over to Bitcoin on the same account.
- Circle: this service is a great one to work with if you would like to receive, store, send, and exchange your coins in one place. This one is only available inside of the United States, and you will need to link your bank account with the service for it to work, so if either of these things is an issue, you may want to pick another service.

- Coinbase: this is one of the biggest exchange sites available, and it is really easy to use for both Euros and USD. You will be able to use your debit card, credit card, bank account, and PayPal account to here to switch your traditional currencies over to digital currencies. It can also provide you with an address to use on your network and a wallet to store the coins if needed.

Some exchanges are pretty basic, and they are just available so you can take your traditional currencies and exchange them out for your digital currency of choice. Others will provide some more services to you, such as help with monitoring your transactions or providing you with a wallet to store your currency. Picking out the right exchange site can often depend on what you wish to do with your currency and how much you want the exchange site to help.

One thing that some beginners don't realize is that they will need to show some proof of their identity before they can get started with these exchange services. They feel that they should remain anonymous at all times on these networks and that this information should not be necessary. However, many countries require that the exchange site collects this information, so you will need to provide it before you can proceed.

Using your exchange site

Setting up an account can be pretty easy, but there are a few things to consider when moving your traditional money over to a digital currency. Each exchange site is going to be different, but we will take a look at how Coinbase works since this is a popular option to use for options like Ethereum, Bitcoin, and Litecoin.

With Coinbase, you can choose between using a credit or debit card, your bank account, or a PayPal account. The bank account can be nice because you will have higher limits when it comes to how much money you can transfer over and exchange for the right coins. If you want to be able to invest a large amount of money, then this is the option that you should go with. However, it is important to note that because you are working with your bank, the transfer is not going to be instant. It will often take at least three to five days for the transaction to be processed. This can slow you down a bit when it comes to getting into the market, so you need to weigh whether that option is a good one for you or not.

Some people like to use their PayPal accounts or their credit and debit cards. These have the benefit of being instant transactions. If you want to be able to get into the market right away, then you should consider using these to help you get started. You will have a limit on how much money you will be able to invest. If you are just interested in getting into the market and trying it out and you do not want to wait around to see how long the bank will take, then using one of these options will help you get started.

Once you have exchanged your traditional currency out for the

digital currency of your choice, it is time to find a place to store it. Coinbase and some of the other exchange sites will provide you with a wallet to use, and you can choose to store the coins there if you would like. If you plan on using the coins right away to make purchases or to send money, these wallets are fine. If you plan on storing the coins for some time as an investment option, it is probably best to find a different wallet.

These exchange sites are often under attack from scammers and hackers who want to get ahold of the coins for free. The longer you let your money sit in that wallet, without having some extra safeguards in place, the easier it will be for these hackers to take the information that they want. For long-term storage, it is much better to move the coins to another location to keep them safe and to ensure that no one else can get ahold of them.

Setting up an account for your digital currencies and making sure that you can get ahold of the currencies that you would like to use can take some time and a little bit of planning. Make sure you look at the various sites that are available because this can help you out a lot. Some are going to have special features that you may want to use with your investing, and with research, you will be able to figure out the one that is best for you.

CHAPTER 6: PROPERLY STORING YOUR CURRENCY TO KEEP IT SAFE

At this point, you have opened up an account with an exchange service to turn your traditional currency over to the digital currency of your choice. Now it is time to pick out the wallet that you will use to store those coins and keep them safe until you use them. Whether you plan to use the coins right away for a transaction or you are going to keep the coins in one place for a while as part of your investment strategy, you still need to have some kind of wallet to hold onto the coins in the meantime.

Picking out a good wallet is going to be important to ensuring that your coins stay as safe as possible. There are always hackers who are trying to get onto these networks and take your money, leaving you with nothing if you are not cautious. There are several types of wallets that you can pick to hold onto your digital currency, and each one will provide you with a different level of security. You can choose to leave your coins in an online wallet, move the coins to a hardware, wallet, or even consider cold storage to keep those coins safe. Let's take a look at each one and how they all work for keeping your coins safe and secure.

Picking out a secure wallet

Once you exchange your traditional currency for your digital currency of choice, you need to have some place to store the coins. You are not going to be able to carry them around in your pocket or your wallet like you would if you went to a bank and exchanged USD for Euros, so it is important to find a good wallet to work with. There are a number of good ones to use, and it often depends on how you want to use these wallets and what features you are looking for.

Many beginners choose to work with Coinbase and just use the wallet that is available there. This can make things easier, especially if you plan to use the money for a transaction right away. The biggest issue with using the Coinbase wallet though is that this platform will provide you with an address, and it is going to contain your name, which takes away some of the anonymity that you were enjoying before.

There are a lot of wallets out there that you can choose from and most exchange sites will make it easy to move your coins over to the wallet of your choice. You want to choose carefully though when it comes to a wallet. This is where you will be storing your coins and if they are not a reputable site, you may end up in some trouble when all of your coins are gone.

In addition to taking a look at some of the neat features that are available with the different wallets, consider looking at their

security. Do they have some extra security features that will help to keep your money safer? Have they had problems in the past with hackers getting onto the network? Do they make you feel that your coins are going to stay safe if you leave them in that wallet? You need to be able to answer these questions and more before you decide to put your coins into any online wallet.

Using a hardware wallet

Leaving all of your coins online is not the best idea when it comes to using your digital coins. If you plan on using them right away and you will perform a transaction right after you exchange your coins, then it is fine to leave the coins in your online wallet. But since many people get ahold of these coins as a way to invest, it is best if you find a more secure way to keep the coins safe.

These online wallets may seem like a great idea in the beginning. You can put all of your coins in one place, and you will be able to access them at any time that you want. But, there are also a lot of hackers who would love to get ahold of this information and use your coins, without having to do the work to get their own coins. If a hacker can get onto the database of the wallet that you are using, they could wipe out your account, and there is very little that you can do about it.

Unlike your bank account with a traditional financial institution, you will not have any protections when it comes to digital

currencies. If someone gets into your account and takes your coins, you are out of luck. And when you are investing your coins, you may be too busy watching the market rather than closely watching the coins in your wallet, making it easier for someone to take the coins.

There is also another issue. Since we are working with a currency that is completely online, there are times when there could be a computer glitch that completely wipes out the whole network, or at least causes some problems with your wallet. If your coins disappear because of a computer glitch and you do not have some sort of proof that you had coins in the wallet, you will be out of luck as well.

This is where having some hardware storage for your wallet can come into play. This is when you move the access key for your coins out of the online wallet and place it on your computer someplace else. Remember that you are not able to have physical copies of the coins that you are using so this access key is like the lock that will tell the system how many coins belong to you at one time. You can save this key somewhere on your computer, usually with a good password, so that if something goes wrong, you will still be the one in control of your coins.

It is a good idea to update this access key any time that there are changes in the coins in your wallet. If you add in more coins or you use some of the coins in a transaction, then you need to update the key to reflect this information.

The hardware storage is a great idea because it keeps the coins on your computer, so they are really easy to access when you need them. And yet they are still off and away from the online wallet so you will not have to worry if something goes wrong with that. However, if someone can gain access to your computer, it is possible that they will be able to get ahold of this information and use your access key how they would like.

Going with cold storage

Another option that you may want to consider when it comes to storing the information for your coins is to use cold storage. This is when you take the information completely offline so that no hacker can get it unless they are willing to break into your home and take the information. Some people choose to place the information on a USB drive and then keep that somewhere safe, or you can even print off your private key and store it someplace safe.

The nice thing about this option is that you have proof of your ownership of the coins as long as you have that private key. Even if a hacker comes in and takes your coins, you will be able to use that access key and get them all back. You do need to remember that you must update it each time that there are changes in your account though. So, if you add more coins into the account, go through and print off that access key again, so it is as up to date as possible.

When you print off the access key, make sure that you are putting it someplace safe. It is not going to do you much good to print it off and then leave it out on your desk or someplace else where it could get lost. You should put it in a safe file, or even in a safety security box at the bank if you plan to keep the coins on hold for a bit as a part of your investment strategy. Then, whenever you need to use the coins again, you will just be able to get that access key and get ahold of them whenever you would like.

Keeping the access key as safe as possible

There are a lot of hackers and scammers out there who would love to be able to get ahold of your digital currency. They do not want to go through the process of investing or using their own traditional currency to get some of the digital currency; they want to just get into your account and take the access key. Once they get ahold of the access key, they can take all of the coins and move them to a new wallet, where you will have a hard time getting them back.

This is why it is so important to take care of your access key and keep it someplace safe. Many experts agree that you should store the access key in several safe locations to make it harder for hackers to take your information. You should choose to have at least two storage places, perhaps storing the information on the hard drive of your computer and then printing the access key out as well, so you always have the proof. Just remember to re-save and re-print the information when it updates or changes so that

it always contains the right information.

Increasing your anonymity online

No matter which type of storage you choose to use with your digital currency, it is important that you try to keep your identity secret and secure when you are online. This is one of the reasons that so many people will choose to work with these digital currencies; they like the idea that they can make purchases and do other transactions online and that it is harder for others to see their information.

Think about all of the different hacking attacks that have occurred over the past few years thanks to improper storage of your personal information. Online shopping is pretty popular right now, and people will often get their shopping done through their favorite retailers online. But to finish those transactions, they need to provide a good deal of personal information including their address, their credit card information, and their names.

Now, the company you shop with online is supposed to keep this information safe in their database. And some of these companies do a good job. But hackers are always interested in finding a way to get ahold of this information to use for their own personal gain. If they can get onto this database, they would have access to hundreds of thousands of credit card information, enough that they could spend it however they would like.

This is not usually an issue when it comes to using digital currencies. You can set up a unique address on these networks, and this can make it hard to trace the information back to you. Add in that the miners are working to create hashes all the time to protect your information, and it becomes very difficult for a hacker to get your information.

But the methods are not full proof, especially considering many governments now want the exchange sites to take in your personal information for tax and other purposes. There are still a few options that you can choose. When picking out an address to use on the network, make sure that it is something unique. It is never a good idea to put your name as the address. This may be something that is easy for you to remember later on, but it will lead a potential hacker right back to you. Come up with something unique, something that is hard to trace back to you, and your information will stay much safer.

If you plan to do a lot of different transactions with your digital coins, it may be a good idea to change up your address on a regular basis. The founder of Bitcoin actually suggested that users would change their address each time that they wanted to complete a new transaction on the network. While this may be a little bit too much for most people, it can help you to make sure that your transactions are hard to trace. Consider changing the address at regular intervals so that you don't have to worry about someone recognizing what you are doing online.

Keeping your coins safe should be your top priority when it comes to working with a digital currency. There are no protections for these coins like you will find with your credit card or with your bank. So, if a hacker can get onto your wallet or your account and take the coins and you do not have any backup available, you may be out of luck and have just lost all of your money. Consider having some backup, make sure that you store your coins somewhere other than online, and always keep the private key on hand and you are sure to keep your coins safe and sound.

CHAPTER 7: HOW TO MINE CRYPTOCURRENCIES

There are several different methods that you can use when it comes to investing in digital currencies. Each of them will provide you with a different level of work and a different return on your investment. The first option that we are going to take a look at is known as mining. Mining can be hard, and it is going to require some special hardware for you to even get started. Most people do not have the hardware or the technical expertise to make mining work. However, for those who are successful, there is a big reward. Right now, when you can successfully mine a blockchain, you will earn 25 Bitcoin, a big prize right now based on the current value of Bitcoin.

The codes that the miners will need to create to be successful, no matter which digital currency they are using, are pretty complicated. This ensures that not everyone can mine on the network and take all of the available coins in a few hours. These codes need to have a specific amount of characters to even be considered, and then those characters need to be dependent on each other so that if one-part changes, all the other parts will need to change as well; this is part of the security that comes with the blockchain technology. Let's stop now and take a look

at some of the steps that are required for the mining process and how it can really help the digital currency network.

How to get started

When it comes to digital currencies, they are designed so that only a few of their coins are released so that the network could remain steady. Each network will only have so many coins available, and no one can create more. However, the miners can release more. Take a look at how Bitcoin works. This network is designed to have 21 million coins available ever. Not all of these coins are in circulation right now though, or the market would collapse. Over time, the miners are going to work with the blockchain to create the right unique codes, and when they do this, more coins will be released into the market. This allows for a steady release of the coins that keeps the market safe.

The ideas behind mining are simple, but it takes a lot of time and effort to complete. This process is important to make sure that digital currencies stay safe and trustworthy with their users. It is in charge of protecting all the transactions that occur on each network through the blockchain record. And without this kind of record, it becomes impossible for users to know whether their payments and transactions have been completed.

There is a little bit of transparency that comes with this network though. Others who are on the network need to be able to take a look at the ledger and ensure that the information is correct so

they can trust it. However, there needs to be some security so that personal information is not taken and used improperly from this ledger. Without both of these pieces, it would be impossible to get users to join the network. This is how the miners can come into play and why they are rewarded well for their work.

At this point, we know how the ledger works with blockchain and how it is so important to make sure that these digital currency networks perform the way that they should. But you may be curious as to why a miner would want to look through this ledger and make it secure for others to use. To keep things simple, when you are looking at the blockchain, you are looking at a general ledger that will list all of the blocks that have ever been used on that network. And each block is going to hold onto all of the transactions of the network, dating back to when the network was first beginning. As a user, you would be able to go through and see each and every transaction that ever occurred on the network, if you felt like doing this.

Of course, this list is going to start becoming long over time, especially as the network becomes more popular and more people start to join the network each day. This means that there are going to be a lot of transactions that will occur. The good news is that these networks maintain their transparency by allowing all users to look at this ledger and see the transactions that are there. This is good news for you if you would like to become a miner because you will be able to use these transactions.

The miner is going to take all of the information that is in these blocks and use it to create a code. They want to keep the transparency there, so the users are still able to see the information that shows up about their transactions, but these codes will hide that information a bit, hiding some of the personal information about the user, so it stays safe. This helps to keep scammers and hackers away from your personal information.

Any time that a new transaction block is finished and then added to the blockchain, the miner will then be able to take that block and create a new unique code to keep the information from that chain safe from others. The code will have some requirement that ensures that it is safe and that it will fit in with the other codes already formed with the other blocks of the blockchain.

The miner will then be able to take that information and use it to create a new hash, following all the rules that the Bitcoin network has created to keep things secure and uniform. The most important thing here is that the hash has to make sense of what information is inside of each block and it needs to match up from the hashes that were already created in other blocks. This way, if someone ends up messing with the hash at some point, anyone will be able to tell that something is not right.

It is important to note that the miner will need to follow a few rules when creating the new hashes for them to receive their payment. This is helpful because it ensures that the hashes all match up, that other users of the system will be able to see when a chain has been messed with and ensures that everyone doesn't

jump on board and use up all the coins in a few hours.

The first rule that has to be followed is that the hash needs to be completely unique. This means that it is not found in any other part of the code and that no one else can claim your code as your own. Another thing to watch out for is that each character must rely on the other characters that are in the code. So, when one character in the code changes, whether it is in your particular part of the code or in another part, it is going to change up all the characters that come after it as well.

The hashes that you create (or the unique codes), need to also line up with all the other blocks that are already a part of the chain. It is like forming a new puzzle where everything needs to fit in with each other. The characters need to all match up so that when anything in the blockchain is changed, such as with a hacker trying to change information, it will make a mess of the whole code and any user will be able to notice the change.

There are a few different methods that the miner can use to make this happen. It is important to note that it will take some specialized hardware and some time to make this work, and it is not an investment option for everyone to try. Many people choose to join a mining group to work on a part of the code, rather than the whole code, and earn money in that manner instead.

Getting your coins

Because of all the rules that come with mining digital currencies, it can be hard to come up with these codes. It is possible for miners to create many different hashes before they find one that will actually work and which follow the rules that the network wants. You may have to work for some time before you get a good hash that is going to be accepted.

However, when you are successful at creating a hash that works, you are going to end up getting a reward for the hard work. Each digital currency will have a different reward for you. Currently, the reward for Bitcoin is 25 coins each time that you successfully complete a hash. With the current value of Bitcoin being over \$11,400, this turns out to be a lucrative method of making money through digital currencies.

There are many reasons why mining is so important for the whole network. First, this process is going to make sure that all of the users on the network can finish their transactions while also knowing that their personal information is going to stay safe at the same time. This is important because it ensures that the user is going to trust the system because they feel secure finishing the transactions without an issue.

The miner is also going to benefit from helping to secure this ledger system because they get a pretty big reward when they are done. With the price of Bitcoin being as high as it is right now, there are many people who want to become miners and earn that

big reward. But it is a difficult task, and most people will not have the knowledge, the time, or the hardware to get it done. If this is something that you are interested in doing, it is important to research the requirements a bit more to make sure you are all set.

These networks have purposely made it difficult to mine these digital currencies. First, they wanted to make sure that hackers and others were not able to get onto the network and change things up, taking away the security of the network. They also wanted to make sure that not everyone would jump onto the network and start mining, taking away all of the coins in a short amount of time. The rules are set up so that these hashes are hard to create, but this helps to keep the network safe and usable for a long time.

The challenge of mining cryptocurrencies

Mining cryptocurrencies is something that takes some time, and not everyone can do it. There is some competition (especially with the price of digital currencies going up so much), and you could easily write a few codes and get beaten out by someone else. Plus, these codes are hard to write in the first place, so it takes up a lot of your time.

Before you can get started with mining these digital currencies, you will need a few system requirements. You will not be able to get the work done with a regular laptop. You will need some

serious power to help and even a few programs that can help you to get the work done. These programs may not always provide you with the right codes, but they can definitely help you to get started and can save some time.

As a new miner, you may want to consider joining a group of miners who work together to create this code. This way, you can concentrate on just one part of the code, rather than having to come up with the whole code. You will then share the profits with the other people in the group. You may not make as much in each shot as you did when doing it on your own, but it is much easier to mine coins when you work with someone else, rather than trying to do all of the work on your own. There are a lot of different groups that you can join for this, but realize that they will probably put you through some vetting process to make sure you will do your part of the work as well.

Mining these digital currencies can sometimes be hard to complete. You want to come up with a code that is completely unique and one that is going to meet all of the rules that are listed for that currency. This helps you to get the results that you want and will keep the network safe and sound. This work is difficult to do, but it is such an effective method to help keep the digital currency network safe, and the reward is definitely worth your time, even if you are splitting it up with others.

CHAPTER 8: THE BUY AND HOLD STRATEGY

The first strategy that we are going to take a look at here is the buy and hold strategy. This is a really simple strategy that you can work with, which is why a lot of beginners like to use this to help them get started. There are not a lot of steps that are needed to make it successful, you do not have to spend all day monitoring the market to make money, and it is a good way to introduce yourself to the various digital currencies and see what they are all about before you jump into some of the other options.

With this trading strategy, you are basically going to purchase the cryptocurrency of your choice. You can wait around and see if the price of that currency will go down a bit before making a purchase, but you can also choose to jump right in and get started if you would like. Then you will just wait out the market for some time, seeing how high it can get while your coins are just waiting. You may want to consider having a hardware or cold storage wallet to hold onto these coins and keep them safe, but otherwise, you do not need to do a lot of work with this option.

The point of the buy and hold strategy is that you are going to hold onto the digital currency until it goes up in value. Once it hits a certain amount, or it reaches some other benchmark that you have selected, you will then withdraw the money and have the profit. Let's take a look at how this works with the help of Bitcoin.

Let's say that you decided to invest in Bitcoin back in February 2017 when the value of this coin was about \$2500 for one coin. You would make an exchange for however many Bitcoin that you would like, and then you would hold onto the money, doing nothing else with it during this time. By the time you got to December of 2017, the value of Bitcoin had increased to more than \$11,400, and the price is predicted to keep going up.

This means that even if you only invested enough money in purchasing one Bitcoin back in February, you would now have made a profit of almost \$9000 in just a few short months. And you can always leave the money in for longer and see where the market goes. As the value of Bitcoin keeps going up, the return on investment is going to go up as well, helping you to make a lot of money in the process.

Of course, there is a time when the market may level out, and there may even be a time when the market, even for Bitcoin, will end up going down in value. If you are not paying at least some attention to your investment, you could easily end up losing money in the process. While these digital currencies are seeing a huge surge in popularity recently and the value just keeps going

up, there will be a time when the value will max out, and you need to be prepared for that, so you don't lose your money.

Getting started with this strategy is pretty simple. You first need to pick out the exchange site that you want to use and then switch out your traditional currency for the digital currency of choice. You can then just leave the coins alone, putting them into the wallet of your choice. Then when the price of your digital currency goes up to the level that you want, you will exchange the currency back to your traditional currency and keep the profit.

One mistake that some beginners make is that they do not watch the market and they do not come up with an enter and an exit strategy. They assume that these digital currencies will keep going up in value all the time, but at some point, they will stop, and there may even be some little dips in the value over time as well. Figuring out the market and seeing when different things will happen will make a big difference in how well you can do.

First, start with an enter strategy. This is the time when the market is low enough that you will enter. Some beginners just choose to start and for this strategy, that is not too bad, but some people will want to make sure that they are getting a good value on the currency before they get going. The enter strategy is a bit more important when you are working on day trading, but it is still something to consider. Take a look at some of the charts that are available for your currency and then look at how the market is doing from day to day to determine the best price for

entering into the market.

You will also need to come up with an exit strategy. This strategy will work in two ways, and it is meant to help you lower the risk that you are going to face. The first part of the exit strategy is where you will get out if the market starts to go down. There haven't been too many times when the market for digital currencies has gone down so far, but it is still good to have this. Sometimes, it is easy to panic when the market starts going down. You may feel bad about losing money, so you stay in the market in the hopes of it going back up. But then the market keeps going further down, and you lose way too much money in the process when you easily could have gotten out much earlier.

To avoid this issue, you need to pick a point that the market can reach and you would still be comfortable with your losses. If you were to lose money, how much would you be comfortable with losing? This should be your bottom exit strategy. No matter how the market does afterward, if the market reaches this bottom part, then you are going to withdraw right away to reduce your risks. If the market ends up going up later on, you can always join back in when you are ready.

You will also need an exit strategy for when you will leave when the market is going up. Many beginners want to just join the market and ride it as high as it can go. This may look like a good strategy right now because the market is doing so well, but what happens when the digital currency market starts to level out and even go down in value a little bit? This can happen overnight and

depending on the market, and how people respond, you could lose all of your money overnight when the value plummets.

It is always best to watch the market and determine the exit plan for when you have made enough profit. This may seem silly because of course, you would like to make as much money as possible, but it helps prevent the issue that we discussed above. Pick the amount that you want, and then when the market reaches that amount, you will withdraw from the market and keep your profits.

Remember, if the market keeps going up after you have taken the money out, you can always take your investment and put it back in, following the same rules again. But it is always better to have that exit strategy on both ends of the board to ensure that your risks are as low as possible and that you don't end up losing all of your money.

When it comes to finding a strategy that is easy to follow and great for beginners to make some good money with digital currencies, there is nothing better than the buy and hold strategy. Take some time to look through the market and consider using the buy and hold strategy to help you get going in the digital currency market.

CHAPTER 9: THE DAY TRADING STRATEGY

Another method that you may want to try out to invest in digital currencies is known as day trading. Day trading has been around for some time, found mostly in the stock market, and you can use it to take advantage of the daily fluctuations that happen inside the digital currency market.

With day trading, you are going to have to work quickly to make decisions and get the most money as possible. Day trading requires that you make a purchase of your stock (or in this case, of the digital currency) and then you need to sell that stock or digital currency by the end of the day. If you end up taking up more time than a day, you are working with a different kind of trading, and it is not going to work with your strategy. For this one, all of the trades, the buying, and the selling, need to occur on the same day.

If you have ever spent some time looking at the market for these digital currencies, you will notice that there is a lot of up and down movement. Over the long-term, the market has been steadily going up, but when you look at the daily changes in the

market, you will see that it does go up and down on a regular basis. These ups and downs often happen many times during the day, and these fluctuations are how the day trader will make their money.

So, how do you use these ups and downs to help you earn an income as a day trader? You will first need to take a look at some of the charts that come with the digital currency that you want to work with. Even though these currencies are still pretty new, there are some great charts that you can use that will let you know how the market is doing and what you can be able to expect on the day that you would like to trade.

From the charts, you should be able to see what the regular lows of the day and the regular highs of the day are. Of course, there are going to be some days when the market will go a bit lower and days when it goes a bit higher, but these charts will at least give you some good ideas of what you can do and how much you can make from this kind of trading.

Another thing that you should take a look at is the averages of the coins. To make an income from day trading in digital currencies, you need to be able to purchase below the average, as far down as possible, and then sell above the average, as much as possible. This will help you to earn some good profit and will reduce your risk, especially if you can purchase the coins way below the average.

Let's take a look at how a trade is going to work. You will first need to open up an account with your chosen digital currency and move in some currency there to work with, just like you would if you wanted to complete any other transactions with these currencies. You can follow the steps that we provided in an earlier chapter to help you get this account set up if you have not done this so far.

Then, you need to take some time to look at the market. Make sure that it is still going up and that it has not gone through a big downturn or that it is flatlining and not going up at all. While most digital currencies are really popular with the general public right now, but there are a few times when you need to wait a bit before joining the market. Even a digital currency as popular as Bitcoin has had a few downturns over the years and you do not want to get into the market to day trade right when the market is about to go down even further.

After you have been able to take a look at the market and some of the charts that go with the digital currency, you will be able to make some important decisions. The best way to make money on this kind of investment is to ensure that you make a purchase at a low point on the day. If you have found the average for the day on your chosen digital currency, it is important that you never purchase the coins when they are above the average. You can purchase the coins at the average, but you will lower your risk the further below the average that you can get.

If you can get below the average, it means that you are getting a

good deal on the currency that you are purchasing. If the market goes up a little bit, even just back up to the average, you will make a bit of a profit in the process. And of course, the higher you can sell the coins above the average, the more money you will be able to make in the process.

The main point here is to purchase the currency at as low of a price as possible and then sell it at as high of a price as possible. With the help of the charts that you are using, you will better be able to predict where these two points are and when you should make a purchase and when you should make a sale to increase your profits.

This is a pretty basic strategy. You just need to purchase the coins when they are provided at a discount, or at a point below the average, and then you need to sell the coins when you get them above the average so that you can make a profit. Of course, this is a very fast-paced investment option, and there are a few things that come with helping you to see results. You have to make the trades quickly before the market goes back down, and for some people, they do not want to spend so much time watching the coins to ensure they make a profit.

There are a few things that you need to remember when it comes to day trading in digital currencies. First, you should make sure that all of the trades are started on one day and that you sell them off before you go to bed for the night. This is pretty easy to remember when you are working with traditional investments, but since digital currencies are open all throughout the world,

the market never has a closing time. You could trade at all hours of the day if you would like because the market is available.

It can be tempting to leave your money in the market overnight, especially if you have a trade that is not going the way that you would like. But this is not a good strategy to go with if you are working as a day trader. Things on the other side of the world can quickly change the price and the value of the digital currency that you are working on. Big news can occur, people may run into issues, or something else comes up, and it can send the digital currency going up or down like crazy.

You are not able to stay up all night long. You are not able to watch for those ups and downs, and this can end up harming you in the long run. A big drastic downturn can happen at night, and by the time you wake up in the morning, you end up losing all of your money in the process. No one wants to deal with this, especially if they are not able to watch for it and get out of the market in time. This is why it is so much better for you to just end your trades before you go to bed at night to ensure that you do not run into this issue.

You need to have a good amount of control over your emotions before you join in with day trading. This may seem like a simple strategy, but it is a fast-paced world that can be hard to keep up with. There are going to be times when the market quickly takes a turn downward, and you are going to lose some money in the process. If you are not able to control your emotions and make sure that you stick with your strategy, you are going to end up

losing even more money in the long run.

In addition, while there are a lot of different strategies that you can pick from to be successful with day trading, you need to make sure that you are sticking with the same strategy. Many beginners will pick a strategy and get started with it, but then will change the strategy that they are using right in the middle of their trade. Sometimes they do this because they think it will help them to earn money or that it will help them to stop some of the losses that they are dealing with. Others may switch between strategies without even realizing it in the first place.

But to get the most out of your day trading, you need to make sure that you pick a strategy and stick with it. If it ends up not working well for you and doesn't help you to make money, you can always change later on for your next trade. But you need to make sure that when you start a trade with one strategy, you can stick with that strategy until you are all done.

Another thing to consider is the cost of your trades. Each time that you switch your traditional currency over to the digital currency and then back again, you will be charged a small fee from the exchange site that you are using to switch the money. This fee is not that much for the occasional trade, but if you are trying to trade frequently, like what happens when you are working with day trading, the costs can add up. It is important to add this to your costs from the beginning so that you can make sure that you will end up with a good profit in the end.

Day trading is a great option for you to choose when you would like to make some good money with digital currencies. These currencies are seeing a huge surge in popularity all over the year, but they do have some fluctuations up and down each day no matter how well they are doing. Day traders can recognize these trends and can use them to make a lot of money. You will need to have the right amount of money, a good understanding of the market, and a good amount of control over your emotions, and you will be successful in no time.

CHAPTER 10: BECOMING YOUR OWN BROKER

There are a lot of people who are interested in joining the world of Bitcoin or other digital currencies. They see that the price of these are going up quickly and they want to be able to jump on the market and make some money in the process. While you can easily trade on any of these digital currency platforms on your own, there are some people who would rather have someone else do it for them. If you have some experience with trading any of the digital currencies, or you are willing to learn, it may be a great way for you to make an income in the process.

There are several ways that you will be able to do this. If you are already a broker or you have already worked in the financial field, you can add digital currencies to your resume. Make sure that you actually have some experience with this though. Many of the people who want to get into digital currencies would love to work with a financial advisor, rather than someone random who states they can do the work. This build up a lot of trust for them and since there are not that many financial advisors who can work with Bitcoin and other digital currencies, you will find a lot of customers.

You do not have to be a broker to offer these services though. While some people will prefer to work with a broker, the financial world has not quite caught up to the popularity of these cryptocurrencies, and there are not a lot of options for them there. If you can showcase your work and that you can provide results to people who want to get into this kind of market, you will find some customers.

You should make sure that you are looking as professional as possible. Putting up a few lines on Craigslist or another site like that is going to make you look really shady and can make it hard for you to get any of the customers that you need. A professional stance can help people to feel more comfortable with working with you and will bring in more clients.

This can mean a few things. First, get a nice website up and running and use SEO to make it easier for others to find you. Even consider a business phone so that you can keep professional and personal calls separate from each other. Your website will want to include information about you, your experience working with the digital currency market, and how you will be able to help others get the same results. And of course, you should talk about how the individual will be able to reach you with any questions or if they would like to start investing. You can even create a resume and some sample trades to the client, so they feel more comfortable when working with you.

Right now, there are not a lot of certifications or anything for

being able to trade in these digital currencies. This can be a good and a bad thing. You can quickly learn a bit about investing in these cryptocurrencies and then add it to your resume, but without the certifications, it does make it hard to prove to the other person that you know what you are doing. Consider making a little portfolio of the work that you have done in digital currencies, showing how you have invested some of your own money with digital currencies, and show this to some of your potential clients to let them know that you have worked in the market and you do know what you are doing.

After you have gotten everything organized together and looking nice, you also need to figure out the fees that you would like to offer to your potential clients. These fees are the way that you will make money on the market, and there are a few ways that you will be able to do this. You can ask for a pretty good fee since the number of brokers who can work in the market for digital currencies is currently low, but you still want to be competitive. Remember that people can do some of this work on their own, so you want to offer them a value, and you do not want to overcharge them in the process.

The first method of fees that you can consider is a flat rate for each trade. You would tell the other person that you will charge them so much each time that you trade into the market and when they take the money out. Sometimes this fee is split into two so that you can get paid on time even if the other person wants to do a long-term investment. You should discuss this fee with your client right from the start, letting them know what the fee entails and whether it covers both ends of the trade or not.

Another option is that you take a commission on what the other person earns. With how much digital currencies are growing right now, this can be a very lucrative option for a lot of traders. With this, once the client takes their money out of the market and has some profits, you will be able to take a percentage of that off the top, to cover your fees and how much you helped them in the process. This provides you with a good incentive to work hard for the other person because the more profit they make on their trades, the more money you will make at the end. The client may prefer this method as well because they know that you are likely to work harder to make them money with this method. Again, make sure that you discuss these fees right up front with the client, so there are no surprises along the way.

There are many other ways that you can charge your client as well. Some clients do not want you to do all the trading work for you because they would rather do this work on their own. Instead, they are looking for someone who will be able to advise them more than anything, someone who will answer their questions, listen to their concerns, and give them some advice on how the cryptocurrency market is doing and what kinds of trades that they should work on.

If you have a good understanding of the digital currency market, you may want to consider doing this as well. Being an advisor can pay well, and you can at least offer it as one of your services. You will not have to spend your time working on the trades with these clients. Instead, you would spend some time answering

their questions and helping them to make smart decisions when it comes to investing in these currencies.

The most important thing that you need to do if you decide to become a broker is to figure out how to provide value to the other party. Just because you know a little bit about digital currencies doesn't mean that the clients are going to just come running to you; they could do the trading on their own. But when you can provide them some kind of value along the way, such as some expert advice or the freedom of time while you do the trades for them, you will find that there are a lot of clients who are interested in your help.

CHAPTER 11: STARTING YOUR OWN BUSINESS TO MAKE MONEY

The next option that we are going to look at for investing in digital currencies is starting your own business. If you have ever thought that you would like to be a business owner, or if you already own your business and would like to be able to attract some more customers over to purchase from you, then the world of cryptocurrencies is a great one for you to consider. There are a lot of people who prefer to work with digital currencies to help them to make purchases, and so they are always looking for businesses that will accept these coins.

Working with digital currencies is not that hard of a process, which makes it surprising that there are not a lot of companies who are willing to accept them right now. There are a few of the bigger companies who are starting to move over, but still, there are quite a few who have not made the switch, and you either have to shop there with your traditional currencies which you will have to find a way to work around this.

Even if you are a smaller business, you will be able to offer your products and services on these digital currency platforms, you

just need to take a few steps ahead of time, and you need to make sure that you advertise enough so that the people on these platforms know about you. There are some great businesses that take these digital currencies, but there are also some scammers out there and being able to show the customer that you are a legitimate company is going to be your biggest challenge.

So, let's get started from the beginning. First, you need to start a business. If you already have your own business and just want to start accepting digital currency through that business, you can skip over this step. There are a lot of great business ideas that you can start with, but make sure that you are selling a product or service that you really enjoy because you may be spending a lot of time with it in the future. Come up with a business plan, get some of the product ordered or created ahead of time, so you are prepared, and set yourself up just like you would with any other online business.

Once the business is started and ready to go, it is time to get it set up with the right payment methods. Remember that you can also choose to accept other forms of payment as well, such as cash, check, credit card or even PayPal. Just because you are accepting a digital currency or two as payment doesn't mean you have to give up on the traditional payment methods. The more payment methods that you accept, the more money you can make on your business.

From here, you are going to go through and set up your own account and address through the digital currency platform. This

is going to work just like it would if you were signing up as an individual, but you will do it for your company. Some people decide to sign up with the name of their business on the account, and this is fine if you would like customers to instantly recognize you. However, if you are worried about your own security and the security of your customers, it may be a good idea to come up with a unique address to use. You will be sending this address out to a lot of people (if the business does well), and you don't really want your name to show up all over the blockchain network for hackers and scammers to get ahold of the information.

After you have signed up for the address that you would like to use, you will be able to use the address for all of your customers, and they can use it to send money to you for the products. Now go back to your website and make sure that you add on a link for accepting the digital currency. For example, if you are looking to accept Bitcoin as a form of payment, you would place a link for Bitcoin right on your website.

You can do this next part on your own, or you can make it automatic if you have the right type of programming knowledge. The customer is going to come to your website and will look around. After filling up their cart, they may be ready to place the order with you and will be taken to the payment page after they submit the rest of the information that is needed. From here, they will probably see a few different options based on the payment methods that you are willing to accept from your customers. The customer will be able to pick out the payment method that works the best for them, whether they would like to use their credit cards, send over a check, or they would like to

work with the digital currencies instead.

If the customer decides to work with the digital currencies, they will click on that link. From here, they will be sent a link to your address so that they can send the money directly to your account. This will work exactly like it would if you were just sending money over to another person, such as sending it to help someone who needed money overseas. The customer would be able to agree to the amount and then would send that money right over, using the unique address that you or the link sent over to them.

Remember that when you are working with digital currencies that use the blockchain, these transfers are going to be instant. Once the customer sends over the money, you will have it in your account within a few minutes at the most. You will be able to check on it right away, and if the money is not there, it means that the customer has yet to send it over to you. There are no issues with processing like there are with credit cards and other money transfers in the traditional world. You get the benefit of seeing that money show up in your account right away, so you can assist the customer right away.

If something goes wrong with the order, such as being out of stock of an item that the customer wants, you will have to go through a slightly different process than you would with traditional currencies. When using a credit card, for example, you would be able to just send the money back to them and reverse the transaction. With the blockchain though, all

transactions are complete right away, and they are final. There are ways around this though. If the customer is not happy with the order in some way and will not exchange it, you would just ask them for their own digital currency address and then send the amount of the payment back to them. It is not really a reversal of the funds, but it gives you the same results.

Once you see that the money is in your account, it is time to send out the order. While you may not spend all day on the computer waiting for orders, you can technically send out this order as soon as it comes in if you would like. There are fewer delays, and you get the peace of mind of knowing that the money is already in your account before you even get started on it. The customer will also not be able to go through and reverse the order so you won't have to worry about that happening either.

Adding in this as a new payment option is going to make a big difference in how much income you can make and as you can see, you will be able to set it up without having to spend a lot of extra time or do a lot of extra work in the process. Many companies will just accept one type of digital currencies, usually Bitcoin, but you can accept as many of these as you would like. The best bet to deal with this is to do some research on your current customers, or at least the demographics that you are trying to reach, and see which digital currencies they are currently using. Picking out options that suite the needs of your customers will ensure that you are getting the most out of this new payment method. It doesn't take up a lot of your time, but it can definitely open up the possibilities for how many customers you can get.

After you have started to accept digital currencies, make sure that you advertise this fact. There are a lot of customers who are interested in making purchases with these digital currencies, and once they find out that you accept those payments, they are more likely to make a purchase with you than with your competition. If you can advertise this fact effectively, you are going to bring in a lot of new customers.

Accepting digital currencies is one of the easiest ways to start your business or grow a current business. Customers want to use the safety and security that comes with these digital currencies, and they are flocking to companies that are accepting these payments. It takes you just a few minutes to get this all set up, and you are sure to see some amazing results with the growth of your company.

CHAPTER 12: OTHER METHODS TO MAKE MONEY WITH DIGITAL CURRENCIES

So far, we have talked about some great ways that you can make money when it comes to investing in digital currencies. There are so many of these currencies that you can choose from that the options for investing are endless, and some innovative investors are sure to find a lot of options that are not even on our list yet. This chapter is going to take a look at some of the other options that you can go with. Whether you are already using some of the methods that we have talked about before in your investment strategy or you are looking for something else than what has been offered, these are some great options to go with.

Invest in a company

There are many companies who have already jumped into the market for digital currencies. They have started to accept digital currencies as a form of payment, just like they do for credit card, PayPal, or even cash. This is great for the company because it allows them to reach a new audience base and it can even put them above the competition. Many customers like the safety and security that come with using digital currency and they may

easily choose a company that accepts these currencies over one that does not.

For this investment option, you would choose one of these companies and then invest in them through the stock market. You would then follow them as their stock goes up as more customers choose to work with this company over another one. You will need to do your research though; just because a company accepts Bitcoin or another cryptocurrency does not mean it is a good company. But if the company is legitimate and looks like it could attract a lot of potential customers and the stock for the company will go up.

Another option with this one is to invest in the Bitcoin company itself. This is an actual company that already has shares on the New York Stock Exchange, and you can purchase these stocks just like you can with any other company. Then as the value of Bitcoin continues to rise, your return on investment is going to keep rising as well. You do need to watch the market though. While Bitcoin and the other digital currencies are doing well right now, they will eventually even out and may fall a little bit in value. If you are not watching the market, you will miss on this and could lose a lot of money.

Invest in the blockchain

One option that is becoming more popular is to invest in the blockchain technology. Blockchain is one of the most important

technologies to run digital currencies, but there are a ton of different ways that this platform can be used. In fact, one digital currency, Ethereum, runs on the idea that investors can place their coins with a platform that they like and help developers to get that blockchain technology out to others.

With how much blockchain is growing and how many new industries are recognizing that it can change the way that they do business, it is no wonder that this is growing so much. Whether you are talking about the financial industry and how they can speed up transactions and provide customer service or about a company that wants to provide insurance or takes reservations, blockchain is definitely going to keep growing into the future and investing in it now will turn out to be very profitable.

There are a few ways that you will be able to work with the blockchain to get results. First, you could be the one who develops the blockchain technology for a specific industry. You will most likely work with a team to get this done because developing an efficient blockchain platform is hard to do, but if you are good at programming and want a good chance to invest your money, this may work well for you.

Some companies provide platforms to the industry after being paid. For example, a large group of banks may decide that they want to have a platform to use and help them provide better customer service and so they will hire developers to help them get the work done. Other times the developers may come up with

an idea and then create the platform before selling it to any business that is interested in that technology.

If you do not have a lot of programming or technology knowledge, there are still ways that you can invest. The developing companies who are working on these blockchain platforms are often looking for investors who will give them money to keep working while they finish up the project. Ethereum is a digital currency that is designed to help bring blockchain developers together with their investors so they can continue to work. You will be able to use this network to put in your investment and come up with the terms of how you will make your money back and how much.

Another method that you can try out is being a salesperson for some of these platforms. Once the platform is done, the developing company will want to find companies who are interested in purchasing this technology. With the help of a salesperson, the companies can make a lot of money in the process. You can work on commissions with these, and that help you to earn a lot of money in the process without as much risk or without having to develop the platform on your own.

Forex trading

Depending on the type of digital currency you are using, you may be able to do some forex trading with that currency. While there are a few more steps to it than this, forex trading can be

compared to trading in money. This is something that can be done with many of the major digital currencies, and if you know how to work the market and can watch how well the trends are doing one way or another, you can use this as a way to invest your money and earn some back. You will have to rely on the currency going up in value, or that you would at least be able to switch the currency over to another digital currency that is doing well to save your investment.

For this to work, you need to watch some of the charts and then purchase your chosen digital currency when the value is below market price. There are some daily, and even weekly, fluctuations, so you need to watch out for this, but getting in as early as possible will make a difference. Then you will hold the money there for at least a few days until the price went back up. Even if the price only goes up a little bit over that time, you will earn a profit.

This method is somewhere in between what you would do with day trading and the buy and hold strategy. It is usually a little bit longer of a trading strategy than day trading, so you need to have the patience to see where the market is going and if it is going to give you the results that you want. However, most people who work in forex trading will get out of the market much earlier than they would with the buy and hold strategy.

The price of different currencies can quickly change so this is kind of a risky option. You need to make sure that the value of your chosen digital currency is going to end up being exchange

out for more money later on, or this is going to lead you to lose money. You also need to count in the fees for exchanging back and forth to the profits that you potentially are going to make. If you are not able to properly read the market and how it is working, or you do not make a big enough profit before withdrawing, you could end up with losing money with forex trading.

One thing that you can try as well is switching from one digital currency to another one. There are times when the market for one digital currency is doing really well, and you decide to do forex trading with it. But then after a few days of being in the market, you find that your chosen digital currency goes down, and you would lose a lot of money if you tried to exchange it back to your traditional currency.

Rather than taking a loss on the money, you can consider taking the coins and exchanging them over to another digital currency that is doing well. From here, you will be able to exchange it over to your traditional currency and hopefully make a profit. This is not always going to work, and you have to remember the extra fees for making exchanges so many times, but sometimes it will help to save you from taking a loss on your investment.

Work with smart contracts

This is another place where you can earn some money if you have a little bit of computer knowledge and you know how to do some

programming. There are many individuals who will get on these digital markets and want to complete business, without having to deal with a third party. But they need to have some contract in place to ensure that both parties are safe in the process. This is where the smart contract will come into play. If you can create a good smart contract that is easy to use and that the two parties can fill in as they need, then you can make some money off the small commissions that they pay you.

To keep things simple, these smart contracts are going to be a series of protocols that will be used to facilitate and even enforce a contract that two other parties have agreed on ahead of time. The contracts are nice in a few ways. The two parties do not have to rely on a third party to write out the contract and enforce it, saving them time and money, and these contracts can execute on their own. Once certain conditions have been met by one or more of the parties (based on what is outlined in the contract), the contract will execute, and the terms are dealt with.

Let's take an example of how this works. Let's say that a tenant and a landlord are using these smart contracts to get things set up for renting an apartment. The tenant may put down a small deposit to hold onto the apartment for a few weeks until they can move in and until the background checks are done. If the landlord finishes up the work and hands over the key to the tenant by the date specified in the contract, the money that the tenant put down will go over to the landlord, and everyone is happy. If something happens and the landlord does not release the key on time, the money will return back to the tenant.

This helps to keep both parties safe. The landlord knows that the money is there and that as long as everything goes through properly, they will get the deposit that they need. The tenant is going to feel more confident about the contract because they can put down their deposit to hold the apartment, but the landlord will not be able to run away with the money and not hold up their end of the deal.

This is just one of the ways that you could use these smart contracts. Many individuals like that these contracts are easy to use and can save them the hassle of hiring a third party. Any time that a contract needs to be drawn up between two or more parties, it could be done by a smart contract rather than using an intermediary.

Since there is a high demand for these smart contracts, it can be a good investment for someone who can figure out how to write them up. When someone uses one of your smart contracts, they are going to pay you a small fee for doing so. This helps you to earn a little bit of money, and it helps the other party because your fee is likely to be way less than they would be paying for a third party to do all of the work. Make a nice generic smart contract that people can fill in with the specifics of their agreement, and you can easily earn a good income.

As this guidebook has pointed out, there are a lot of different options that you can choose from when it comes to investing in digital currencies. Some of them allow you to use the coins directly to make your investment, and others have you use some

of the technology that comes with these digital currencies to make your money. It is easy for anyone to get into the digital currency market, and making money through these investments is not as hard as they may seem.

CHAPTER 13: TIPS FOR PICKING YOUR WINNING STRATEGY

Investing in digital currency is a great way to make some money, no matter who you are. It allows you the opportunity to get into a market that is exploding right now, much more than any other company out there. And even though the market for some of the digital currencies is so high right now, it is still just the beginning, and you can still get in relatively early. Other companies who have seen this kind of growth in the past are now big names that are expensive to purchase and won't make as much profit unless you have a lot to start with. But digital currencies are still rising and will continue to rise in the near future.

With that being said, it is a good idea to have a strategy in place to help you to get the results that are needed for success with digital currencies. It is not enough to just put your money into the market and hope that it all works out well for you. For some people, this might work right now while the market is doing so well, but you have to remember that investing in the cryptocurrency market is just like any other investment. There are going to be ups and downs and a lot of movement, and you need to have a plan in place to deal with this.

At some point, these digital currencies will start to stabilize some more and depending on how various governments around the world start reacting to these coins, some of them may start to fall. Having a good strategy in place can help protect you if this starts to happen, making sure that you still get out of the market with a profit, even if the market ends up spiking back down.

As a beginner, you may be uncertain about the steps that you should take to pick a good strategy and see the results that you want, no matter which digital currency you decide to go with. All of them work in a similar way, and you will be able to use this information to help you pick out the winning strategy that you want. Whether you are working with smart contracts, day trading, the buy, and hold strategy, or another option, it is important to have a plan in place. Let's take a look at some of the top tips that you can use to see results in your investment with digital currencies.

Pick a good currency

The first thing that you should do when looking into this kind of investment is to pick out the right type of currency. There are thousands of these currencies available for you to choose from and not all of them are going to do all that well when you get started. It is not possible for all of these cryptocurrencies to do well in the market; some will keep on going, and some are going

to fail pretty quickly. You want to make sure that you are going with a digital currency that is going to do well, one that will stick around for a long time and can actually make you some money.

There are still a lot of good digital currencies that you can work with. Bitcoin is a great example, but you do need to be careful with this one getting oversaturated too quickly. There is also a lot of other cryptocurrencies that are not as well-known but which will be able to make you some good money in the long run if you give them time and invest in them the proper way.

The important thing here is to do your research. Each digital currency is going to have a different market and will work in a different manner. You want to make sure that you research specifically how to invest in that digital currency and even the type of market so you can determine if this digital currency will stay around for long enough or not.

There are a lot of great digital currencies out there that you can use. Bitcoin is a popular option, but there are a lot of other options that you can go with, and since they are not as popular and well-known, they can sometimes provide you with the most return on investment as well. No matter which digital currency you would like to work with, it is important to do your research ahead of time so that you pick out a good option that will last for a long time, that will continue to grow, and that will actually make you money.

Remember that there are thousands of these digital currencies that are available on the market. Many developers are hoping to create a currency that will take off and will earn them some money in the process. But the market is just not big enough for thousands of these currencies to survive. However, just because these currencies are not big names doesn't mean that they aren't profitable. This is why doing your research is so important; it will help you to determine the difference between digital currencies that are going to fail quickly and the ones that are going to make you some money, even if they are not as well-known as Bitcoin and some of the others.

Stick with your strategy

There are a lot of great strategies that you can choose when it comes to investing in the digital currency market. All of them can help you to make some good money, but you need to follow a few rules to get the most out of your work.

The worst thing that you can do when it comes to working with a strategy that you pick is that you would get into a trade and then switch up strategies. This is common with beginners who see that the market is not doing all that well in the market and they want to change things, or perhaps they don't really understand the strategy they are working with, so they make mistakes. Before you pick out a good strategy, make sure that you go through and fully understand the rules that go with each strategy. And then when you get into the market, make sure that you keep with that same strategy until the trade is done. You can

always change up the strategy that you are working with after the trade is done, but do not switch the strategies in between a trade or it will make a mess.

Learn how to keep the emotions out

No matter what kind of investment you choose to go with, it is important that you never let your emotions get in the way. You can be the most level-headed person in the world, someone who usually things through their decisions and makes sure that they are doing what is best for their money, but trading and invest can really mess with the emotions. And as soon as those emotions get in the way, you are going to end up with a mess.

If your emotions get in the way, it is a lot harder to think through the decisions. You may get caught up in the market and may make split-second decisions that are not the best for you, just because they may have made sense at the moment. Your emotions may make you stay in the market too long and cause you to lose a lot more money than if you had just placed your stops in place and listened to them. Or your emotions may get in the way when you are making money, and you will end up losing money because you couldn't just take your profits and walk, and you end up staying in the market until it started going down.

This is why it is so important to come up with a strategy and then stick with it. You will then be able to follow the rules that come with that strategy, and you will be able to make great

decisions without letting the emotions get in the way. In fact, these strategies help you to get the decisions made long before you even get into the market. Then, no matter how the market behaves, you will be able to do a trade and get out of the market without the emotions getting in the way.

Use your stop points and don't deviate

We talked a bit about these stop points earlier on, but they are so important to help you make smart decisions with your cryptocurrency investment that they are worth mentioning again. These stop points are the two points when you exit the market, no matter how well or poorly the market ends up doing later. Sometimes, in the heat of the trade, you may be tempted to stay in longer than your stop points, but these are meant to help lessen your risk and to keep your profits as high as possible.

The first stop point that you should work with is the profit loss stop point. This is going to be the point where you will get out of the market and just accept your losses if the investment reaches this point. No one wants to take on a loss, but this stop point is important because it keeps the emotions out of the game. If you see that the stock is going down, you may start to panic and decide that it is best for you.

It is common for some beginners to get caught up in the whole trade that you are working on. You may think that you can stay objective, that you will be able to make decisions no matter

what. But when you see that the market is starting to go down and that your investment is not doing well, will be able to stay as objective in the long run? For most new investors, they will see the numbers going down, and they will hold onto the stock, hoping that it will rebound again, and they will make the money back. The issue here is that sometimes the market will not rebound and if you keep the money in the market, you may find that you lose out on more than you can afford in the long run.

Setting the stop point is important to limit the amount of loss that you are dealing with. You should set a stop at the point where you would be comfortable losing that much money if the market turned. Right now, the market is doing well, but it is best to plan so that you don't lose out on more money than you can afford.

In addition, you need to make sure that you are setting a stop point for profit gained as well. With how much the market is going up, this may seem a little silly. But it is still a good idea to have in place. This is going to be the point where you will pull out of the market and take your profits when the market gets there. You should withdraw at this point, taking the profit that you earn. You can always get back into the market, but this option ensures that you do not stay for so long that the market turns, and you lose all of your profit.

Stay anonymous on these networks

While the blockchain is a pretty secure piece of technology that helps to build up trust in the network and will keep your information pretty safe, it is a good idea to keep your information as private and anonymous as possible. You will have to give up some personal information on most exchange sites now, especially if you live in the United States, but outside of that, there are some ways to remain anonymous.

First, when picking out the address that you want to use on these networks, do not put your personal information. Putting your first and last name as the address just opens you up to others being able to easily find your information. It will not take a hacker long to see what transactions you have completed and how much you earned in the process. They would then be able to use that information to find your account, take money, or to perform identity fraud as well.

In addition, it is best to not give your name out to others on this network. Your address should be enough, and if you picked out the right kind, it should not have any of your personal identification information. It is important to remember that these digital currencies are available all over the world so anyone can get on. These networks also don't have any screening options, and they are really not there to protect you at all. This means if you are not careful, you could be giving away personal information to someone who would misuse that information for their own gain.

Always be careful with who you are talking to online and on

these digital currency networks and make sure that you can keep your personal information safe. Most retailers and others who you may be working with are going to be happy just receiving your unique address to complete the transaction. If someone is asking for more information than that, you should remember that your anonymity is important online and keep that information to yourself.

Store your coins in cold storage

As we mentioned a bit earlier, it is important that you have a few different storage methods when it comes to keeping your coins as safe as possible. There are a lot of hackers and scammers out there who would love nothing more than to be able to get into your account and just take the coins. And since these digital currencies do not work the same as the big banks or other financial institutions that you usually work with, if the money disappears and you don't have a backup of them to prove the coins belong to you, then you are just out of luck.

You have the potential to make a lot of money when you are working with digital currencies. Depending on what strategy you use and how well the market is doing, you can easily double, triple, or more your money in a short amount of time. And if all those coins are just sitting around inside of your cryptocurrency account, it is going to look very tempting to a hacker who just wants to get hold of the coins, without the risk or time consumption of investing.

This is why it is important to move your coins around and place them either in hardware storage off your online account or in cold storage off your computer altogether. You will basically just receive the key and will then be able to store that someplace safe. If something happens to the money in the account, you will just be able to pull out the key and get it all back in your account. This can be helpful whether a hacker gets into the account or there is some kind of computer glitch along the way.

This kind of storage is important no matter what kind of investing you choose to do. If the coins are going to be in your account for more than a day or two, it is time to place them in cold storage for safe keeping. You never know what will happen and it is better to be safe rather than lose all of the coins. Remember to update the key on a regular basis as well. As you earn more coins, or you use some of the coins, you need to make sure the key updates with this information, so you get the accurate amount back into your account if something goes wrong.

Ask for advice from a broker when needed

No matter how much you research the niche of digital currencies, there is still going to be a lot that you can learn. This is a growing market, and it is still fairly new so even those who have been in it for a couple of years are still learning new things each day. As someone who has just entered the market and hasn't had much time to invest and look around, you are likely to have a lot of questions along the way.

It is much better to ask these questions than to ignore them and hope they are not a big deal. The digital currency market is a great one to get into right now, but it is a volatile market and could turn around any day. This is why it is recommended that you find a broker who can work with you and answer any questions or concerns that you have on the market. At the very least, consider finding a friend or someone else who has worked in the market for some time who would be happy to answer your questions and help you get along in this kind of market.

Working with digital currencies is a great way to put your money to work for you. There are a lot of digital currencies out there right now that you can choose to work with, which gives you a lot of opportunities to pick your investment. Make sure to follow some of these tips before you get started to ensure that you are making good decisions and that you will make a good profit when trading in digital currencies.

CHAPTER 14: LEARNING HOW TO AVOID THE SCAMMERS

There is a lot of commotion that comes from the world of digital currencies. A lot of people throughout the world are excited about joining into these networks. They want to be able to trade with people all over the world, they want to be able to invest, and they like all of the security that comes with using these digital currencies.

Because of all the benefits that come with working in digital currencies, it is likely to stay popular for some time. Every day the value of some of the biggest digital currencies are going way up, and this makes them even more valuable to those who haven't joined yet. All of this popularity is perfect for you as an investor. It means that you are going to have plenty of opportunities for the value to go up and for you to earn even more money in the process.

The biggest issue that comes with these digital currencies is that there are a lot of scammers out there. Digital currencies are still pretty new, and while so many people want to join the market, many of them don't really know how these markets work. They

will jump right in on the opportunity without even paying attention to what they are doing. All this excitement means that scammers are going to be able to find a lot of potential victims, and they can easily take that money and become rich off those who are not paying attention.

You really need to be careful with scammers when it comes to working in digital currencies. First, since it is so easy for people all over the world to get on these networks and there really isn't any vetting process, it means that anyone can get onto the network and do what they want. Scammers can get on just as easily as anyone else can, so you have to always watch out.

In addition, the blockchain technology can sometimes make it easier for the scammer to get away with what they have done. Transactions on the blockchain cannot be reversed. The only way that you can get your money back is if the other party sends it back to you, and do you think that a scammer is really going to send that money back to you when they are done? If you end up sending money over to someone who is a scammer, that money is gone, especially in the world of digital currencies so protecting your investment and your coins can be really important to ensure that you actually make money in the process.

The good news is that there are some different methods that you can use that will help you to avoid these scammers and keep your coins safe. When you are in the market to invest in these digital currencies, you want to make money, not to make the scammers rich.

You can never have enough security

Your digital currency, whether you are working with Bitcoin or another option, is a very valuable asset right now. There are a lot of hackers and scammers who would love to be able to get ahold of these coins, whether you are holding onto just a few of them or you have made some good money. As an investor, it is a good idea for you to add in some digital security, as much as possible, so that your passwords and your wallet are as safe as possible.

The first step that we will do is make sure that the passwords on your wallet are as safe as possible. You should have some form of backup in place to help keep the coins safe rather than just leaving them in your online wallet, but you still want to protect the online wallet as well. Remember that while the wallet is a convenient place for you to leave your coins, you need to realize that depositing coins there will not be the same as when you deposit the money into a bank. There isn't anyone who is watching this money or an FDIC that will be able to guarantee the funds from being lost. You have to be the one who takes the right steps to keep others from accessing your coins.

The safest type of wallet that you can use is one that you have taken offline and placed onto the hardware of your computer. You want to make sure that your computer also has some good security in place and is protected by a password to keep others off. Hackers will sometimes try to get onto your computer as

well, and if they are easily able to access your hardware wallet, then they can take the money from there as well.

Of course, you can choose a digital wallet as well. This is an option that many people will choose because they like to make sure that they can access their coins easily without the hassle that the hardware wallet and the cold storage have. These are often overseen by a third party, and you will want to go through and research the company before using them. If they have recently had a data breach or some other issue, they are probably not the best option to choose.

If you do choose to use a digital wallet, make sure that you only put in the amount of coins that you are willing to lose or that you are going to use right away. It is never a good idea to keep all of your coins in the online wallet at all times. As an investor, it is fine to use some coins in there if you are going to invest them right away, but the longer that you keep the coins in the wallet and the more coins that you leave in that wallet, the easier it is for a scammer to get ahold of the money.

Watch out for the exchange service you are using

It is not possible for you to go through and produce these digital currencies on your own. The options that you have include selling some items to someone and accepting Bitcoin as a form of payment, become a miner and get some coins, or you will need to go onto an exchange site and switch your traditional currency

over to the digital currency that you want to use. Most people are going to use the third option to help them to get their digital currency because it is one of the fastest and most convenient methods.

You have to remember that not all of the exchange services for digital currencies are going to be the same. There are some really great exchange platforms that you can choose to work with. These may charge a small fee to do the exchange, but it is not that much, and you will not be cheated from the money that they take from you. However, there are some services that have just sprouted up in the hopes of taking your coins and running away. These are sometimes easy to find and sometimes a bit harder to figure out. If you pick out the wrong exchange service, you are going to feel like you were fooled, especially if this company just takes the money that you are exchanging and then run off.

The best way that you can avoid this issue is to make sure that you really know about the service industry. When you know how this service industry is developing, you will have a better chance to compare and contrast the services. This makes it so much easier for you to find the companies that are the most reputable, the ones that will charge you the lowest fees and will not end up running off with your money. It is really hard to join any of the networks for these digital currencies without using an exchange site. Making sure that you pick out a company that can do the work and won't take your money is the best way to ensure that you can actually get started on your investment without losing money in the process.

Knowing how the technology works

To do well with investing in digital currencies, you need to make sure that you understand the technology that helps to run it. These digital currencies rely completely on technology, and if this is something that scares you away a little bit, then this may not be the best investment option for you to choose. But by learning at least a little bit about technology can help you to work on the currency without falling into any traps that a scammer is trying to set.

You do not have to spend all of your time becoming an expert in the technology of these digital currencies. Unless you want to start with mining, it is not necessary to learn all the different facets that come with these digital currencies, but knowing some of the basics, like what we have discussed in this guidebook, can help you out. The more information that you can learn about these digital currencies, the harder it will be for a scammer to try and trick you out of your investment.

So, before you decide to start investing in the cryptocurrency market, you need to learn as much about this market as much as possible. The more that you know about this kind of market, the easier it would be for you to spot some of the hackers and scammers who are trying to steal your money.

Purchasing online

There are many times when you will want to make a purchase online using your digital currencies. This is one of the benefits of being able to use these digital currencies in the first place. If you are going with one of the major retailers that have started accepting digital currencies, such as Overstock, you will not have anything to worry about. However, there are a lot of individual sellers that you can work with, and while many of these are going to provide you with a good experience, there are also some that are just looking to trick you out of your money and will run as soon as the transaction is done.

Any time that you want to do a transaction online, you have to do your due diligence. This means that you need to take some time to do research ahead of time to make sure the seller is reputable to use ahead of time. Looking at reviews can help you to get a good idea of the seller you are working with and will help you to prevent some of the bad sellers right from the beginning.

You can also take a look at some of the correspondences that you have with a seller. Most scammers are not going to look through their emails and make sure that there is proper grammar. Most scammers are going to send out emails and other correspondences that are full of errors and other mistakes. This is a good hint that you are working with someone who is not legitimate.

If you are talking with a seller and at any time you feel uncomfortable about the transaction, it is time to walk away. You do not have to finish the purchase, and you can pick out the

sellers that you would like to work with. Often your intuition is going to provide you with the best guess on whether a seller is someone good to work with or not.

Other methods to use to avoid scammers

There are a lot of other things that you can do to avoid the scammers and make sure that they are not able to get ahold of your personal information and of your digital coins. Some of the other things that you can do to ensure that your information and your coins will stay safe online includes:

- Never let someone else get remote access to the computer you are using. This includes support staff for your exchange site or anyone else that you are working with. Once these individuals get ahold of your computer, they will be able to do whatever they would like, and it is hard to get them off before they get damaged.
- Never hand out your 2FA codes. No exchange site is going to ask you to hand these over so don't.
- Double check that the channel you are using for support is considered a legitimate site before you try to send over funds.
- Before sending any money to another person, you need to

search for some reviews on that person. There are often some good reviews online about the different businesses that you want to work with so make sure they are reputable before sending the money.

- If you get a link in your email, it is always the best if you can copy and paste these links rather than clicking on them. This is true even if you recognize the website. There is the possibility that a scammer would have placed in a hidden link or redirected you without you realizing it.

- Any time that you are emailing or looking through a website, you will want to look for any grammatical errors. Many times, scammers are not going to take the time to proofread the right way, so their emails will be full of mistakes.

- Be aware if you find an email that says someone else has sent you some money. Often this is going to send you to someplace where you will accidentally send money out of your account, and then you can't get the money back.

- No exchange site is going to ask you for your password through email or through calls. If someone is asking you for this information, then you know that it is a scammer.

- Make sure that you are regularly scanning your computer with an antivirus. This will ensure that the scammer has not been

able to put malware on your computer without you noticing.

- The operating system and other programs on your computer need to be kept up to date. This will ensure that all discovered malware and other issues will be prevented.
- If you ever notice anything unusual, it is important to scan your computer and talk to the exchange site that you are using. They will be able to help you figure out what is going on and can maybe prevent a scammer from causing more damage ahead of time.

Scammers are always interested in getting ahold of the money that you have in your wallets. They like the idea of just being able to take your money and run. And without any type of security in place or another agency who will go through and protect your information, you could be out of luck if you end up falling victim to one of these scammers. Make sure to follow some of the tips in this chapter, and you will be able to keep the scammers away.

CHAPTER 15: THE ROLE OF THE GOVERNMENT

As these digital currencies keep on growing, the governments around the world are starting to take notice. There are just too many people who are joining the market. Some of these governments are just allowing the currencies to grow and are not doing much with them. On the other hand, there are a lot of governments who are taking notice of these digital currencies and are deciding how they will grow in the next few years.

Each country is going to react to these digital currencies in a different way. Some, like the United States, are starting to put up some new laws to prevent tax evasion from a lot of people who may be on these networks. Other countries are taking a different path, opening the doors to these digital currencies to ensure that they can keep growing and even help out with their own economy. Let's take some time to look at some of the ways that different countries are responding when it comes to the rise of digital currencies.

Australia

As the value of Bitcoin and some of the other digital currencies are starting to grow, the Australia government is starting to take a look and try to determine how they should react. Right now, most of those in government believe that Bitcoin and the other digital currencies should be regulated, at least enough for the government to tax the profits that are made from these currencies.

They do recognize that this can cause some issues. As there is more regulation on these currencies, it is more likely that people will choose to go off the grid and hide themselves. This is easily done when you are working with digital currencies, and it can be hard for governments and other agencies to find these individuals if they decide to go off the grid.

Right now, the Australian government believes that the best way for them to overcome this challenge all while adapting their system so that the digital currencies can be included in the tax payment system is to embrace the blockchain and some of these digital currencies. If the government decides to embrace this kind of technology and allow it to happen, it would be easier to collect tax payments from their citizens who use the system.

China

The Chinese government has taken a completely different outlook on these currencies compared to some of the other countries. Rather than putting a lot of regulations on these

digital currencies or even just letting the currency go without changing anything, the Chinese government decided to take the blockchain or the distributed ledger that is behind these digital currencies, to help them out. Right now, the Chinese government can use this blockchain technology to issue electronic invoices and to collect taxes.

The Chinese government was one of the first countries to recognize that the blockchain could be used to help bring the whole country together. Before this, they were having some trouble with those who lived in rural areas of the country being able to do banking or being connected when it came to finances. With the help of blockchain, these people were able to get ahold of their money and help out with commerce, even if they lived far away from a city and couldn't get to a local bank.

China did develop their own blockchain, so it is a little bit different than what you would find with the blockchain that is on Bitcoin or other digital currencies. However, it is just as efficient and has allowed the Chinese a better way to financially bring together the whole country.

India

As of April 2017, a committee that included central bank and government officials in India were ready to propose some new regulations that were related to the use of digital currencies inside of the country. This committee was formed because there

was a bit media flurry after the state minister suggested that India was going to ban Bitcoin as well as some of the other digital currencies that are growing in popularity throughout the world. Many people began to ask the government to clarify its position on these currencies, and it made it difficult for people to know the future of digital currencies in their country.

The point of this committee is to answer some of these questions, and they are in charge of making the decisions that could form regulations on digital currencies within the country.

Many people are hopeful that digital currencies will be allowed to flourish in India, but the past references by the government to these digital currencies is not all that promising. A few years ago, the Reserve Bank of India released statements that warned the citizens of India against using these digital currencies at all, and some of the statements from the recent committee have stated that digital currencies are causing them some major concerns.

The United Kingdom

In November of 2017, the UK government began a search to learn more about digital currencies, with the main focus of these currencies being a payment method and not as an investment source. After receiving a good amount of replies about how these currencies functioned, the government released a document that was meant to not only summarize the submissions that they

received, but it also outlined the views of the government and what they planned to do in concerns of these digital currencies. Some of the things that were released in this document include:

- The UK government intended to make regulations to prevent money-laundering through digital currency exchanges and to prevent any criminal use of these networks.
- The government will work to create some voluntary standards to protect the consumers who are using these digital currency networks.
- The government would look more into digital currencies and the technology that comes with them to find how it can be beneficial to others. It is estimated that the government will increase their funding of researching this technology up to ten million pounds to make this happen.

Basically, there are going to be a few things that are going to happen in the United Kingdom when it comes to digital currencies. First, the UK government is going to work to prevent any criminal activities from occurring with these digital currencies. This is meant to protect the government from individuals who try to hide their money for tax purposes and can help the citizens to stay safe when using this kind of network.

However, the government is not trying to close these digital

currency networks. They still want people to use the networks, legally and ethically. In addition, the UK government recognizes that the technology behind Bitcoin and other digital currencies is very valuable and can have a variety of uses. Because of this, the government is working on funding to learn more about this technology and how it can be used to benefit many different industries and groups.

The United States

Another country who is considering putting some regulations on the use of these digital currencies is the United States. Many government officials are looking to add in some new laws to make sure that people are not using these digital currencies to avoid making tax payments at the end of the year. The biggest worry is about some people investing their money into these networks, making a good profit, and then just not telling the government about this money, so they don't have to pay taxes at all.

Right now, there has been a movement to require some of the major exchange sites in the United States to start collecting personal information on all users, such as their names, addresses, and even social security numbers. This way, the government will be able to find out if someone made money with these coins and if they are paying taxes on them. This would affect those who choose to get on these digital platforms to invest and make money, but it wouldn't really affect those who switch the money over just to make some purchases online, and

they want to use a cryptocurrency.

However, there are many people who are going against this. One of the benefits of using these cryptocurrencies is that there isn't any government intervention present on the networks and that the user can stay anonymous. If the government starts collecting personal information on the users on these networks, it makes it harder to enjoy both of these benefits and still use the network. The government has already started to implement some of these new rules, and it is likely that you will have to provide some personal information before you can use the network.

As you can see, there are a lot of different ways that the government is responding to these digital currencies. Some are trying to ban the currencies altogether, some are putting on regulations to ensure that no one tries to do tax evasion, and others have already taken the technology that comes with these digital currencies and created their own to improve their financial institutions throughout the country. The way that the government in various countries respond to these currencies will make a big difference in how much they grow and stay around in the future.

CHAPTER 16: THE BEST CRYPTOCURRENCIES TO WATCH IN THE FUTURE

If you are interested in getting started with investing in digital currencies, it is important that you take the time to find the right currency that will give you results. There are hundreds of these currencies around, but it makes sense that most of them are not going to produce the results that you want. Most of these currencies are too small or unknown that they will disappear in a short amount of time, and it is not worth your time to even look at them. With all of the options available for these currencies, how are you going to choose the one that will actually make you money in the process? Let's take a look at some of the top digital currencies that you should be watching out for if you want to see a good return on your investment with cryptocurrencies.

Litecoin and Bitcoin

As we discussed a bit earlier, Bitcoin was the first digital currency to be developed during 2009. It came about at the perfect time, providing an alternative to the traditional currencies that were failing at the time. This currency was available online, provided a way to spend money without the

help of a bank or a central government deciding how the currency will work, and it made transactions easy and instant. It also used the technology of blockchain to keep it safe and secure when doing purchases online. It has seen a huge increase in popularity and value lately, making it a great investment for most people.

Litecoin is similar to Bitcoin, but the reason it was developed was to avoid some of the issues that Bitcoin has faced over the years. For example, you will find that Litecoin has been changed around so that it can generate blocks on the chain faster and handle more transactions, helping it to keep up with the growing demand of digital currency better than Bitcoin can do.

Bitcoin Cash

Recently, there has been an official split that happened in the Bitcoin network. While Bitcoin Cash is technically considered a competing coin online, most people don't realize that there are differences between Bitcoin and Bitcoin Cash. Bitcoin Cash became its own digital currency because there was a divide in the idea of how Bitcoin should be used. A big minority saw that there were going to be some issues with using Bitcoin in the future and the best way to solve this was to split up and start with their own form of digital currency.

Bitcoin Cash sees some benefits from this split. First, this split has renewed a bit of the excitement for Bitcoin in the market,

and it even helped to raise the price of both coins in the process. On its own, Bitcoin Cash is becoming more popular, and investors are starting to take a look at it, at least more than before. It is even possible to get ahold of these coins without having to complete any purchases in the process.

The developers of Bitcoin Cash have made it easy for users to get their new coins. If you already have a private key for Bitcoin, you will be able to use that same key to access coins with Bitcoin Cash using their regular Bitcoin. This makes it easier to switch over if you would like.

If you have never used Bitcoin in the past and want to be able to work with Bitcoin Cash, you can do so with one of the current cryptocurrency exchanges that are available. Coinbase is one of the most popular sites available, but they are not accepting Bitcoin Cash right now. For the most part, you will need to go and get some Bitcoin first and then switch it over to get the results.

Altcoin

Altcoins are going to be any digital currencies that are not considered a part of the Bitcoin network. This can include a ton of different cryptocurrencies. Some of these are really great options that will be good investments for you to enjoy while others are not worth taking your time to look at. Knowing which altcoins are going to be good options for you can help you to get a

great return on investment.

Ethereum

One of the biggest altcoins that you can choose for a good investment is Ethereum. This company will also rely on blockchain technology to help it work, but it will use the technology in a different way. When working on this kind of altcoin, you will work with a coin that is known as an ether, which you can then transfer between accounts, so you can compensate the other person when a transaction is completed.

You will find that this kind of currency is going to be used in a slightly different way than Bitcoin is. Instead of using the ether to make purchases of most goods and services, the ether is often used to help fund developers who want to create their own blockchain technology. You can invest in their company and their ideas using the ether and then will receive compensation when they make a sale later on.

Ripple

Ripple is another option that you may want to consider when looking for a digital currency to invest in. Ripple is considered a type of open payment network where currency can be transferred. It is a new idea, but the goal of working on this network is to allow the users to break out of the financial

networks they are currently on, like banks and credit cards, and use the technology that comes from blockchain to get their transactions taken care of quickly and without any delays.

According to what Ripple writes on their website, the company that was in charge of this coin wanted to use this currency to address the issues of the flow of money in most current economies. There are plenty of people who have a lot of frustration when they work with financial institutions and other banks. We can finish a transaction in a matter of seconds, but it takes days before the bank can get the transaction to go through. But most people feel like there is nothing they can do about this issue. Ripple has a goal of building up a digital currency that is decentralized, one that doesn't cost a lot of fees and is easier to use compared to our current financial institutions.

There are going to be some differences and some similarities that come with Ripple and Bitcoin. Both of these are considered digital currencies that are relying on mathematical formulas to run them and both are going to have limited coins that will ever be in circulation. It is even possible to use both networks to move coins from one account to another without the delays and fees that are found with traditional banks.

However, it is important to note that there are a few differences that come up between these two networks. Ripple is often seen as a complement to Bitcoin, and not one of its competitors. There are even options on the Ripple platform for using Bitcoin. And you will find that the network for Ripple will be a bit

different than what is found with Bitcoin because it is designed to transfer currency in many different forms, whether you are working with a digital currency or a traditional currency.

If you are looking to invest in digital currencies, it is important to pick out the best one so that you get a good return on investment. Try out a few of these popular cryptocurrencies to help you get going on the right track with your chosen investment.

CONCLUSION

Thanks for making it through to the end of this book, let's hope it was informative and able to provide you with all of the tools you need to achieve your goals whatever they may be.

The next step is to decide which type of digital currency you would like to work with. This is going to help you to determine the best type of investment style that you want to work with. Each cryptocurrency is going to be different, and having time to research and truly get to know how each one works will ensure that you are getting the most out of your investment options.

This guidebook took some time to look through all the different aspects of trading in cryptocurrency that you need to know. While cryptocurrencies are fairly new, they have garnered a lot of attention, and many of them are gaining in value each day. This makes it the perfect opportunity for an investor, either a new investor or one who has been in the market for some time, to make money if they have the right strategy.

Inside, we have taken a look at what cryptocurrencies are all about, how to open up your own account to get started, what determines the value of the different digital currencies, and even some of the best trading strategies that you can choose when you are ready to make money with this lucrative new currency. While most people are jumping onto Bitcoin and other digital currencies to do safe and secure transactions, you could jump on and take advantage while making some money with the help of the information and tips in this guidebook.

When you are ready to get started with investing in digital currencies, no matter which one you want to check out, and you want to make sure that you are prepared to get the most out of your investment, make sure to check out this guidebook!

Finally, if you found this book useful in any way, a review on Amazon is always appreciated!

BITCOIN

Guide to Cryptocurrency Trading and Blockchain Technology

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INTRODUCTION

Congratulations on downloading this book and thank you for doing so.

The following chapters will discuss everything that you need to know to get started with investing in Bitcoin. There are a lot of people who are just starting to hear about this great currency and who are deciding whether or not this is a currency that they want to support. Because of this, it is the perfect time to join the market and start making money on the value going up. There have already been enormous increases in the value of Bitcoin over the last year, and as the news spreads, it is likely that this value will continue to go up.

Investing in a currency that is so new can be a little bit daunting. You may have been considering it for a long time, but are worried that you will not know how to get started. With the help of this guidebook, you will be able to start out with investing and become a professional in no time.

In this guidebook, you will learn all the tips and tricks that you need to help you get started with investing in this digital currency. We will look at what Bitcoin is all about as well as how the blockchain technology helps this currency to run. We will look at some of the strategies that come with this type of investment, such as the buy and hold strategy and day trading as well as how to reduce your risks so you can get the most out of your money.

Investing in Bitcoin is a new and exciting world. Make sure to check out this guidebook to learn more about how to invest in this currency so you can make the most money possible with as little work as you can!

There are plenty of books on this subject on the market, thanks again for choosing this one! Every effort was made to ensure it is full of as much useful information as possible. Please enjoy!

CHAPTER 1: THE HISTORY OF BITCOIN

Before you can join in on a new investment and put your money into it, it is important that you learn what it is all about. And when it comes to these digital currencies, they are so new and so misunderstood, that there is bound to be a lot of information out there that is not all that accurate concerning them. While many people are jumping onto the idea of Bitcoin and the other digital currencies, and while they can be a great way to make some money, it is important to take the time to learn about these currencies so that you can make the best decisions for you.

To start, let us look at what Bitcoin is all about. Bitcoin is known as a digital currency, which basically means that this is a currency that is traded only online. You will not be able to go to an ATM or a bank and withdraw some of these coins, but you will be able to make purchases and send money with it all around the world. Users have even found that exchanging their traditional currency over to Bitcoin is relatively easy as long as they can sign up for an exchange site. Despite all of the benefits of using these currencies, you will never be able to go and print off the money and use it around with you. All of the transactions that you do with this currency need to be done online.

Bitcoin was the first of its kind and was released in 2009. It is unknown who created this currency, but it was done under the name of Satoshi Nakamoto. This person or group of people worked on creating white papers that described how Bitcoin was supposed to work to get it started, but they never revealed who they were and they are not sticking around and taking control over Bitcoin. Once Bitcoin was up and running, this group or individual has stopped being heard from.

From Bitcoin, many other currencies started to form. Some of them focused on fixing some of the bugs that came with Bitcoin and some would be completely different. For example, Ripple is a popular digital currency that will send money around the world, whether you are relying on traditional currency or digital currencies. Ethereum is a digital currency option that offers a free open-sourced platform of blockchain to make it easier for developers to work with and create their own applications of the blockchain. These are just a few of the differences that you will find when it comes to the various digital currencies that are out there.

At the time, and still even today, Bitcoin was a leader in the industry. There had been other times when a digital currency was tried out, but often it was supported by a big company who wanted to give it a go. They may have had trouble with the ledger working right, trouble with showing enough transparency, and sometimes they just had internal trouble that made it hard for the currency to stick around and survive. At the time that Bitcoin

came out, many people had given up on the idea of having a digital currency that would work for more than just gaming.

However, Bitcoin was a new idea that caught on quickly. Most experts believe this is because it came out at the perfect time in history to be accepted. It came out in 2009, right after huge economic crashes happened all throughout the world. People were frustrated with how their governments had been handing money all this time and they liked the idea that they could use a currency that was simple and did not have that government control at all.

While Bitcoin has a slower start getting going and it was associated with the black market and some illegal activities in the beginning because of the nature of its anonymity, it has quickly caught on in the mainstream and now there are people all around the world who have decided to start using this currency to help them to send payments, make purchases, receive money, and even to invest. The value increase in Bitcoin has been crazy over the past few years, and people are taking notice.

There are even a few big companies who will now accept Bitcoin as a form of payment and it is even possible to go into some local stores and present the barcode for your private key to pay for things using Bitcoin. It will take some time for some of these big name companies to catch on, but you will find that it is something that is happening pretty quickly and you already have some different options when it comes to using Bitcoin for some

of your purchases.

Over time, there have been some issues that have arisen with Bitcoin, which is to be expected since it is one of the first of its kind. One of these issues is a big split that came and ended up creating Bitcoin and Bitcoin Cash. Those who stuck with Bitcoin liked that it was secure and that there were so many Bitcoin that could be released into the market at one time. They basically did not think that anything should be changed when it came to using the Bitcoin network.

On the other hand, those who went with Bitcoin Cash felt that there needed to be a few changes to ensure that this currency was going to stick around and stay relevant, especially since there is now so much competition on the market for these currencies. They wanted to go through and speed up transaction times (Bitcoin takes about ten minutes for a transaction, which is fast but much below what the other currencies can do), change how many coins could be in the system, and basically fix any of the bugs that were found with Bitcoin.

These two systems are pretty similar and will follow a lot of the same rules. If you are a Bitcoin user and have some coins in Bitcoin, you can always switch over to Bitcoin Cash. You can do this without having to worry about the exchange rate or anything because they wanted to give people a choice about which option they went with.

As you can see, the world of digital currencies may be new, but there are already a lot of things that are changing in it. Bitcoin has a lot of competition now that it did not have in the past and while it continues to grow, the way that it reacts to this competition and the way that it evolves will determine whether it is a currency that sticks around for the long-term.

While these digital currencies have only been around for a little bit and they are still new, it is pretty obvious that right now, Bitcoin is the one rising to the top. Bitcoin is designed to hold onto your security, to be transparent, and to be easy for its users to work with so that they do not have any reservations about working with an online currency. While most people are not used to all of the ideas that come with a digital currency because they are used to working with traditional currencies, digital currencies have a lot of benefits that are drawing people in each day.

You will find that Bitcoin, as well as many of the other digital currencies, will have a few variations from traditional money. If you do a lot of online shopping, Bitcoin will be very similar in the way that you can use it. This means that you can use Bitcoin and other digital currencies to make purchases and to trade. However, there are some differences between the two currencies, and most agree that these differences are what makes Bitcoin so successful.

For example, you can use Bitcoin to complete all your transactions without all the fees. There also are not any issues

with a central authority who is trying to control the currency or messing around with it for their own agenda, and you can use Bitcoin as a form of payment for various goods and services with many people and businesses throughout the world.

The point of choosing to work with Bitcoin is because it was created to be used in the digital world, and it is run by a community, rather than the government. With this currency, it is not going to matter where you live, you can join the network as long as you have an Internet connection and some money to exchange. There are even different methods that you can use to earn the Bitcoin so this opens the door for a lot of people. There are some people who choose to mine the coins and earn them that way. Some will open up their own business or sell a product and will accept Bitcoin. And others will just take their traditional currency and switch it over to the Bitcoin.

So, a big question that a lot of people have about Bitcoin is why they would want to choose to use this currency rather than working with their traditional currencies? They may wonder whether or not dealing with a digital currency is the safe and secure method of doing transactions. There are actually a few reasons why a lot of people around the world are choosing to switch over to a digital currency instead of using their traditional currency. The first reason is that people like the idea that this currency is decentralized. They like the idea that they can use a currency, make money, and complete purchases without the government having to be involved all the time. This can take out some of the control and corruption that often happens in the world of money.

At this point, it should be pretty easy to see that there are a lot of different benefits that you can get from digital currencies and that they are a bit different compared to what you worked with in the past with your traditional currency. You can work with this kind of currency online, there is the blockchain technology that makes this network safe and efficient to work with, and you will never have to worry about a big government agency trying to take over the currency and mess around with it.

In fact, there are some rules that have been added to Bitcoin to help make sure that it works the right way. For example, there is only a set limit of Bitcoins that were ever created; no more can be added ever. Now, only a few were released originally to start out the program, and the miners will be able to do their work to release some more Bitcoin over time. But overall, you will find that this helps to keep the Bitcoin market stable because you will not have to worry about over inflation of the currency.

The next question that a lot of people have when it comes to using Bitcoin is what it is based on. When we take a look at traditional currencies, the currency is based off what a government agency states. So, when it comes to the USD, the dollar is based on the gold standard. What this means that, in theory, you would be able to take a dollar bill to the bank and then exchange it out for that much gold. This is not something that you can technically do, but it is the basis for how the dollar works in the United States.

Bitcoin will work on a different idea than this. Instead of being based off something like the gold standard, Bitcoin will be based on an online code, so it will completely rely on mathematics, as well as what the programmer added to the code, to tell you what the worth of the currency is and what it is based on.

Now, you will find that the value of Bitcoin will be based on the idea of supply and demand. There are a limited amount of coins on the market right now, which is the supply, and there are so many people who are interested in owning some of the coins. It does not matter if they want to use the coins for investment purposes, to show, or to do something else; they all want to get into the market and this is the demand. When the supply of coins is low and the demand is high, then the value will change for these coins. Any time that you see the value of Bitcoin go up, it is because there is a growth in the demand, without a growth in the supply, of these coins.

The idea of Bitcoin had been tried in the past. There were a few other times when the idea of digital currencies was tried out, but none of them seemed to be able to last. They were not able to build up the trust that was needed on these kinds of networks (remember that they are supposed to work without a third party or a government agency interfering with them), and many times the government was able to go after them as well.

It was not until Bitcoin came around that things began to change. And the biggest reason that it has been successful and has avoided some of the problems that other previous digital

currencies had to deal with is that this network used the blockchain technology. This was a really neat technology (which we will talk about more later), that was able to help keep the Bitcoin network safe, secure, and transparent all at the same time. Without this kind of technology, it is unlikely that the Bitcoin network, or any of the other digital currencies that are available right now, would have succeeded.

Due to the big success of Bitcoin, there are many other digital currencies that are available on the market now. None of them have been able to see the success of Bitcoin as a whole, but they have each come around and filled in a niche that was needed at the time.

How much is Bitcoin worth?

One of the biggest questions that you may be asking if you are interested in getting started with Bitcoin is how much it is worth. You do not want to get into this kind of investment if you are not sure that you will be able to make some good money in the process. Just like when you work with any type of currency, the value that is associated with Bitcoin will go up and down and these variations can happen from day to day. These variations are quite big right now because this currency is so new and a lot of people are just now learning about it. As more of these people start joining the network after hearing about it, the value of Bitcoin will go up.

You will notice that while the general trend of Bitcoin is going up, there are a lot of fluctuations that occur in the market. This can happen from one day to the next, but there are times when it will occur from hour to hour. This means that you will need to watch the market a little bit and look over some information ahead of time to ensure that you understand how this currency has been reacting and to determine if the value is good enough for you to join in.

For example, the value of Bitcoin at one point in October of 2017 reached \$6000 USD. Back in February of 2017, the value of a Bitcoin was only \$2500. This is a huge increase in less than a year and while there have been some major dips that have occurred during this time, they have not really stopped the value of Bitcoin from going up. Before you decide to invest or trade in Bitcoin, look up the Bitcoin exchange rate to find out exactly how much Bitcoin is worth when you are ready to get started.

Why do people like Bitcoin?

There are a lot of reasons that people choose to work with Bitcoin rather than a different digital currency or rather than working with their traditional currencies. First, people like that these currencies are not regulated by a government entity at all. They can work with their own money without having to rely on how well the government can handle the money. All of the Bitcoin networks will rely on a mathematical code to help it take care of everything that is going on in the network, helping to add some safety and security.

Some people like to work with Bitcoin because it is a great way to keep their personal information private when they are shopping online. With all of the hackers being able to get online and steal your personal information, it can be hard to know whether your information is safe online. The Bitcoin network is set up so that you can keep your information safe while doing various transactions online.

People who use the Bitcoin network also like that they can make lots of transactions instantly. While bank transfers will often take up a lot of time, up to five days in some instances, Bitcoin, with the help of blockchain technology, will be able to do this in a few minutes. You no longer have to wait around and hope that you can get your money at some point before you send out a product. You will be able to receive that money right away and then it is all set for you.

These are just a few of the reasons that people enjoy working with Bitcoin in their daily lives. They like the freedom of being able to be in charge of all their money in a way that is just not possible when you are working with traditional currencies.

Traditional currency versus Bitcoin

You may be curious as to how traditional currency and Bitcoin will be different. These two actually have quite a bit in common

when it comes to spending them and using them to purchasing things, as long as you realize that working with Bitcoin will be completely online. If you just plan to use Bitcoin as a way to send and receive money online, it will look a lot like online shopping for you.

However, there are other ways that Bitcoin will be different compared to traditional currency. First, you will not be able to print off paper copies of your coins and then take them to the grocery store to use them. These coins are all part of a mathematical code online and that is where they will stay. You can print off the private key that unlocks your account and your coin total, but you can't print off any of your coins.

One big difference that a lot of people like about Bitcoin is that it is not controlled by a central authority, a bank, or a government agency. After the Big economic crash throughout the world of 2008, many people have stopped trusting their government to help them get the results that they want with their money. They do not want the government to have anything to do with their money in the first place. They like the idea that they can keep their information online private and that it is not controlled by any government. They figure that the money must be a lot safer this way.

And of course, people love that they can use this network and still keep their personal information safe from others and that it is a secure network. You have to go through and make sure that you are taking the proper precautions to keep those accounts

safe, but if you do, this can be one of the safest methods to use to make payments online without having to worry about someone stealing your information.

As you can see, there are a lot of benefits that come with working with Bitcoin. It is a currency that will work well for everyone and everyone will have their own niche to get into. Make sure to check out some of the other cool things that you can do with the help of Bitcoin.

CHAPTER 2: WHY SHOULD I WORK WITH BITCOIN?

Just a few years ago, the idea of Bitcoin was relatively new. There had been a few attempts at a currency like this, but for a variety of reasons, including that they were not decentralized, these currencies never made it out of their niche markets. Many people thought that it would be impossible to set up a digital currency online that people would enjoy and would have all the benefits that they were looking for.

When Bitcoin came out, it was like the beginning of a new era. People loved all the benefits that came with it and how easy this whole thing was to work with. Even people who have never spent much time online were able to get on the network and see some amazing results.

This chapter will spend some time taking a look at the benefits, as well as some of the downfalls, that come with using the Bitcoin network. This is a great network to go with, but it is important to look at both sides of things to ensure that you know what you are getting into.

The benefits of Bitcoin

There are so many benefits of working with Bitcoin, which is probably why so many people are jumping on board and giving it a try. They like all the freedom that they can get with these coins and that they do not have to worry about a government agency interfering. They like that they can pick from a lot of options to help them earn money, to send money, to make purchases and so much more. These digital currencies may be pretty new, but they are here to stay thanks to all the cool features they possess. Let us take a look at some of the great benefits that come with using Bitcoin in your own life.

Easy to use

Using Bitcoin has never been easier. You simply need to have some traditional currency ready and get onto an exchange site. Then, as soon as you set up your preferred payment method, you can exchange out for the coins that you want. As long as you have a good Internet connection and pick out a wallet, you will be able to use your coins on the network. It is really that simple!

Secure

You will be surprised at how secure working on the Bitcoin network can be. All of us have spent some time doing online shopping, but we also know that this can be a little dangerous to

do because there are so many hackers and scammers out there who are trying to get our information. We have to be smart and learn how to take care of things on our own if we want to be safe and usually, it is a gamble whether someone will get our personal information or not.

This is not such an issue when you are working on the Bitcoin network. You will be able to choose a lot of the security that you are working with, which will keep your information as safe as possible. You can choose a Bitcoin address that suits your needs, you can send transactions without having it linked up to any of your payment methods, and you can choose how much information you are willing to give out.

The blockchain technology is really good at its job. It will allow you the opportunity to know that your information and money are safe on the network. We will talk about this a bit more in future chapters, but this technology is taken care of by the miners, who will ensure that codes are placed on your transactions so no one can find your information, even if they are looking right at the ledger.

Fast

Working with banks can be a challenge. They are nice in some regards, such as when you would like to get a mortgage, but all the red tape and the hassle can make it hard to appreciate these institutions. And when you have to wait between three to five

business days for any transaction to go through, no matter how big or small, it can become really frustrating to deal with.

The reason that it takes the bank so long to process transactions is that it has its own ledger. Each bank and financial institution will have their own ledger that they have to spend time sorting out, and this can take them some time. When you use the blockchain ledger on Bitcoin, you are not going to deal with this issue. This is because the blockchain is one ledger throughout the entire system. It can do all this work in a few minutes, rather than a few days.

This is good news for a lot of people. If you want to make a purchase, the seller will know that the money is in their account within a few minutes, rather than in a few days. If you want to send money to someone else, they will be able to get it fairly quickly. This can be a nice change from all the waiting that we are used to doing with our traditional banking system.

Low transaction fees

If you have ever tried to send money to a friend or family who lives far away, you are probably sick and tired of all the fees that you need to deal with. Banks and other financial institutions will charge large fees to help you get the money switched over to someone else, and this can sometimes make it prohibitive to even send money in the first place.

When it comes to working on Bitcoin, you will have very low transaction fees. This is because the blockchain can send and receive money, regardless of where you are in the world. You could be right next to the person or you can send it over across the world. The blockchain technology will be able to take care of it for you. Usually, the only transaction fees that you will need to pay include the fees to exchange your traditional currency over to Bitcoin.

No government interference

Throughout the world, people are getting sick of having their government around, messing around with their money. After the economic crises that recently happened, you may be worried that the government will keep having issues with the currency. They do not think that the government is doing a good job with running the money and that the recent economic crisis is a result of all that.

When you are working with Bitcoin, you will be able to work with a currency that does not have a government agency that is controlling it. This is a great thing for those who are not interested in letting the government get ahold of their information or control their currency.

Thanks to the blockchain technology as well as the coding that puts Bitcoin together, it is possible to get great results with security and all the features that you need without having to

worry about how a government will mess around with it. It does take away some of the safety and help if something happens to your money, such as the FDIC backing in big banks, but you can protect yourself if you take the right steps.

The ability to keep your information private

You will be surprised at how easy it is to keep your information private when you work with the Bitcoin network. Unlike some of the other payment options that you do online, you do not have to spend all your time worrying that someone can get ahold of your information and use it in any way that they want. You will be able to pick out your own Bitcoin address which can be as unique as you would like and that is the only information that will show up when someone tries to see what you are up to.

This means that your bank account, your credit card, and your online wallet will never be associated with a transaction that you are using. Instead, you will only need to present the other person with your Bitcoin address, and then the transaction can be complete. Even if a hacker does get onto the network or into the wallet, if you took the right precautions, you will find that it is hard for them to get ahold of the information that they need.

Many ways to invest your money

If you are looking for a great way to invest your money, then

there is no better option to go with than the Bitcoin network. There is an investment opportunity for each person who wants to get started. Whether you are getting started as a beginner or you are just looking for a new way to expand out your portfolio, you will find the right investment option for you in Bitcoin.

For those who are looking to find a good investment strategy that can last for the long-term and will not require as much work out of them, working with the buy and hold strategy is a great option. It really does not take all that much work, as long as you watch the market on occasion, and you will be able to make a good amount of money as long as you are willing to hold onto the coins for some time

Now, if you would like to earn money in the short term and earn a lot during that time, then you could also choose to work with day trading. Day trading has a lot more risk to it, but it does provide you with a way to take advantage of the constant ups and downs that come with the Bitcoin network.

These are just two of the options that you can choose when it comes to investing in the Bitcoin network. You can take a look at how to invest in the Bitcoin network and even how to start your own business and then accept Bitcoin as a form of payment. And as this currency grows more and more, you will be able to enjoy more ways to invest your money.

The blockchain technology

One of the best things about working with the bitcoin network is that you will get a chance to work with the blockchain technology. This technology is so amazing because it is basically the reason that people actually trust the Bitcoin network. Blockchain technology is basically a ledger that will ensure that all the transactions inside of the Bitcoin network are taken care of and that you are getting everything in order. It also provides the transparency and the security that are needed on a network like this. We will talk more about the blockchain technology in a future chapter.

The negatives of using Bitcoin

There are a lot of benefits that come with investing and using Bitcoin for your own personal needs. You will be able to make a good amount of money, you will be able to make purchases without someone trying to steal your personal information, and you can do things with a certain amount of anonymity. These are all great things when Bitcoin is compared to using traditional currencies.

With that being said, there are a few negatives that come to using Bitcoin. It is a great way to do a lot of things, but you also have to be aware of the dangers and the risks of using this currency. Here we will talk about some of these dangers and risks so that you can be fully prepared ahead of time.

Lots of hackers to deal with

As the value of Bitcoin and other digital currencies go up, there will be more and more hackers who get on the network. These coins can be really valuable, and there are super interested in being able to get on the network and steal someone else's coins, rather than doing the work or using their own traditional currency to get the coins that they want.

This is why it is always a good idea to backup the coins that you have in a safe and secure place. Unless you plan on using the coins right away to purchase something or to use them for day trading, it is best to consider hardware storage or cold storage so that it is much harder for a hacker to get ahold of your coins and take them as their own. Once a hacker gets ahold of your coins, it can be nearly impossible to get them back.

No one to help if something goes wrong

If something goes wrong with your investment, if a hacker gets into your account, or there is a big bug that goes through the Bitcoin network and wipes out your account, where are you going to turn? As the Bitcoin network is set up right now, you have no option. There is not a big entity that is controlling the network, one of the major benefits that come with it, so you have nowhere to turn to help you out.

Luckily, there are a few steps that you will be able to take to help you keep your investment safe, but you are the one who has to take the initiative to do them. If you forget to do the work or you just do not think about it then there is no one to help you when it gets too late.

Ability to have a crash

There is a big possibility that this market will end up crashing on you at any moment. It is not as secure as some of the other investments that you may have and without a long history to support it, it can be a scary investment to work with. With that being said, Bitcoin is a great way to make money, but you do have to realize that there are times when the market may crash on you, and you will not be able to predict it ahead of time. If you are not ready to be a big risk taker in your investments, then it may be best to pick out an option that is not in cryptocurrency.

Nothing to support it

Unlike working with traditional money, you are not going to get the benefit of having something to support this money. With traditional money, it is usually held to some kind of standard, such as the gold standard, or there is a big bank or financial institution that will help to make sure that the money does well. If you put money into a bank account, it will be there when you want to withdraw it. There is some security in that system.

However, there is not this expectation when you are working with the Bitcoin network. You have to watch what is going on and protect yourself, or you will be out of luck. There are a lot of people who are surprised when the market crashes, when they are hacked or when something else goes wrong and they have no one to turn to to help them out. They just lose the money and have no recourse to help them, which can be scary to a lot of people.

As you can see, while there are a lot of benefits that come with using Bitcoin in your life, there are also a lot of negatives that can come as well. It is important to see both sides of the spectrum because it will allow you to fully understand what you are getting into when you are joining the Bitcoin network.

CHAPTER 3: UNDERSTANDING BLOCKCHAIN – HOW THIS LEDGER KEEPS THE NETWORK SAFE

In this guidebook, we have spent a little bit of time talking about Bitcoin and how it all works. We have also mentioned the great blockchain technology that you can use as well. This blockchain technology is relatively new since it was released at the same time as Bitcoin, but there are so many neat things that you will be able to do with it. Most people do not fully understand how this technology will work and may feel that it is the exact same thing as Bitcoin, but this is not true. They are two separate items, and while blockchain helps to keep Bitcoin running, it is a platform that can run on its own.

To keep things simple, the blockchain technology is basically going to work like a ledger that holds onto information about all of the transactions that happen on the Bitcoin network. This ledger started on the first day that Bitcoin was ever used and will continue to record all of the transactions that come around until the day Bitcoin is no longer in use. When you can understand how blockchain does its job, it becomes a lot easier to understand how Bitcoin works.

The blockchain will be like a database that will work to store all of the transactions that occur on the Bitcoin network. The blockchain can also be designed to hold onto anything of value, such as contracts, if they are used on Bitcoin as well. However, it is important to notice that blockchain will be more than just a database because there is just so much more than can occur with it. Yes, the blockchain is in charge of storing information, but it can also do a lot of other things at the same time.

Let us take some time to compare the blockchain with some of the traditional databases that you work with so that we can understand how it works. With a traditional database, a transaction would be added to the ledger, usually recorded by some third party such as a bank. The bank or other third party would be responsible for recording all of the information and they would also need to be able to create trust between themselves and the others who are using their ledger.

For example, if you have a checking or a savings account and you choose to leave some money in that account, you are doing this action because you trust that the bank will keep it safe and that it will record the amount that you added. The bank will record each time that you put more money into the account, as well as recording when you take money out or you spend it.

You are even trusting of paying for services and making money transfers out of your account. You often do these with credit card companies and other banks. You know that both of these companies will take out the right amount of money from the

account and then will transfer it to the right person in the process. Of course, your seller will need to trust the system as well. If the seller does not trust the system, then there is no way that you would be able to leave the grocery store after putting the bill on your credit card.

Over the years, we have become trusting of credit card companies and even banks when it is time to deal with our transactions. We have learned how to trust them to handle all of these transactions and to only take out the money amount that you authorized for that particular transaction. We have also learned that we should be able to trust that our personal information is private and that these institutions are doing their best to keep the information hidden. This type of trust is a lot, but it is something that banks and other financial institutions have worked hard to earn over the years.

One thing that you will notice when you work with the blockchain technology is that the blockchain is actually able to provide you with the same level of security that you have come to know and expect from your traditional financial institutions, without needing to have any type of third party present at all. Bitcoin was the very first of the digital currencies who started to use this technology because it made it easy for the network to control the transactions that were going on and to keep the information of its users safe and secure.

So, what does this mean for you? What it means is that you can use the blockchain to complete transactions throughout the

world and it does not matter where the seller is located in relations to you. You can receive money from others if you would like. You can even use the blockchain to help yourself invest online. This technology can work for all of these transactions. This is because the technology will take in all the information for the transactions and then move it between the two parties almost instantly. Then, when it is all done, you will be able to go through and look at the ledger to see where the money was transferred to, or to look at any of the transactions that have occurred on the network, without the worry of personal information being exposed.

Of course, the blockchain technology is really amazing and what we just talked about is only one thing that this type of technology can do. There are actually a lot of industries who are looking at the blockchain now and considering how it could be adapted to help them out. Banks, insurance agencies, hotels, trading and so much more could benefit from using the blockchain ledger in so many ways.

How this blockchain technology works

From here, we want to have an idea of how the blockchain technology will be able to work. Since blockchain and Bitcoin were released together at the same time, we will concentrate our efforts on how this blockchain can help the Bitcoin network so that things are a bit easier.

When you are working with the blockchain, each transaction, once it has been completed, will be put on a block in chronological order as you complete them. Once you have done enough transactions to fill up one block, then this block will be sent over to the permanent record and you will be given another block to fill up. The amount of time that it takes to fill up a block will depend on how many transactions you complete. If you only use Bitcoin on occasion, then you will take a long time to fill up a block. For those who do a lot of day trading or who do other things on the Bitcoin network often, your blocks will fill up fast.

Each block that you fill up will hold onto all the information about a transaction that you do on this network. These blocks can hold a ton of information so you will take some time to fill them up. When you are looking at one of your blocks, you would be able to find a variety of transaction types including identity, digital rights, currency, and property files. With all these options, it is no wonder that the blockchain could be expanded out to a lot of other areas.

So, the next question that you may have is how you get one of these blocks to use. This is really easy to do. As soon as you join the bitcoin network and switch your coins over, you will already have a new block. You do not need to do anything else. This new block will start out with your very first transaction and then it will hold onto a specific number of transactions after this. Each person will take a different amount of time to fill it up. Those who are really active on the Bitcoin network may end up going through a lot of blocks and will have a long chain. Those who only use it occasionally will have shorter chains.

After you have completed the number of transactions that you need to fill up the block, that particular block will go under the control of the miners and then be sent off to be part of the permanent record for Bitcoin. This is a good thing because it means that your transactions will be secure and will be kept away from hackers or others who may want to get your personal information or who want to try and change things that are on the network. Once you are done with one block, the system is automatically going to send another one over to you. You will continue on with this process the entire time you are on the Bitcoin network.

You will be able to take a look at the blockchain at any time that you would like. Anyone who is on the Bitcoin network can look at this ledger, whether they want to double check a transaction that they did at some point, or they are interested in checking how secure the blockchain is. This is how the blockchain can maintain not only its security but also its transparency to the users. Nothing is hidden on this network except the personal information of the users and you can check it out at any time.

Working on the blockchain may seem a little bit complicated at first, but it is really not so bad. In most cases, it really will not matter all that much to you how this system works. It is there and it keeps your information safe online, but it will work without you having to do anything. So, if it seems overly complicated, then it is not a big deal if you do not understand what is going on with it.

As the blocks that you complete are getting ready to join the networks main blockchain, the miners will start doing their work. These miners are responsible for assigning unique codes to the blockchain to keep transactions, and the information that is contained inside of them safe. There are some rules that miners will need to follow to get this done, such as the new code needs to be dependent on the code that is already found in the main blockchain, but it helps to ensure that no one can mess around with your currency.

How to use this technology in other places

There are a lot of different applications for this type of technology. While it is most often associated with the Bitcoin network, there are a lot of other ways that it can be used as well. In fact, the digital currency Ethereum is designed to help expand the blockchain technology by providing a basic platform of blockchain that is open sourced, making it easier for people to get started. This technology is bound to keep moving us into the future so it is important to get ready for it.

The first application of blockchain technology is how it can be used in banks. Since we already know that blockchain is used to maintain the records on the Bitcoin network, it is easy to imagine some of the ways that the banks and financial institutions you use would be able to benefit from it as well. There are already several of the larger banks in Europe who are

coming together to share a blockchain ledger so that they can speed up transaction times in many countries.

This is just one example of how blockchain technology can be used to benefit other industries. Pretty much any industry that has to hold onto information of value would be able to benefit. From insurance companies to polling stations and even reservation companies like hotels and cars could develop their own platform of blockchain to speed up transaction times and to provide better customer service.

As you can see, there is a lot to learn about the blockchain technology. It is a neat piece of technology that, while being really successful at helping Bitcoin to become the big powerhouse it is today, also has the ability to take the world by storm in other applications as well.

CHAPTER 4: GETTING YOUR FIRST COINS

Now that you know a little bit more about Bitcoin and how it works, it is time to learn a bit more about how to get your first coins. It is impossible to join the network and send money, make purchases, receive money, or even invest if you do not go through and purchase the Bitcoin to make this happen. This chapter will take some time to look at the steps that you should take to get started in Bitcoin.

The first thing that you need to do is take a look at the exchange site that you want to use. There are several of them available, but most people in the United States will rely on Coinbase to help them out. This one can handle the exchange of traditional currencies over to Bitcoin, Litecoin, and Ethereum and may be expanding into more. So, if you want to work with an exchange site that is easy to work with and will be able to help you to expand out your portfolio later on, then this is the one for you.

To get started, you will simply open up an account with Coinbase and get it all set up. You do not have to provide too much information for this to work, but it is possible that this may

change. The United States government is starting to ask for more information from these sites in the hopes of stopping issues with money laundering. Right now, you're able to just set up with your name and telephone number, which is used to help verify the account.

Once the account is all set up, you will be able to take a look at your dashboard. This is a really great place to look around for a bit because you will be able to see what the current value of the coins are and take a look at how it has changed over the past year. This is a good place to look around and become familiar with before you decide to start trading.

Before you can exchange your traditional currency over for Bitcoin, you need to take the time to set up your banking information. This is basically going to tell Coinbase where it can take your traditional money from and then it will give you the coins that you are looking for. There are three main payment options that you can use through Coinbase and they include your bank account, a credit or debit card, or you can make payments through PayPal. Each of these has their own benefits and negatives and it will depend on your overall goals and how much you want to be able to transfer at a time for which one you will choose.

The first option that you can choose from is your bank account. This is a great option to work with because it is easy to set up and can provide you with a level of security when you are on the Bitcoin network. You are also able to transfer more money when

you use this option compared to the other two. On the other hand, the transactions will be much slower. Since you are working directly with your bank, it could take three to five days for you to be able to get the coins.

Using your bank account is the best option if you want to be able to work with larger amounts of money to start your investment. However, if you are working with strategies like day trading, you will be disappointed in this option because it will take way longer than you have. But for those who want to do long-term investing, such as the buy and hold strategy, then using your bank account may be a good option.

You can also choose to work with your credit card, debit card, or PayPal account. These are all going to work in a similar manner so it is not important which one you are wanting to choose. You will find that day traders and those who want to get into the market quickly will go with these options because they can get the money moved over quickly. However, if you have a large sum of money that you want to invest, then you will not be able to use this method.

Of course, you can choose to use more than one method of payment with Coinbase, you will simply need to go through and set all of them up. There are different steps that you will be able to follow when you are setting up each account so make sure to follow the steps that Coinbase lays out.

Once you have changed your traditional currency over to Bitcoin, you will need to store it in the proper wallet. Coinbase does provide you with a wallet to use, as well as a Bitcoin address. If you plan to use your coins right away, it is perfectly fine to stick with these. It is important to note though that Coinbase wallets are online wallets and they will provide you with a Bitcoin address that simply contains your full name. This is not very secure so if you want to go with a long-term investment strategy, it is probably best to pick out your own wallet and move the coins over to that one.

Picking out a good site

One thing that you need to consider when you are ready to get some new coins to start your adventure with Bitcoin is which site you will use. Coinbase is one of the most popular options to use and you will certainly see some good results when you choose to go with it. However, if you are looking for something that is a little bit different or you want a site that offers the ability to exchange more than Bitcoin so you can diversify your portfolio, then it is time to look at some of the other exchange sites that are out there.

You need to be careful when you are picking out the exchange site that you want to use. First, take a look at all the features that come with the site that you want to use. There are different features that you can get depending on which site that you go with. You should take a look at what each site has to offer and then choose the one that is right for you. Some may be better

equipped to help you out with day trading, some may have nice wallets to work with, or there may be another benefit that draws you to it.

You also need to look at the security that is on the exchange site. You want to make sure that you will be able to trade around your currencies without someone else taking a look at your information and stealing it. Most of the exchange sites have been successful so far at protecting their users, but always be on the lookout for when that may change.

And finally, really take a good look and research the option that you use for exchanging currencies. There are many good exchange sites to go with, but there are also a lot of ones that are not that good. Since Bitcoin has become so popular, it is not surprising that there are a lot of scam artists who are willing to rob you of your money. If you are not careful, you could end up trusting an exchange site that takes your money and never gives you anything back in the end.

Picking out a good exchange site is the first step to making sure that you will have a great journey with the Bitcoin network. Consider working with a site like Coinbase or something similar so that you do not run into any problems along the way.

CHAPTER 5: PICKING OUT THE RIGHT WALLET TO KEEP YOUR COINS SAFE

There are a lot of things that you can do when you decide to join a digital currency network. You will be able to send money to other people, no matter where they are located throughout the world. You can make payments and purchase items from anyone who is willing to use Bitcoin as an accepted form of payment. You can also choose to invest in these currencies and see how much money you can make in the process.

Because of all the neat things that you can do with Bitcoin and some of the other digital currencies, there has been a lot of popularity throughout the world and the value of Bitcoin is steadily going up each day. And when there is a currency, or anything of value, that goes up as much as we have seen with these digital currencies, there will be a lot of hackers and scammers around who are just waiting for you to mess up and they can get your coins.

If you choose to store your coins and not use them right away, you need to make sure that you are picking out the proper wallet to hold onto them. A wallet is basically just the place where you

store the coins you have until you are ready to use them. You are not able to get a physical copy of these coins and you will not be able to carry them around, so having someplace to hold the code that opens the coins is very important.

Preparing yourself for these hackers is so important. Since there is not a governing body that takes over on these networks (part of the benefit of working with these digital currencies), you have no one to talk to if you do lose your coins. If a hacker gets ahold of them and you do not have your key safely stored to bring up, then you are just out of luck. Or, if you choose an online wallet, which is one popular option with those who are just getting started with Bitcoin, then you may also have to worry about something going wrong with the computer system. If the system crashes and your coins disappear, you will find that it is hard to get them back without the right key in place.

The good news is there are several different wallet types. The one that you want to go with will vary based on how you want to store the coins, how long you want to store the coins, and which trading strategy you would like to use. Let us take a look at some of the different wallet types that you can choose from so that you can make the right choice for your needs.

Online wallet

Most beginners will choose an online wallet. They do this because it is really simple to use. You even get one of these from

Coinbase when you join the site and get your first coins. The coins are all online and ready to use whenever you want and you do not have to take any extra steps to start using the coins.

While this may be the most convenient method to use for storing your coins, it is definitely not the most secure method. With this method, you are running the risk that something will go wrong with the online world, and that is not the best thing to do when we are talking about an investment.

The first issue is security. While most of these online wallets have been pretty secure so far, there is the risk that a hacker will eventually be able to get onto them and steal your coins as well as your personal information. And if the hacker can get this kind of information, you are stuck because there is no one there to help you out on these networks.

You also have to worry about whether something goes down on the system. If you are using the coins and the system miscalculates something or the system goes down and your coins disappear, who are you going to contact to get this fixed?

For those who are planning on doing day trading or who will trade out some coins to make purchases right away, the online wallet will be just fine. For those who are dealing with some of the long-term investment strategies, it may be best to go with a different wallet to keep your coins safe. If you do plan to use the online wallet for some reason, it is best to search around. There

are many different types of online wallets that you can choose from and some will offer better features or more security than others.

Hardware wallet

Another option that you may want to go with is the hardware wallet. This is a good option when you want to add a little more security to your coins than what you can get with the online wallet, but you still want to be able to easily access the coins whenever you need them. It may not provide as much protection and security as you can get when you are using the cold storage, but it will do the job and is far better than working with an online wallet.

With the hardware wallet, you will move your coins offline and place them on your computer. You may place the wallet somewhere on your hard drive of the computer or on another device like a USB card. Then when you have some new coins, you would just redirect them there. You will have to add in a few more steps any time that you wanted to use the coins, but if you do not plan to use the coins right away, this can be a good option.

Now, you do need to take a few precautions because there are a few ways that hackers and others can get the information. Make sure that you password protect your wallet so it is harder to get ahold of. Consider updating your operating system to the newest

version and keeping the antivirus up to date. While this method is much better than using an online wallet, you are still leaving your coins online. If a hacker can gain access to your computer, they will be able to gain access to the coins so take the right precautions ahead of time.

Cold storage

Another option that you can choose to go with is to store your coins in cold storage. This is one of the most secure options, but it will also take some extra steps to get your coins back online so that you can use them. This is often a strategy that those who want to use the buy and hold strategy will use because it is almost impossible for the hacker or someone else to gain access to these coins.

With cold storage, you are basically moving your coins offline. You will be able to print off the private key that accesses your coins and then you will need to store this key someplace that is really hard for someone else to find. Do not leave the code out somewhere on your desk where someone can take it or where it can get ruined. Placing in a safe, fireproof box, or somewhere similar can make sure that you know your key is as safe as possible.

When you are ready to bring those coins out again and use them, you will simply scan that private key back into the computer, and they will all be there in your wallet again. This does take a few

extra steps, but it is really hard for someone to get ahold of the coins.

This is the best strategy to use if you are working with the buy and hold investment strategy. You are not really going to use the coins for at least a few months while you wait for the profits to come in, so it is not as important that they be in an online wallet and easy to access. When you store them in the cold storage, they are completely off your computer, making it really hard for something to go wrong with the technology or for a hacker to get ahold of them. Just remember that when you add more coins to your investment, you will need to reprint out the private key so that it has all the information that you need.

Backing up your data

Now, all of these options can be great ways to store the coins that you are using on this network and each one will have its own benefits for you to use. It is important to take a look at all of them and decide which one will provide you the best benefit and which one will work with the way you plan to use the Bitcoin network.

If you would like to add on another layer of security to the coins that you are using, it is a good idea to backup with a few of the wallets. There is nothing wrong with having an online wallet and then also storing those coins in a cold storage as well. This way, the coins are still readily available to you when you want, but you

will be able to recover them if something does happen on the network.

It is good to have at least two places where you are storing your private key for the best results. And any time that you want to add more coins or you want to spend the coins, make sure that you update all of your wallets. If you miss doing one, you could end up with a key that is out of date when something goes wrong and you will have nothing to help you with it.

As you can see, there are several different types of wallets that you can work with. If you plan to use your coins right away, then it is probably fine to work with an online wallet because the coins will likely be gone before anything can happen and it would be a waste of time to move them somewhere else. But if you plan to do a long-term strategy with investing, it is best to consider going with the hardware storage or the cold storage to ensure that your coins stay safe and sound.

CHAPTER 6: CAN I REMAIN ANONYMOUS WITH BITCOIN?

One thing that a lot of people are curious about when they get started with Bitcoin is if they can remain anonymous. They like the idea of being able to go online and keep their identities hidden from the world. They like the idea of being able to do purchases and other things without having to worry about a hacker or a scammer being able to get ahold of their information and use it against them. And they even like the idea that they could do all this without the government necessarily being able to trace it back to them.

This is one of the good things about Bitcoin, but also one of the things that gave it a bad name when it first got started. There were many people who would use Bitcoin to do some things that were illegal. They would perhaps purchase or sell items that were not allowed in their countries. They may try to do money laundering. Or they may choose to do something else that is illegal and then be able to hide behind their anonymity to stay safe.

While most people will use the fact that they can remain

anonymous on Bitcoin as a way to stay safe from others, there have been a number of people who have tried to abuse this. This is a big reason why the United States is trying to get a bit harder on those who use these kinds of currencies. The IRS is worried that people will get on these networks and then try to launder money away from them. Because of this, the IRS is making it harder to keep the government out of Bitcoin and making it harder to keep your identity hidden from others.

How is the government doing this? They are looking to pass some laws that will require exchange sites like Coinbase to provide information on those who use the site. This may seem like a good way to prevent money laundering, but it takes away some of the benefits of using Bitcoin and could end up turning people away from using the Bitcoin network. For now, if you do use this network to make money or to invest, you must tell the IRS what you made for an income.

With that being said, there are some ways that you can keep hidden when you are using this network and it will be really hard for someone to figure out who you are. The first thing that you should do is make sure that you are picking out a good Bitcoin address to use. If you put something that easily describes you or that has your first and last name on it, then you have pretty much given up your anonymity online because they can figure out who you are.

You do get a choice to pick out your own Bitcoin address in most cases so it is a good idea to pick out one that is completely

unique to yourself and would be hard for others to figure out. If you have trouble coming up with a unique Bitcoin address that you want to use, it is fine to go online and search around for a site that can create one for you. Then you can make sure that it is random and that no one can figure out who you are.

Another thing to consider is that if you are using Coinbase, they will provide you with a Bitcoin address. However, this address is not going to keep you anonymous. They will use the first and last name that you provided to them and that is it. So, it will be really easy for someone to figure out who you are. You can always use Coinbase to exchange out the currency that you are using, but then consider picking out a different wallet with a different address to ensure that you can keep your identity hidden.

In addition, it is a good idea to switch out the Bitcoin address that you are using on occasion. If you are really active on this network, it is not a good idea to keep the same address all the time. This will just make it easier for someone to find you because they can trace all your transactions back to one point. In fact, the developer of Bitcoin suggested that people who used this network would change their address after each transaction to keep things safe. While you do not need to go that far, you can consider changing it on a regular basis, especially if you are someone who uses the network often.

It is also a good idea to never share the information about your Bitcoin accounts with other people. This can include any passwords that you are using and the address of your account.

This should only be shared with the people who will receive money from you. Otherwise, you are risking yourself and your identity in the process.

As you can see, there are a few different things that you can do to ensure that you remain hidden when you are online. In the future, this may change a little bit but for now, you can keep yourself hidden when you use the Bitcoin network.

Coin swapping

Another thing that you can consider doing if you plan to use an online wallet, is to find one that does coin swapping. This will ensure that you are always receiving some new coins and that it is much harder to trace the transactions that you are doing over time.

What happens with these wallets is that you will place some coins in the wallet, whether you get them from an exchange site, someone sending you coins or some other method. Then, while they are in the wallet, the wallet company will take all the coins from all its users and place them in a pot. Then they will randomly give you some coins back. These coins will equal the same value that you had before, but they will have different codes with them that make them harder to trace later on.

This can add in another level of security when you are working on these networks. It will be really hard for a hacker to find your

information because your information is always changing. There are a few different sites that offer this so it may be worth your time to check it out.

With that being said, you must be careful with these. First, you want to make sure that the site is legitimate and that they will actually give you coins back when you submit yours, rather than trying to take off with the coins and give you nothing in return. You also want to make sure that the website is secure. It is not going to help protect you much if the site that is supposed to trade around the coins gets hacked and all your information is out in the open anyways.

What happens if someone finds out who I am?

The whole point for some people, of getting on one of these networks and trying to use Bitcoin is that they want to make sure that they are anonymous and that no one can figure out who they are. They are tired of having to worry that their credit card information or other personal information will be stolen by someone and used against them. There are countless times when we listen to the news and hear that one store or another had a big breach and all the personal data was at risk. Even Equifax, a big credit reporting company has fallen under attack, leaving millions of people exposed.

These issues can take years to fix up. You may never be able to get things fixed depending on how the work and how much

damage did they cause. While online shopping may be convenient and may make life a lot easier, it can take years for these things to clear up and it is often up to debate whether that convenience is worth the price.

You can avoid some of these problems when you are working with Bitcoin, but you have to be your own advocate. Remember that there is nothing that you can do once the coins are gone if you did not back them up properly or if you messed up in some other way with them. This can be a hard lesson but is something that you just need to expect if you will work with this currency since there is no one running it.

You need to take care of yourself and make sure that you are keeping your information private. Do not give it out to anyone else, or they will have access to your information and can basically do what they want with it. It is best to just keep the information to yourself, no matter how close you may be to the other person.

With that being said, if someone does happen to get onto your account, or someone finds out who you are, it is best to take steps as quickly as possible to avoid worse things happening. If your coins are already gone and you did not have them backed up at all, then you are too late and there is nothing you can do at this point other than learning from your lesson.

But if someone has gotten your information, or there is a breach

with your online wallet and you feel that someone may try to get onto your wallet, then it is time to take some changes as soon as possible. The first thing that you should do is consider getting a new Bitcoin address and wallet and then transferring all your coins over there as soon as possible. That way, when the hacker does try to get your coins, they will already be removed and they will not be able to figure out where they went.

Once you do this, consider backing up some of your coins to one of the other wallet types that we talked to, either the hardware storage or the cold storage. This helps you to keep the coins as safe as possible and if something does happen and they disappear, you will be able to get the private key from those other wallets and then use that as a proof that the coins belong to you and then they will come right back to you. It is that simple.

CHAPTER 7: THE PROCESS OF MINING IN BITCOIN

As you will see as we go through this guidebook, there are a lot of ways that you can make money when you are working on the Bitcoin network. We will look at some of the best options that you can go with when you can invest in the network, but before we get too far into that, we will take a look at the first method of earning money on this network known as mining. While there are a lot of benefits to the mining process, both for the miner and for the whole network, and while it can sound really appealing, it is important to note that it is hard work and you will need to spend some money to get started and have some technical background to make it happen. However, if you are successful, you will find that mining in Bitcoin can be really profitable. You can earn 25 Bitcoin for being successful and for helping to keep the Bitcoin network safe and secure.

As a miner, you will be taking those blocks from the blockchain that we talked about earlier and providing them with unique codes that will hide the information that is in the codes to keep them safe. These codes are pretty complicated to make and have a lot of rules that go with them. They need to include a specific number of characters and these characters need to be dependent

on each other. This means that if at any time one of the characters is changed, then it will change all of the other characters in that code as well.

This may seem like a big hassle, but it is part of the process to ensure that a hacker is not able to change things or steal information. If the hacker tries, it will mess with the whole code and everyone will notice.

Now that we have taken some time to introduce the idea of mining, let us go through and look at it in more detail so you can see what is involved in this process and whether or not it is the right option for you.

How to get started with mining

When the network for Bitcoin was started, only a few of its 21 million coins were released. This was done to make sure that there could be more coins released later as the network became more popular and it would keep the network steady. There are only a few Bitcoin coins available at the beginning and it is impossible for anyone to go and make some more. At this point, unless the code is changed, there are only 21 million coins. Some are in circulation and being used right now, but not all of them are.

To get to these coins and get them released on the network for

others to use, the miners will need to go in and do their job. When they can successfully write out a code to protect a block, they will then receive some new Bitcoin as a reward. These miners will then use the Bitcoin in some way and they are then released into the market. Of course, there is a lot of work to doing this job, but that ensure that not everyone tries to become a miner and so all the coins are not released right away.

Now, as you can imagine, the process of mining will be so important to ensure that this network stays safe. There are many people throughout the world who are using this network and they are using it to make money, transfer money, and to spend money. These transactions will be placed onto the blockchain and at some point will become part of the permanent record. Without the blockchain being there to record these transactions, it would be pretty impossible for a user to figure out whether a transaction went through or not.

While it is important that the blockchain has some transparency so that the users can trust it, it is also important that it has some security and protection. Since anyone who is on the Bitcoin network can take a look at the blockchain ledger, it is a good idea to hide some of the information or else someone could easily track you and take all your personal information. No one wants to work on a system was a hacker or someone else can get on and see all their transactions just laid out. They want to ensure that when they are on the network, their transactions are as safe and secure as possible. They also want to ensure that nothing on the ledger can be changed so that no one is going through and making a mess on the network.

What happens now?

So, we already know how the blockchain ledger works and why it is so important to the whole Bitcoin network, you may be wondering why the miner would want to spend their time learning the ledger and making sure that it is secure for others. Basically, in Bitcoin, the general ledger will be a list of blocks that were used in the system. These blocks will hold onto all the transactions that every user has ever done since the beginning of the Bitcoin network. You would be able to go through this whole list to explore the transactions that have occurred at any time on the network.

When you look at the blockchain, you will be able to see each transaction that occurs every day. Over time, this list becomes really long because more people join the network and a lot of transactions will happen. Anyone who is on the Bitcoin network will be able to see this ledger and take note of the various transactions, so technically anyone who is interested can choose to be a miner.

For the Bitcoin system to continue working and attracting more people over to using it, there has to be some process so that there is transparency in the ledger. Transparency means that people can tell whether the transactions on the ledger are legitimate or not. But there also needs to be some security so that a hacker or someone else can't come on and see and use all

the information that is there. The miners will come in at this point to make sure that the ledger is transparent as well as secure at the same time.

When a user has finished filling up their block, it will go to the miner, who will be able to use the information in the block to create a new code that keeps that information safe. The miner will be able to then add their formula to the main blockchains formula so that they all fit together well.

The job of the miner is to work to create a new hash for that block. There are a few rules that Bitcoin has created to make sure that the blockchain is staying secure and uniform and that a hacker is not able to get in and cause trouble. The important thing for the hash is that it has to make sense for the information that is inside of the block or it will not work. It also needs to match up with the hashes that were created before it. If one letter or character in the blocks code changes, it will change everything. If one letter or character in any part of the blockchain changes, it will mess everything up as well. This makes it easier for all users to tell when someone has been messing around in the blockchain and trying to make changes that are not authorized.

Now, it is important as a miner that you learn what properties are required to make a good hash. You will first need to take a look at the information that is inside of the block to see what is there. The hash will be created off this information. Now, that is just the starting point. If creating hashes were that easy,

everyone would be doing it and the coins would be long gone by now.

Another rule that you need to follow is that your hash needs to be unique. No one else can claim it inside the blockchain, or it just will not work. In addition, it is important that each character in the code can rely on the other characters that are present as well. This means that if any character is changed, the others that are after it will change as well. This is a pain to create, but it can certainly show up a mess if a hacker tries to mess with the formula.

Basically, the miners can create these hashes as a way to keep the network as secure as possible. There are several different ways to do this, but most miners will rely on some software to help them out. There is way too much competition for this to just try and do it all by hand. Having some software that can automatically go through and create some codes can help you to increase your speed at getting it done.

Why would I want to mine?

We have already discussed how mining is important to the Bitcoin community. Without the work that the miners do, it would be impossible for anyone to trust the network. Without a big agency controlling this currency, and without some help from the miners, it would be so easy for a hacker or someone else to get onto the system and mess around, and it would be

hard for the users to find out. Thanks to the work that the miners do, it is now possible to go through and see these changes if they happen, which can be a big deterrent to the hacker.

As you can imagine though, the rules that govern how to make these hashes can make them a little bit difficult to create. You can sometimes create a few different types of hashes and find out that none of them will work, even if they follow the rules of Bitcoin. There is a lot of competition out there and if someone beats you to making a code, then you do not get the reward. With all this work, Bitcoin realized that it needed to provide some kind of incentive to make sure that people would do the mining.

When a miner is successful at doing their job and they get a hash that is accepted, they will earn 25 Bitcoins. With the value being more than \$10,000 for Bitcoin right now, this can be a huge profit. You do need to consider how much time you will actually spend working on this, the processing power you will need to use, and so on, but this can be a great way to earn a good amount of money if you are successful.

The process of mining is important for everyone who decides to use the Bitcoin network. First, it makes sure that the users on this system will be able to complete all the transactions that they want while knowing that their information will stay secure. This builds up some more trust in the system because they will get all the security that they need from using a bank or another financial institution, without needing that middle person.

In addition, the miner will be able to benefit because by securing the ledger system, they will get a good reward. Of course, since each Bitcoin is worth so much right now, you can imagine that there are a lot of people who want to get into the mining process. Anyone can give mining a shot, but remember that it is not meant to be easy and not everyone will succeed.

This is the main reason that Bitcoin set up all the different rules to go with the mining process. They knew that if there were not some rules in place, everyone would become a miner and there would not be any more coins available to mine in a short amount of time. It also adds in another level of security to the ledger process so it benefits. The rules may make it harder to create a new hash, but it helps to keep the network safe and the miners to make a good deal of money in the process.

What do I need to get started with mining?

If you are still interested in getting started with mining, it is important that you have the right things to make it happen. You can't just randomly start making these hashes and hoping that it will all just work out. Without the proper tools in place, it will be impossible to create any hashes or to get started with mining in Bitcoin at all. Some of the things that you should consider getting when you are ready to start earning a profit on Bitcoin include:

- **Software:** There are several software programs online that you can choose from that will make it easier to make your hashes. You will be able to pick which kind of software you would like to use, but take a look at some of the reviews and information about it to ensure that you are picking a good one.
- **Hardware:** It is a good idea to get a computer that can keep up with all the work that you need to do to mine. And most miners who are serious about this endeavor will choose to go with a system that is specifically for mining. This makes it easier for them to do the work and still do other things on the computer. You should also make sure that you have a computer with plenty of memory, with lots of graphics cards, and a good speed on it so you can get the work done.
- **Cool room to work in:** It is a good idea to place your system somewhere that is nice and cool. This mining will take up a lot of processing power, which can mean that your system will get overheated in the process. When you leave a nice fan on the system and you leave it in the basement or in another cool room, you will ensure that your computer can keep up with all the work.
- **Bitcoin wallet:** You will need to sign up for a Bitcoin wallet if you would like to get started with mining. This will hold onto the coins that you earn through this system.
- **Enthusiasm and knowledge:** It can be difficult to work with mining and not everyone will see the success that they want. This is why you need to have some passion for what you are doing. When you can bring passion into the game, it is much easier to enjoy what you are doing, and you are more likely to enjoy the work that you are completing as well.

Consider a mining pool

If you are interested in getting started with mining, it is a good idea to consider joining a mining pool. It takes a lot of time and resources to work with mining on your own and for someone individually to get a code done without help can be impossible. It is very likely that you could spend a lot of time working on this process and using a lot of power and money, and then not make anything because there is just so much competition.

When you join a mining pool, you can share in the resources that you have. Each person who joins the pool will be given a part of the equation to work on, rather than the whole thing. When you are done with your part, you will be able to submit it in and get more parts as you want. Then, if the group gets a successful hash that is accepted, they will split up the reward based on how much work each person did on it.

Working in a mining pool makes it much easier. You will only have to look at one part of the equation at a time, rather than all of it. This takes up less time and less computing power so it is more cost effective as well. You may not earn the whole 25 Bitcoin on your own, but this often provides a steadier source of income that is easier to get compared to trying to go for the whole reward on your own.

Mining is a complex process that is so vital to how well the Bitcoin network will work. Without the miners, it is hard for

people to trust Bitcoin and the blockchain because all their information is right out in the open. But, while mining can be a rewarding task, it is really complicated and has a lot of competition. Learning how to make it work can take some time and dedication, but it does help you to earn money and is beneficial to everyone who decides to use it.

CHAPTER 8: THE BEST WAYS TO INVEST IN BITCOIN

There are a lot of different reasons that people will choose to join the Bitcoin network. Some people are interested in getting to this network because it allows them to make purchases anywhere in the world quickly and anonymously so they do not need to worry about someone getting ahold of their personal information. Some people just like the idea of being able to work with a currency that does not have the government controlling it. But the main reason that people will choose to work with Bitcoin is that it is a great source of investment.

There are many ways that you can invest in Bitcoin and make good money. You have to be smart about what you are doing if you really want a chance to be successful. While working in Bitcoin investing seems like a great idea in the beginning, you still need to pay attention to the investment and treat it just like any other investment that you may choose to go with. Simply jumping right into the investment and hoping that it works out will lead you to a lot of failure and disappointment.

With that said, there are many different ways that you can invest

in Bitcoin, which can make it a great investment for many people, no matter what their style of investing may be like. If you are interested in making some money with Bitcoin, take a look at some of these great investment opportunities that you can use today!

Day trading

Day trading can be a good way to earn money while investing in Bitcoin, as long as you are smart about how you do the process. With day trading, you are basically taking advantage of all the ups and downs that occur in the Bitcoin network. While the value of Bitcoin is going steadily up, for the most part, there will be a lot of little variations that occur throughout the day and the successful day trader will be able to take advantage of this.

To start with day trading, you will need to take some time to watch the charts and the history of Bitcoin. What you want to find here is the market value, or as close to it as possible, for Bitcoin. Then, when you see that the value of Bitcoin goes below that market value, you will purchase your own coins. Then, you will keep watching the market for that day. When the value of the coin goes up to market value or above, you will exchange back out for your traditional currency and keep the profit.

This is a fast-paced investment type. You are required to make your purchases and then sell your position all on the same day. In some cases, you will need to do this a few times each day. You

will not make a ton of money on each trade, but when you do this many times during the week, it can add up to some good profits.

There are a few things that you should keep in mind when it comes to trading this way. First, it is very quick and you must be able to devote a lot of time to make day trading work. If you do not have time to watch your trade after you have purchased into the market, then it is best to pick one of the other trading methods. In addition, remember that you need to do your trades all in one day. Never end the day with money still in the market. Day trading strategies will not work with this and you could end up harming your investment. You are not able to control what happens in other parts of the world who are trading Bitcoin, and keeping your position overnight can end up being a costly risk that you can't control at all.

The buy and hold strategy

The buy and hold strategy is one of the most effective strategies for you to use, and it does not require a ton of work to get results. This is the method that most investors, especially newer investors, will use to get results because it is such a basic idea that has been relatively successful over the years.

With this method, you simply need to set up your own Bitcoin wallet and then transfer some traditional currency over to Bitcoin. Then you just leave the coins there for some time. You will not touch the coins and you will not use them at all. You just

wait until the value of the Bitcoin goes up. If you look at the history of Bitcoin, it has gone steadily up over the years, so if you are willing to leave some money in your account for a year or more, you will be able to earn a good amount of money.

Let us look at how this can work for you. In February 2017, the value of Bitcoin was around 2500 a coin. If you purchased one coin at that time and waited until December of 2017, the value of the Bitcoin went up to \$19,000. If you traded out at that time, you could have made a huge profit, even off just one coin. Now, there was a downturn that caused the coins to fall in value pretty drastically in January of 2018, but it is not rebounding and you could trade in February of 2018 and the value is around \$11,000.

As the market looks right now, it is likely that the value will continue to grow over time. You can still get into the market and as long as you pay attention to what is going on in the market, you could make a lot of profit with very little work involved. It pretty much just involves an occasional look at the market and at the news and you are set.

Of course, you need to pay attention to the wallet that you are using. Since you are not planning on using the coins for at least a few months, it is probably not a good idea to go out there and keep it in your online wallet. Instead, choose to go with hardware storage or cold storage to ensure that a hacker is not able to get hold of your coins.

Selling your own products

There are a lot of people who are interested in going on the Bitcoin network and making purchases. They like the idea of being able to purchase things that they want or need online, without any interference and with the assurance that they can keep their personal information safe. But, while the list of companies that accept Bitcoin is growing, there still are not that many available.

This can be a great opportunity for you to jump in and make some money. If you have a product or a service you are willing to sell, why not consider accepting Bitcoin as one form of payment. You will simply need to have your own website and then add a Bitcoin payment button on. Then, you sign up for your own wallet (using the steps that we gave above). When a customer decides that they want to purchase a product or service from you, they just click on the button. You can send over your Bitcoin wallet address and they can send their payment instantly.

This is a great way to complete transactions. It only takes a few minutes to send the money and then you can send out the product or complete the service for the individual. And if you do not necessarily need all of the money right away, you can combine this with the buy and hold strategy above. This will allow you to hold onto the coins for a little bit and watch as the value grows and you make more money in the process.

Investing in the blockchain

While this one is reserved for the Ethereum network, it is possible for you to invest some of your money in the blockchain. We spent a little bit of time talking about this blockchain earlier, looking at how it helps to maintain the transparency and the trust that Bitcoin and other cryptocurrencies are looking for to keep users on the network. Without the blockchain, these digital currencies would have no changes of being successful at all.

With that being said, the blockchain is a really powerful piece of technology that can take on so much more than just what it does for digital currencies. There are so many different industries that would be able to benefit from this kind of technology. The problem is that a blockchain platform needs to be developed to handle all of this so that these industries can get all of the benefits.

Creating these blockchain platforms can take a lot of time and can be expensive. There are some developers who can create them, but they need to have some funding to help them get it all done. As an investor, you can provide them with the funds that they need to get results. You can work on creating a contract with the developer that will allow them to get the funding they need to start but will also ensure that you can get your money back and more at the end of it all.

Work on smart contracts

This is a really neat thing that has become really popular now that a lot of digital currencies are on the market. These smart contracts are designed to self-execute, so that you and the other party can make your own agreements and both parties know that they are safe, without having to deal with the hassle and costs of a third party.

These contracts will execute the terms that are inside of them as soon as all the conditions are met. The conditions can vary depending on what agreement you and the other person come up with together. You can choose to do a contract for a lease, for purchasing a home, for selling something, and so much more. The ideas of how to use these smart contracts are pretty limitless, as long as you can get it all put together.

Let us take a look at an example of how this can work. We will look at the example of a landlord who is trying to rent out an apartment to a tenant. They could use a smart contract to help them out. The tenant may be nervous about giving the landlord a deposit before they are even in the apartment because they do not know this person at all. But the landlord will require this deposit before they will even start the background check and get everything moving so they can prove that the tenant will be a good one and to ensure they will have some money if something goes wrong with the rental agreement.

These two parties can use a smart contract to help protect

themselves. The tenant would be able to put the deposit into an escrow that is supported by the smart contract. The landlord would then have so much time to figure everything out, run the credit report, do a background check, and so on. If they finished it in time and handed the keys over to the tenant, then the deposit would be released to the landlord. If for some reason they did not do this, then the money would be released back to the tenant and they could just go find another apartment.

Both parties under the smart contract example above were protected, but they did not have to rely on a third party at all. Lawyers and other third parties who run these contracts are often expensive and there may be times when you want to do a transaction that no one else knows about. This can be hard to do if you are working with a third party.

These smart contracts can eliminate the need for that nosy and expensive third party. You will both still be protected and the smart contract would take care of all the details for you, but you will know that both sides are as protected as possible.

As a developer, you can make some money off these smart contracts. You simply need to create your own template or be available to create a custom smart contract for your customers. You can charge a small fee, which is usually way less than someone would have to pay to the lawyer or other third party participants, so they will be happy to work with you.

Get into the stock market

This method is not as popular to use as some of the others, but it is a way that you can make some money with Bitcoin. With this option, you will be able to purchase stock of the Bitcoin company and then receive dividends from that stock, just like with any other company that is on the exchange. You will be able to look at the New York Stock Exchange and find that you can work with the Bitcoin company. For those who are new investors or who have been working in the stock market for some time and want to stick with this, then this is the best option for your needs.

You still need to use some caution when working on this strategy. It may seem like Bitcoin is the hottest thing right now and that there is nothing that will make the stock go down in value. There have been a lot of different individuals who jump into the market and are disappointed when things do not work the way that they want. Just like with any of the other stock investments, make sure to do your research, decide on a strategy of attack before joining, and always watch the market to look for signs that things may not go the way that you would like.

As you can see, there are a variety of investment options that you can choose from. You will need to research each one to find the one that will work the best for you. You can pick from a strategy that can be profitable but does not require a lot of work or you can go with a strategy that will require more work but will hand you the profits right away. All of these are successful strategies that have been used in the Bitcoin market and are sure to help

you get the results that you want.

CHAPTER 9: HOW IS BITCOIN BEING RECEIVED AROUND THE WORLD

We have spent quite a bit of time discussing Bitcoin and how much it has been growing in the past few years. It seems like everyone is talking about them and that the value and the popularity are going through the roof. While Bitcoin seems to be one of the most popular with the biggest amount of growth, there are plenty of other ones that are doing well at the same time. All this growth and popularity is a good sign for how these currencies will do in the future. It seems like just a short time ago when the idea of Bitcoin and a currency that could be run online without the help of a government or another big agency would have been laughable. Today, most people have heard about one of the digital currencies, and it is usually Bitcoin, and many have decided to find their own path of joining this currency.

Because of this, there has been a huge amount of value growth for these various coins, which in turn makes even more people want to give them a try. While there are plenty of people who are interested in joining the market to send money transfers or to make purchases, there are a lot of people who are capitalizing on this growth and using it as a way to make good money. There is

so much growth with the Bitcoin network that a whole strategy, the buy and hold strategy, is dedicated to just getting some coins and holding onto them as the value of the Bitcoin increases. Looking at the history of Bitcoin, doing this for even a few months could result in thousands of dollars in profits based on how much you initially invested and how high the value goes.

All of this is amazing news for investors. Everyone likes to hear about an investment option that allows them to make a lot of money without having to do a lot of work in the process. However, there are also a lot of governments who are seeing how quickly these currencies are growing and they are starting to get worried about what this can mean for them. And in some places, these governments are starting to add on new rules and regulations concerning how these coins can be used.

One of the biggest concerns that a lot of governments and countries have for these currencies is that users can get onto the network and do what they want without anyone having an idea of who they are. Users do not necessarily need to provide their name, address, or even their personal information of any kind. This is good news if you would like to get onto the Bitcoin network and not let anyone know who you are. However, this could be an issue for some governments.

The first issue that many governments have about this anonymity is that this makes it so much easier for illegal activity to occur. Someone who lives in one country may have the opportunity to legally get ahold of an item. Then they may decide

to ship it over to someone in another country where that item is not allowed. But because of how Bitcoin and its network are set up, it could be hard to figure out who either of these individuals is ahead of time, and sometimes you can't figure out one or more parties after the fact. This could lead to a lot of issues down the road such as illegal sales of drugs, firearms, and more going across borders. This is not something that the government is too happy about.

Another issue that some governments are worried about is that people will get on this system to launder money. The United States is one of the big ones that are worried that someone will go onto the network and make a big profit from their investments, and then will choose to not report those profits when it comes to tax season. Or they may worry that an individual can get on the network and then hide their information so that they are charged less when it comes to tax time. This is why the United States and a few other countries are starting to make it a requirement that exchange sites have to list information about all their users to help make sure that the money is all accounted for.

There are also different methods that are used to ensure that Bitcoin, as well as some of the other digital currencies, will be allowed in that country. Some are pretty open to this kind of currency and the technology that it will bring in. They may do this because it will help to make things easier between two countries when they want to trade together. There are also those countries who are not happy with these digital currencies and who are adding a lot of restrictions meant to harm these

currencies.

There are a lot of examples of countries doing one or the other of these two scenarios. Let us take a look at some of the ways that various countries are reacting to see how it is affecting the way that Bitcoin is used.

First, we will take a look at China as well as a few of the Asian countries around it. In China, the use of these digital currencies is pretty restricted. This is because the Chinese government took a look at how the blockchain worked and they decided that they wanted to create their own government-backed blockchain to work with. They wanted to be able to create this and not have a lot of competition to go with. China is doing this because it has recognized that their banking system is not that good in some of its rural areas, which slows down trade in the country. So, as a way to fix this, China decided to develop its own version of the blockchain to help facilitate this trade and to make banking and other transactions easier for everyone in the country.

Now on to Switzerland. In this country, you will find that people are taking a slightly different approach to using the Bitcoin network. One of the largest banks in this country, which is known as Falcon, has embraced the trend of Bitcoin and how popular it has become. This bank has embraced it so much that it allows some of its clients to store and trade Bitcoin right from their trade holdings at the bank. In addition, the government in that country opened up regulations to make this easier for the bank to do. While this is a step in the right direction, customers

are still using a bank to do their trading so there will be a few added fees as you are working with that intermediary.

You will find that there are many countries who are not quite as open to the idea of working with Bitcoin, but they are not as restrictive as you will find in China and some of those countries. Let us take a look at some of the things that the United States has done in response to the new digital currencies.

The United States has responded, but not in necessarily a good way when it comes to these digital currencies. In fact, some of the regulations that the United States has decided on will slow down the growth of these currencies because they will take away some of cryptocurrencies best benefits. To start, the United States has taken steps to place sanctions on other countries, such as North Korea and Russia, in the hopes of getting them to do a better job at monitoring Bitcoin transactions in their countries. The point of these sanctions is that the United States is trying to make sure that illegal weapons and weapons that are used for terrorism are not being traded around using this currency. Whether or not those countries will really be able to do anything is still to be determined.

This is not where the United States will stop though. There is also some legislation going on to figure out who is earning money on these currencies and whether or not they are reporting it at tax time. The legislation will be aimed at the popular exchange site known as Coinbase, and it could spell disaster for how these digital currencies do in the future. The IRS wants to

ensure that people are not going to Coinbase, as well as other exchange sites and then making money without ever reporting their earning. It is possible that you could go on one of these exchange sites with whatever name you want, and since they do not really ask personal questions about you, you could earn money on Bitcoin and other currencies and not tell anyone, and it would be very hard for anyone to know.

If the United States does push through this kind of legislation, it could be hard for users of these sites. Coinbase, as well as many of the other sites, would be required to start collecting personal information on all their users before they could use the site and get the coins that they want. Then, Coinbase would be held accountable for sending this information, along with information about how much you traded on the site, to the IRS at the end of the year. This would allow the IRS to see who is on the network and find out how much you made. If you did not pay your taxes, the IRS would know right away and you would get in trouble.

The point of this may be to help avoid tax evasion and to make sure that people are not trying to money laundering, but it could cause some issues with these digital currencies. When this is passed, it takes away some of the anonymity that people enjoy on this network and it takes away from the decentralization that is so important for how well Bitcoin and those other networks can perform. And if these two important benefits are gone, it will be harder to convince people in the United States to join on the network.

As you can see from the three examples above, there are different ways that each country is taking in this currency. Some are choosing to create their own blockchain and eliminating the competition completely so that no one can use Bitcoin or other currencies in that country. Some are deciding to embrace this technology and are even implementing it into their banking system to make things easier, faster, and more efficient for the customer. And then there are some in the middle who are not necessarily trying to stop the growth and use of Bitcoin and those other currencies, but who are passing various legislations so that it is harder to enjoy and get all of the benefits out of these networks. Each of these countries is not only reacting in different ways, but they will see very different results when it comes to how Bitcoin will do in the future.

CHAPTER 10: ARE THERE WAYS TO STAY SAFE WHILE USING BITCOIN?

There are several things that you can do to make sure that you are keeping your Bitcoin investment as safe as possible. Remember that while Bitcoin is a great way to invest your money or to make purchases without anyone knowing who you are, there is some risk. There are hackers and other individuals who would be more than happy to take your coins and your personal information. And since there really is not anyone who runs this network, it can be hard to know how to keep your information safe.

Luckily, if you are proactive and work hard, it is possible to protect your personal information even on the Bitcoin network, but you need to be the one to do it. Let us take a look at some of the different things that you can do on the Bitcoin network to ensure that your investment and your personal information stay safe.

The first thing that you need to do is pick out a good Bitcoin address. You can pick out the name that you want to use unless the site like Coinbase give you one. Even if you are using one of

these exchange sites, you can always switch over and use a different wallet to hold onto your transactions and this is a much better option to go with.

When you are choosing the Bitcoin address that you want to use, it is important to go with something that will be hard to trace back to you. Picking out an address that has your first and last name on it will make things really easy for a hacker to figure out. There are various sites that you can use that will make up a random address for you if you are not able to figure one out on your own. Store this information in a safe place so that you know the address, but no one else can figure it out.

If you are planning on using Bitcoin to do a lot of different transactions, then you may want to consider changing that address on occasion. The founders of Bitcoin suggested that the address should be switched after every transaction on the network. It is not necessary to do it that often, but the more often you change out your address, the more security that you can add in.

Another thing that you can do is make sure that you are not sharing any of your personal information online. This includes information about your account and how to get in. If you make a purchase or are accepting money, the only thing that you need to provide is your Bitcoin address. If someone is asking for your password or other information to do the transaction with you, then it is best to walk away.

You should always be careful about the transactions that you are doing with Bitcoin. There are a lot of reputable companies out there that will accept Bitcoin now and these are safe to use. But if you are doing an individual transaction, you have to be careful. Once the money has been sent, it is impossible to get it back unless the other person sends it to you. You do not have a bank or another institution who will be able to step in and refuse the payment if you have a complaint. You need to be careful with the people you interact with to make sure that your personal information is safe and that you are not sending money that you will never receive the product for.

You should also consider making sure that your anti-virus is up to date. Remember that because of the rise in the value of Bitcoin, there are a lot of scammers and hackers out there who would love to get ahold of the coins and use them. If you are not careful, you may find that they can get on your computer and take the coins, and it may take awhile before you would even notice. Having an updated operating system and working with a good antivirus will make sure that all your information stays as safe as possible.

It is important that you are your own advocate when you are working on the Bitcoin network. There are a lot of people on this network from all around the world and if you are not careful, you will run into the few who are out to trick you. Follow the steps above, be careful of everyone you work with on the network, and never give out your personal information and you will be as safe as possible on this network.

CHAPTER 11: TIPS TO INCREASE YOUR RETURN ON INVESTMENT

Investing in Bitcoin is one of the best investments right now. It will probably see a downturn at some point in the future as the people who are interested in it kind of levels off, but right now, interest is high and that means that there is still a lot of money to make.

However, since Bitcoin and cryptocurrencies are still relatively new, it is hard to say how they will react in the future. While they look promising right now and a lot of experts believe that the value will just keep going up, it is hard to know what to expect with these coins because they are so new and we do not have a lot of history to look back on.

What this means for you is that you need to take some steps to ensure that you can reduce your risks, because there are risks no matter how well Bitcoin is doing. Some of the things that you can do to increase your return on investment and to reduce your risks when you decide that it is time to invest in the Bitcoin market include:

Pick out a good strategy

Earlier in this guidebook, we discussed some of the best strategies that you can use when you are ready to get into the Bitcoin network. All of the strategies can work well, as long as you use them the proper way. Sometimes you may lose money, but overall, they will provide you with the return on investment that you are looking for.

With that said, you need to pick a strategy and stick with it. If you are constantly changing your strategies around or you do not know how to properly use your strategy, then you will lose money. One of the biggest reasons that beginners are not able to do well in the market is because they will get started with one strategy and then in the middle of the trade, they will try to switch over to another strategy. This is never going to work so make sure to pick out the strategy that you want to go with and then stick with it.

Pick the stop points

Before you get into the market, it is a good idea to pick out your stop points. These stop points are basically the points when you will enter and when you will leave the market. The enter point will be determined by the strategy that you decide to go with and how the market is doing at the time. But it is so important that you set up a start and a stop point for each trade that you do.

The stop point for losing money is really important. This needs to be placed where you are comfortable with losing money if the trade does not go how you would like. Once the market reaches that stop point, you will cut your losses and get out of the market. If you stay in, it is possible that the market will keep going down and you will lose a lot more money than you are comfortable with. Set that stop point from the beginning and then never deviate. You may lose some money, but it is much better than trying to save yourself if the market goes even further down.

You should also consider working with a stop point for the amount of profit that you are fine with earning. This may seem silly, but it ensures that you do not stay in the market for too long and then end up losing your profits. Once the market reaches this stop point, you will pull out of the market and take your earnings before planning on your next move.

The whole point of adding in these stop points is to make sure that your emotions never come into play. When you add the emotions into the mix, you are basically allowing your money to fly away.

Keep the coins secure

If you are planning on using the day trading strategy, you are probably fine with using the online wallet. You will not be storing the coins for a long period of time anyway so it is not that

big of a deal if you leave them there to make trading easier. A hacker most likely will not have a lot of time to get into that wallet before you trade back out again and switching to a different wallet may slow you down and make you miss out on some good trades.

But, if you plan to hold the coins for any length of time more than a few days, then it is a good idea to pick a secure wallet. Keeping your coins in the online wallet may be the most secure option, but it is also the easiest one for hackers to get into and have fun with. Using the hardware wallet or the cold storage that we talked about earlier can make things easier.

Keep the emotions out of it

As soon as you let your emotions get into the game, you will start losing money. This is true no matter what kind of investment you choose to go with. Your emotions will take away all of your critical thinking skills and will make it impossible for you to think through your decisions in a logical way. Instead, you will be too focused on how to regain your money, or how to make more money, that you are sure to miss out on some important signs before the worst case happens.

Think about if you were in the market and you started to see that it was going down, or at least away from what you were hoping. This can be a really emotional time, especially if it is your first trade. You will want to try to reverse this and may go against the

advice of your strategy in the hopes of earning your money back. Unfortunately, the opposite thing usually happens. You will end up staying in the market for too long and when that happens, you will lose a lot of money, rather than a little money.

This is why it is so important to pick out a strategy and stick with it no matter what. When you are already in a trade, the emotions can quickly overpower you and it is hard to keep them in check. But if you already have a plan in motion and you stick with it no matter what, you will be able to make money or limit your losses, much better than what you can do all on your own.

Ask for help from your broker

As a beginner, it may be a good idea to ask for help from a broker. This is a new type of investment and a lot of people are still unsure how it will work. The cryptocurrency market is really volatile and figuring out when it will stop going up so much or when it will level out can be hard for someone who has never really invested in anything before.

Working with a broker can help solve some of these issues. These individuals have worked in many different types of investments throughout the years and they know how to read the market pretty well. It is best to go with a broker who has worked in Bitcoin, or at least in cryptocurrency, in the past because they have really taken the time to learn the market and they know the best strategies to use to get the most money. A

regular broker will probably work as well if you are not able to find one in your area who has worked with digital currencies, but working with one in this particular niche is the best bet for success.

There are some choices when it comes to picking out a good broker to work with. It is often going to be a matter of personal preference as well as a matter of how much money you are willing to spend on this individual.

If you would like to work with someone who will work with you the whole way, someone who will hold your hand during the investment, then you need to go with a full-service broker. These brokers will be able to sit down and explain the market to you and some will even take over the investment, under your direction. This can help you to make money in this market, even if you are low on time or you have no idea what you are doing. Keep in mind that these types of brokers will charge more money because they are doing a lot more work for you.

Another choice that you can go with is a broker who is just there to help, but who lets you still do all of the trading. These individuals will offer you advice when you ask for it and they often have some tools or platforms that you can use to make things easier. However, you will be the one who does all of the trading, not the broker. This puts you in the top position and can save you money while still allowing you to get some help.

Diversify your portfolio

This guidebook spent a lot of time talking about Bitcoin and how to invest in it. Once you are making some good money in Bitcoin, it is a good idea to consider diversifying your portfolio to include other digital currencies.

It is really risky to put all of your money into one investment type. If something goes wrong with that investment, you could stand to lose everything. But if you spread out your investments and expand out that portfolio, you will be able to protect yourself more without having to worry about one failing so much.

Investing in Bitcoin can be a great experience. There are a lot of different ways that you can invest your money, which can make it the perfect option for so many people. Follow the steps above and you are sure to reduce some of the risks that come with this kind of investment, making it easier to make money in no time.

CHAPTER 12: WHERE BITCOIN WILL GO IN THE FUTURE

The world of Bitcoin is quickly changing. There are a lot of people who are just starting to learn about this amazing currency and all the neat stuff that it can do for them. Some people love that they can get online and do shopping or send money without having to worry about people knowing their personal information or without having to worry about the high fees or long transaction times. There are some who like that it opens up the world to them when it comes to customers to work with. And others like to get into Bitcoin because it is a really good investment opportunity. As more and more people start to learn about Bitcoin, it is likely to change in even more ways.

The Bitcoin network may be really popular, but it is pretty new, as is the whole world of cryptocurrencies. Bitcoin was the very first of these and it was not released until 2009. While it is relatively new, it has seen a lot of growth over the past few years and it has been the basis of many other digital currencies that are now popular as well. The issue is that since this currency is so new, people worry that it may run into some issues in the future. Since it does not have a lot of history to go from, many people worry about what it will do in the future.

The first issue that a lot of people have with this currency is whether or not it will be able to handle the interest that is coming its way. There are a lot of people already who have jumped on board and are excited about this currency, but there are still a ton more who could jump on at a later time. Even with the amount that have joined so far, there have been some issues with the currency, including a huge decline in value of the coin from almost \$20,000 a coin to \$10,000 between December 2017 and January 2018, and the market went even lower to almost \$5,000 before it rebounded back up. No one knows for sure if this will happen again in the future or if Bitcoin will ever go back up because there is nothing to compare it to. This could make it really risky for a lot of investors.

Some people are worried about how various governments will react to Bitcoin and the other digital currencies and how this will impact how these digital currencies are doing. In China, many citizens are no longer allowed to use the Bitcoin network because China developed its own blockchain to help take care of customers in the banking system. This ended up causing a big crash in the market as so many people left. The United States is trying to put stricter rules up to prevent issues like money laundering, which could result in something similar as people forego using digital currencies since some of the benefits are gone.

There is also the issue of Bitcoin being split up because of different ideas of how it should be run. Some people like Bitcoin

and feel that it needs to stay exactly how it is. They do not want to change up the technology that helps to run Bitcoin and they do not want to make it possible to add more coins to the network either. On the other hand, there are some people who believe that a change is needed so that Bitcoin can keep up with the growing demand that it has. These are two clashing thoughts that ended up in a split. This split resulted in Bitcoin and Bitcoin Cash.

Bitcoin continues to grow at a fast pace and while there is a lot of upward movement that comes with this currency, there is also a lot of volatility that comes with it as well. All of this volatility will make it hard to invest in the currency because it is hard to figure out where the market will head next.

There are also a lot of other digital currencies that are coming out on the market and providing some competition. Some work in a different manner than Bitcoin and some work to improve on some of the issues that are present with Bitcoin. Not all currencies will work in the same way that Bitcoin does and some people may decide to leave this network and choose to go with a different one. For example, Bitcoin is known as a payment method, but Ripple can help send money anywhere like an exchange site and Ethereum is good if you would like to work with the development of blockchain technology. As people start to learn more about those currencies and choose to use them, it is likely to affect Bitcoin in some way.

One of the biggest problems that Bitcoin, as well as the other

digital currencies, are facing, is regulations from governments. While Bitcoin is decentralized and does not require any help from a central government, this has not stopped governments in various countries from setting up rules and requirements for users of digital currencies in their countries.

Each country is different, but let us look at how the United States government is reacting to these currencies. In the United States, the government has started to step in and require that all exchange sites receive personal information on any user who is switching out their traditional currency for a cryptocurrency like Bitcoin. This personal information is supposed to help prevent illegal activities like money laundering and tax evasion, but it takes away some of the benefits that come with using digital currencies; namely, that the user wants to go on the network and remain anonymous.

The United States has also required that some of their allies put more stringent rules and regulations on digital currencies in other countries to help prevent the sales of illegal arms and other materials. The United States is just one of the countries to go against Bitcoin and other digital currencies in this manner and the more rules and regulations that are placed on these currencies, the harder it will be for users to join the networks and this can definitely affect currencies like Bitcoin.

Overall, it is estimated that despite some of these setbacks, currencies like Bitcoin will keep growing. There is a large market for these currencies and there are many different benefits to

using these currencies. The large increases that are occurring now are not likely to continue in the future, but it is unlikely that these currencies are going anywhere soon.

CONCLUSION

Thank you for making it through to the end of this book, let us hope it was informative and able to provide you with all of the tools you need to achieve your goals whatever they may be.

The next step is to get started on your journey with Bitcoin. There is so much to love about investing with this kind of currency, and once you get into it, you will be surprised why you did not get started before. This guidebook will provide you with all the information that you need to get started with this investment option so that you can make the most money possible.

The world of Bitcoin is growing quickly and while many people may not fully understand where it is going yet since it is so new, it seems like it is here to stay. Whether you are looking to use it as a way to send and receive money without having to risk your personal information or you are interested in starting out with an investment, Bitcoin as opportunities that will work for everyone.

As a beginner, you may be uncertain about how you are supposed to get started with this great investment opportunity. You may want to read this book as much as you want to get the full understand about Bitcoin and the blockchain that helps run it, how to mine, and some of the best ways to invest and reduce the risk of being in the Bitcoin network.

Remember, though, that putting your money on the clouds will expose you to risk. So, secure your money with a well-built wallet with strong layers of security.

As clear as it can be, Bitcoin really is a life-changing innovation that helped improve how people view money. It has also helped businesses grow and have more secured transactions. Thousands of investors around the world have earned (and still are earning) by investing in Bitcoin. Of course, by using the right strategies, they are able to make Bitcoin a money-making machine.

The Bitcoin and blockchain technology just prove that there is so much for our economy to improve. It is a revolutionary innovation that we must learn and adapt.

We encourage you to continue getting knowledge about this beautiful technology. Your learning should not end here.

Finally, if you found this book useful in any way, a review on Amazon is always appreciated!

BLOCKCHAIN

Understanding the Technology of Bitcoin and Cryptocurrency

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INTRODUCTION

Congratulations on downloading Blockchain: Understanding the Technology of Bitcoin and Cryptocurrency and thank you for doing so.

Unless you've been living under a rock, you've at least heard the whispers and rumors about this great money-making machine called the Blockchain. And, if you're like most people, you probably raised an eyebrow and viewed the whole thing with a bit of skepticism. No doubt you mentally pushed it aside and labeled it, as many others did, as another Ponzi scheme or type of fraud where little tiny con artists were waiting in the back of your computer to steal all your hard earned money away.

It's logical that one would think that way. When the technology was first introduced, it was definitely something new and foreign to the world. And that's is the normal knee-jerk reaction received when someone approaches with a too good to be true story.

But then, you probably heard the fantastic reports that surfaced at the end of 2017 about the meteoric rise in the value of cryptocurrencies. No doubt, the reports were staggering and it peaked your curiosity so you decided to do a little research. That was when you uncovered the fact that large corporations like Amazon, JP Morgan, and Microsoft were either entering this market in a big way or seriously considering it.

If this is your story, by now you've come to realize that the Blockchain and cryptocurrency is not some fly by night scam. They are real and legitimate and people all over the globe are realizing that it could be a real goldmine if they choose to invest.

Yes, it is something new, something exciting, and some may even consider it to be something mysterious but this new technology is here to stay. It is a world phenomenon that has taken hold and brings with it a wealth of opportunities that have yet to be revealed.

If this, or something like it has brought you to this book today, then you're ready to know the secrets behind the Blockchain. Unlike hundreds of other books on the market, this is not a book designed to convince you to invest, nor is it a book geared at convincing you of its worth. Within these pages, the goal is to discuss the inner workings of the Blockchain, to give you a foundation of knowledge that you can then use to determine where you will most benefit from Blockchain Technology and use it to your advantage. Here, in these pages, you will learn...

- A history of currency and why it needed to be changed
- The original intent of Blockchain Technology
- Why so many are adopting it
- The basic mechanics of the Blockchain
- Its future prospects and how it will impact society
- Alternative uses for the Blockchain
- What is Blockchain mining and why is it needed
- Bitcoin – the one that started it all
- Where it might be used in the future
- The legal side of Blockchain Technology and what you should know

Just like anything new that is introduced to the world, there is a lot of conversation about the Blockchain, some of it good and some of it not. However, it is essential that you learn as much as you can about it. Every year, more and more people are openly adopting it and incorporating it into their everyday lives. Even if you choose to not invest in the Blockchain Technology, the chances that it will have an impact on your life at some point in the future are very real. To that end, we hope that you embrace that opportunity when it comes to you so that you can benefit just as many others have from this new technology that is embracing the world.

There are plenty of books on this subject on the market, thanks

again for choosing this one! Every effort was made to ensure it is full of as much useful information as possible, please enjoy!

CHAPTER 1: UNDERSTANDING THE BLOCKCHAIN STRUCTURE

Bitcoin Technology has now been here for nearly a decade. What was once viewed as a crazy fad is now being embraced worldwide. But much of the public has been cheated out of learning the real underlying value of this technology. Because the majority of the focus on the Blockchain is presented in the world of finance, the idea that the Blockchain is used to transfer money, to make purchases, and to invest in some way obscures the real value and the myriad of other uses behind it.

It can go well beyond what has been presented in the market today. One could think of it in the same way the world viewed the iPad when it was first released. What can you do with it? The potential for the new little device was hard to grasp because the world had never seen anything like it. It wasn't until people began to realize that its value went far beyond a basic calculator and it was the applications that one used that would determine its true value. Through these applications, the device could perform whatever actions the user wanted it to do.

This is the power of the Blockchain. It speaks to its unlimited

potential and therefore will be a major asset in the global scheme of things. To appreciate this more deeply, we need to compare what life was like before Blockchain Technology and how that technology is changing the way we do things today.

Life Before the Blockchain

Introducing Blockchain Technology into an economy is not new. In fact, we have introduced this technology repeatedly when it comes to handling finances. We happily accepted the debit cards and credit cards based on the old system without complaint and in time the masses will converge on Blockchain Technology soon enough and those instruments represented our first step into the digital world.

When you think about the immigrant who has traveled to a new country to get money to support their families back home, you begin to understand how things are in need of changing. He works hard and when it is time for payment he happily accepts his pay and heads off to the Western Union office or his local bank to send the money he has promised his family. For this convenience, he has to pay a fee, which usually takes up a substantial amount of the money he plans to send.

Think of another scenario. Most of us have bank accounts where we deposit our money. When it is time to transfer money into our accounts it mysteriously takes “3 to 5” business days before it is credited and available for us to use. Strangely enough, if you

go to the supermarket and use your bank card to pay for your purchase, by the time you get home, the money is already deducted from your account.

This is often the case, which can create quite a bit of frustration. The money leaves your account instantly but it doesn't arrive at its destination for several days, even if the destination is right across the street? There have been many answers to explain this. The banks will readily say it is slowed down to prevent fraud, others will say that it is a way for the bank to make a little money on the interest.

Whatever the reason, it seems that the person who loses out in these situations is the one that the money belongs to. In this system, the banks get to use your money before you do. The cost to the world for this and other charges that these middlemen demand is actually gradually eating away at many people's chance to get ahead. But all of these situations are only possible because of our use of digital currency. While this feature provided us with speed and convenience it also gave the middleman more opportunities to use our currency for their own advantage.

Why was the Blockchain Needed?

Understanding why we needed the Blockchain can be easier when you take a look at history. Before any current economic system, people transferred value through the barter system. The

system was pretty simple, you have an asset that someone else wants and you trade it for something you want of equal value.

As people began to understand the importance of this type of transaction, they began to keep records of the trade in some form of a ledger. Our current economic system is really just the next evolutionary step in the barter system, only now, everything is done electronically and all the data is kept on a single centralized server. One common location where it can be easily accessed when needed and only certain approved or permissioned parties can use it.

What do we mean by an electronic barter system? Basically, it is a network of users with special permission to update the ledger. For example, you go to the bank to make a deposit, the money you receive is a trade for the work you've performed (barter). The teller (permissioned user) makes a record in the system that you deposited a certain amount of money in their care.

Another example, you want to buy a home, you go to the bank and make an agreement to borrow the money to pay for the house. The loan officer makes a record of the money you borrowed in a different electronic ledger. As you make payments on your debt, the ledger is updated until the debt is finally paid off, at which point the deed and ownership are transferred into your name.

Of course, in our more technologically advanced world, things

became a little more complicated. Now, we have these networks where all of the data related to any transaction is continuously maintained online. The person with the permission to access, retrieve, or update them is the person with the control so even if it is your money, you do not have control over what happens to it. If you want to send currency to your grandmother, you basically have to get permission to send it.

As we go through the pages of this book, you'll learn that there is a huge difference between Bitcoin and the Blockchain. Everywhere you look there are rumors being spread about Bitcoin. Some say that it is just a fad and will soon fade away while others claim it will eventually fall into disuse for one reason or another. The reality is you cannot ever uninvent the technology that Bitcoin is based on. So, if in the future, Bitcoin disappears from use in the same way the barter system eventually faded out of use, the technology can and will still be practical and will continue to be used in a myriad of other ways.

Like all technology, the Blockchain has already evolved beyond Bitcoin and every day more and more uses for a decentralized platform continue to be created. Since its first introduction, the technology has been studied, analyzed, and researched from all angles. Its adaptability makes it possible for practical uses to continue to be introduced and as more of the world becomes comfortable with it, the Blockchain will gradually become a part of everyday life in much the same way the computer, the Internet, and the Smart Phones have.

Now that it has made its debut and we've learned its potential, it will be difficult for the world to let it go. Imagine having to go back and live life with the horse and buggy, or relying solely on a landline for communication. In today's modern world, that lifestyle would be viewed much like a disability. We'd be seen as the person walking down the street with the white cane, slowly tapping our way to our destination.

In reality, we have yet to completely grasp the full potential of the Blockchain and probably won't ever be able to. One thing for sure, the Blockchain is here now and those who are reluctant to embrace it will find themselves soon falling behind the rest of the world as they continue to make giant leaps into the future while the rest of them are standing on the corner with their white canes in hand.

What Problems Does Blockchain Technology solve?

Like many people, when you first heard the word Bitcoin or Blockchain it was in the context of a means of disrupting the current financial system we are under. The general concept was that because users could now execute transactions without that pesky middleman in the mix, which could save you a lot of money on those expensive fees. As more and more people come to realize this one simple fact, they would flock to Bitcoin like birds flying south for the winter.

While all of that is true, there is a lot more to Blockchain

Technology than just the ability to save on transaction fees. Right now, there are more than 1,300 different cryptocurrencies, all using the same technology to solve many more problems than those related to money.

Because the technology can be used to record any type of transactions, the problems it can solve are quite extensive. Here are just a few examples:

Supply Chain Management: It can be used in any situation where there are a number of different participants in collaboration to manage the production or supply of goods.

In any supply chain, there is a manufacturer, a shipping department, warehouses, distributors, and finally retailers. If all the logistics were kept on a Blockchain then you would have a trail that could easily be followed detailing every step in the history of a product.

Blockchain Technology in this type of situation works better than a centralized system because of its ability to cross many organizational boundaries. It creates an environment of neutrality between all different parties but allows information to flow from one area to the next freely.

But that doesn't mean that all information is shared this way. Once it is decided which information should be shared with each

person, it is encrypted on the Blockchain so that only those with the correct keys can get the data they need.

The general idea behind the Blockchain is to eliminate the middleman. By doing so, you give more control over transactions back to the people engaged in them. When you think about how many ways we use the middlemen in our day-to-day lives you might even be surprised.

Of course, there are the obvious ones. Banks are used to validate and approve loans. Title companies are there to validate and approve properties. And Legal professionals usually validate and approve contracts of all kinds.

Using this middleman often creates problems of their own. These types of transactions take a lot of time to process, often cost a lot of money in the form of fees, they are more vulnerable to hacking or technological failure, exclude the users from much of the process, require special skills to process, and are more likely to be prone to error.

For the most part, we've accepted these limitations without question, but the Blockchain is an effective means of solving many of these issues.

By using what is called a consensus protocol, The Blockchain system automatically will reject any changes to a transaction

without the agreement or the consensus from the other nodes (or computers) in the system. This means that there is no single point of failure that could compromise or cripple the system like the weaknesses found in a centralized system.

Those participating in that community must validate any transactions completed in that database. As a result, it is almost hackproof, eliminates the middleman, and it cuts down much of the transaction costs and a lot more.

We don't know yet all of the problems that the Blockchain can solve but we do know that the ability to perform peer-to-peer transfers of currency, data, or other forms of value will continue to find uses in many more applications in the future.

Security

When it comes to security, the Blockchain itself might prove to be even more valuable than what it was designed to protect. We'll learn more about how the Blockchain works in the next chapter but here, we'll discuss why it gives your assets a lot more protection than the current centralized systems. Before we can really grasp the risks associated with protecting assets, we need to understand the difference between a public Blockchain and a private one.

A public Blockchain, like the one Bitcoin uses, is a system

designed to record transactions so that they are accessible to anyone who wants to read or write them. This means that anyone can create and publish a transaction as long as they can solve the cryptographic puzzle.

With Bitcoin, for example, since there is no single user that is responsible for verifying any transaction, all those in the system must follow the same algorithm created for verification purposes. The user who solves the problem first is the one that will be rewarded by payment of a set amount of cryptocurrency. Once the transaction is verified, it is added to a block and then work on the next block begins.

Private Blockchains, however, are primarily used in financial settings and limits access to only those who can read or write on its ledger. Markets that use the private Blockchain are those where multiple parties contribute to the transaction but may not have real trust with each other. Some examples of this type of situation may include property assets, commodities, and private equities.

It stands to reason that private Blockchains will be much more secure than the public Blockchain as fewer people have access to the data stores. However, when it comes to ensuring security, we need to take a closer look at the system's architectural aspects.

Whichever type of Blockchain is used, there must be a consensus obtained from the users within the system. This means that each

node must be able to communicate with all of the other nodes in the system. In the private Blockchain, users have control over who is allowed to access a node and which nodes are allowed in the system.

There are a number of reasons why a node may be pushed out of the system. They may not maintain enough activity to keep up with the other nodes; any node that transmits incorrect data will be rejected, and a node that restricts the flow of information may also be rejected. This creates one level of security in preventing fraudulent transactions from being uploaded into the system.

Another feature that makes the Blockchain more secure is the hash feature, which we will discuss in more detail in the next chapter. When a block is formed and added to the chain, a hash is created and assigned to that particular block. Part of that hash must include the hash from the previous block in it. This means that no other hash code can apply to that particular block and no other block can use that same hash code. Any attempt to change it would affect all the transactions in that block and every one afterward. A flaw that would readily be identified and would literally stop the chain in the process

The combination of a decentralized system, the hashes, and the other protocols that are implemented with a block, makes the Blockchain much more secure than any centralized system that most institutions are now using.

Transparency

As we see more and more corrupt individuals continue to try to “beat the system” the demand for more transparency in business becomes increasingly louder. Over the years, businesses have often used non-transparency as a means to cheat or overcharge its customers. However, the Blockchain has addressed this issue by making all activity transparent.

By conducting business on the Blockchain, every transaction performed is time stamped and kept on a ledger for everyone in the system to see. While the identities of the individual participants are encrypted along with other proprietary details, the fundamentals are openly displayed on the ledger. The fact that information can only be added to the Blockchain and can never be deleted ensures that the details of all the data will remain there in perpetuity. Nothing can be altered or hidden from view.

Interestingly enough, any transaction made on the Blockchain has to be accessed and executed with public and private keys. No matter what you try to do these keys are linked to a wallet, which, in turn, is linked to a personal identity. The very permanence of the data trail makes it resistant to manipulation and fraud.

Right now, those using the Blockchain have completely different purposes in mind but there is a certain level of consistency when

it comes to transactions. By making it much harder to conceal the details of any action done on the Blockchain, you can be confident that there is less risk of fraudulent activity involved.

How the Blockchain will affect institutions and traditions

No doubt, there is a great deal of hype when you're talking about the Blockchain. Whether or not it will be the panacea of the future remains to be seen, but no one would deny that the potential is there for huge changes in many of the institutions and traditions that we have become so accustomed to.

The Blockchain's first introduction through Bitcoin paved the way for many changes in the future. Its potential for a major impact in economics is undeniable. The absence of a decentralized server means that transactions can be made from one person to the next eliminating the need for the middleman, who for years has been taking the lion's share of profits from work done by other people.

For example, a writer for a book will usually have to go through a publisher to get his work published and then sold in bookstores. The publisher would take huge sums of money from the sale of the book and the writer, the creator of the book receives only pennies on the dollar. With this system, the writer has little control over what happens to his creative work.

However, with the Blockchain, working in conjunction with other digital programs, the writer can create and publish his own book and create a smart contract with distributors and retailers, cutting out the publisher altogether. This same rule can also be applied to the music industry and other similar areas of creative art.

Financial Services

We've already talked a little about how Blockchain Technology can impact the financial industry. By allowing for peer-to-peer transactions, the middleman can be completely cut out of the picture, saving the users huge amounts of money in fees and other charges.

This would cut out the need for banks and other financial institutions and can cause a great deal of disruption in the process. As of now, the financial services industry is the primary source for moving money from one place to the next, storing value, lending, trading, and basically keeping track of all monetary assets and their transactions IN THE WORLD.

With Blockchain Technology, this could all change. As more people find out they can do many of these transactions themselves, we can expect to see a gradual shift away from the current system and move more towards monetary coins like Bitcoin, Monero, and ZCash and more.

Insurance Industry

The Blockchain allows for any type of asset to be tracked and traded in cyberspace. Each transaction will contain the identification of the asset, its validation, and all the ownership rights connected with it. When all of this documentation can be uploaded to the Blockchain without the need of a mediator, the tables will have turned.

Rather than waiting for weeks for an insurance claim to be settled, smart contracts will be able to execute claims in a matter of minutes without having to submit an endless trail of paperwork.

Already, many insurance companies are using their own insurance apps that will allow individuals to purchase a new insurance policy, make their payments, and have all the details and requirements on it uploaded to the Blockchain. If there is an accident or an injury, the individual could go directly to an approved doctor to be treated and have the claim managed almost instantaneously.

Smart contracts can self-execute payouts when previously agreed upon conditions are met and can keep track of the results of any claim they have already submitted all via the Blockchain.

The IoT (Internet of Things) will also play a significant role in the

insurance industry because as more and more consumer possessions need to be protected they will have to adapt. Today, a great number of devices and gadgets function because they are connected to the Internet and are capable of transmitting data in all forms. Insurance companies will have to expand their reach and create a means of protecting data that is completely stored in cyberspace.

Shipping and Handling Industry

Right now, the shipping and handling industry is huge and tedious. It is not just matter of logistics, but as each item passes from one phase to another to get to its final destination, it leaves a huge trail of paperwork behind. Sales contracts, legal agreements, bills of lading, port documentation, credit letters and so much more; all of it must accompany the cargo on its journey.

As these shipments go from one place to the next, each person handling them usually asks for a cut of some kind. If payments are delayed so are the shipments that go with them. This can cause major complications in the shipment process, keeping it from reaching its final destination.

By making payments through the Blockchain, the whole process can be simplified. Once the information relating to each shipment is sealed in the block, there is no way to change or delete it. This means that the system can't be altered to show

preferable treatment from one industry to another in the form of bribes or additional fees that are not customary.

Initially, it is expected that the Blockchain will be used in financial areas of the shipping industry. Rather than letters of credit and sales agreements, there will be smart contracts detailing the movement of each shipment. Once implemented in one area they will then gradually spread to other areas, eliminating the need for massive amounts of paperwork to be kept.

There will be many advantages to adopting the Blockchain.

- Processing of orders will be faster
- Fewer mistakes
- Full transparency
- Better security
- Less costly

The Music Industry

Already, there is a great deal of interest coming from the music industry to adapt the new Blockchain Technology. The idea of transferring digital artistry from one person to the next with the help of smart contracts could prove to be huge.

For example, if you create a song and upload it onto the Blockchain, you can also create a smart contract that stipulates how the recipient will use it. If they plan to use it for their own listening pleasure, the cost may be minimal but if they plan to use it in a movie, the Smart Contract could detail the conditions that must be met to have access to it. Whether you want to charge a higher fee, put a time limit on it, or some other stipulation, the idea of managing your artistic content can eliminate the need for expensive managers who are constantly working out deals to get ahead.

Already, we are seeing music Blockchains appearing where musicians can directly license and collaborate and sell their own music completely free from the middleman, who could take away much of the money they worked for.

Since it can store any type of data on a decentralized server, it is literally impossible to alter, few people will be able to get away with stealing the music and take credit for it.

Of course, if you give a little more time, there are many other reasons for institutions to use the Blockchain and every day more and more people are finding new ones. Its ability to transform us into a more modern world is just an afterthought. The real power behind the Blockchain is the equalizing that will result. When more people are getting a fair wage for the services they provide, more equalizing of the distribution of the world's

assets will be the natural result.

How Technology Make Our Institutions Faster and More Cost Effective

While there are still some people who are determined to think that the Blockchain is just a new fad that will eventually fade away, more are beginning to realize its true potential. It's no secret that Blockchain Technology is giving us a whole new way to do things. The more we can secure these digital relationships, the more likely the Blockchain will eventually become the primary support of all sorts of industries.

CHAPTER 2: THE BLOCKCHAIN TECHNOLOGY – HOW DOES IT WORK?

Even with all the talk and the hype floating around about the Blockchain, it is not always easy for the average layperson to understand exactly how it works. If you are new to the concept and struggle to get the full sense of the Blockchain technology, then chances are this simpler explanation will help to clear things up for you.

To begin with, many people have the mistaken idea that the Blockchain actually started with Bitcoin. This conclusion is logical since it was with Bitcoin that the whole concept was launched and revealed to the public. However, even before Bitcoin, the Blockchain was quietly making its debut in cryptography many years before. The idea of using it for other aspects in the real world was not even considered at the time.

The Nuts and Bolts of the Blockchain

The first introduction of this new technology actually appeared as far back as 1979 in the domains of cryptography and data

structure. It was not called the Blockchain at the time but instead was named after its creator, Ralph Merkle and was then called the Merkle Tree.

Its function was to verify and handle data that was being transferred between different computer systems. It began as a peer-to-peer network whose sole responsibility was to validate the data to ensure that none of it had been altered as it was transmitted from one computer to the next. Its secondary task was to ensure that it was not corrupted with any false data that may have been inserted later on. In simple terms, the Merkle Tree was to ensure that integrity of any data that was being shared in the system.

In its early formation, the Merkle Tree was primarily used to create a secure chain of blocks, which consisted of a series of data records with each one connecting to the one before. By doing so, the most recent addition to the block would actually contain a complete history of every transaction that happened prior to its creation.

To understand how this actually works let's take a closer look at some of its features and what it was expected to do.

- It keeps a record of all data and transactions uploaded to the ledger.
- It uses a distributed system to verify every transaction.

- Once a transaction is verified, it is added to the Blockchain and can never be deleted or altered.

OK. Now that we know what it is supposed to do, let's see how it actually accomplishes it.

When you hear or read anything about the Blockchain, you'll often hear the word "key" in relation to any transaction. Keys are actually an encrypted code that carries within it your unique identity. Each transaction receives a public key and a private key. These are combined to create a unique digital signature that unlocks the data stored inside that block. The public key is the one used to identify who the originator of the transaction is and the private key gives them the authorization to perform certain agreed-upon actions on the account.

You might think of the public key as your address to the home you live in and the private key is what opens the front door and gives you free access to what's inside. With both keys, you can transfer funds, make withdrawals, or perform any other actions with the digital property contained inside. Without the private key, nothing in relation to the content of the address will be accepted. This is why it is so important for you to keep your private key safe. Anyone who has it also has the power to control those assets.

Every transaction requires an authorization. So, if you want to send a transfer to someone, you need to have that other party's

public key. For example, if you want to transfer funds then you need the private key from your account and the public key from the second party. You would use your private key to authorize a transfer to the address of the other person. The amount of the transfer will then be deducted in the public ledger and once validated, it will be recorded that the funds no longer belong to you but now belongs to the next person. This will be time-stamped and given a unique coded identification number (or hash).

Once everything is verified, the completed transaction is broadcasted to every node in the system so they too can acknowledge that the transaction occurred. The result is that every transaction will have the exact same data: your digital signature (both public and private keys used together), a timestamp, and a unique identification code.

Each transaction is then connected to the one before, thus creating a chain that you can go back and refer to for as long as it takes.

What is Decentralization?

One thing you hear repeatedly when people refer to the Blockchain is the term decentralization. This is the key feature that makes the Blockchain resistant to tampering, fraud, or corruption.

All of these actions are completely peer-to-peer. Every time a transaction is completed it is not loaded up onto a centralized server but instead is added to the ledger, duplicated, and each copy is sent to thousands of locations around the globe. Bitcoin, for example, has about 35,000 nodes working in its network. To alter anything on the network, you would have to change it on every single node in the system, therefore, making it practically impervious to destruction. As a result, any attempt to modify it would not be very successful because as soon as the node you alter is not in sync with the rest of the nodes it would be rejected from the system.

This does not necessarily mean that the system is foolproof. As with all other things that offer security, there are already hackers and cyber criminals of all kinds, all working diligently to find ways to circumvent these security measures and given enough time, they will eventually succeed. However, the system as it is now does offer a lot more security than our current centralized method of doing things. The fact that there can be no single point of failure, for now, creates a situation where there is little incentive to try to “cheat” the system as the amount of effort required to accomplish that task may not actually be worth it.

There is also the concept of the “51% attack,” which means that if for some reason 51% of the nodes in the system were to validate an erroneous transaction it could still get approved and uploaded into the ledger. While that is highly unlikely, it is still a possibility for some dedicated hacker.

This may sound very technical for the average person. There is a lot of industry jargon that can sometimes sound confusing but as this new system continues to be adopted into the global scheme of things, more people will come to understand its fundamentals much better.

Another negative that many people point out are the transaction fees that are often associated with each transaction. While they do not cost as much as the regular institutions that now conduct these funds, they can be a drawback for some people. And finally, there are no “instant” transactions. Unlike with banking and other online transfers of funds that happen within seconds, when making a transaction on the Blockchain you could be expected to wait as much as an hour for the approval to come through.

It is expected that in the future as there are more tweaks to the technology these validation times will be much shorter. The hope is that they will be able to get approvals at the same rate as VISA approves credit card charges now processing more than a thousand transactions per second. Today, Bitcoin’s top speed is around 7 transactions per second so they have a long way to go to compete with the biggest credit giant in the world.

What is a Block?

You may now be wondering what exactly makes up a block on the Blockchain. Is it a single transaction, a list of codes, or is it

perhaps a type of ledger? These are logical questions when you are new to the Blockchain Technology.

According to Investopedia, A block is a collection of files with data pertaining to the actual network. It is a system of creating a permanent record of the most recent transactions. In essence, a block is simply a page out of the digital ledger on the network.

Let's try to break this down a little more simply. Every time a transaction is performed and information uploaded into the system, a permanent record is recorded detailing what was sent, who sent it, and who received it. All of the transactions along with the accompanying data performed within a certain given time period is compiled together to create a single block. Each block is linked to the previous block completed before it creating the Blockchain.

The Blockchain is the complete record of all transactions made on the network and the block is like taking one page out of the ledger. It is not a single transaction but instead a group of transactions made within a specific time period. For example, Bitcoin creates a new block every ten minutes on average so a block would be all the transactions made globally within that time period. Each block contains the history of the blocks before it and once it's added to the chain becomes the foundation for the next block to be formed.

To validate and complete a block, a complex mathematical

problem is linked to it, which must be solved. Miners work to solve these complex equations (done by computer) to earn a reward. Once it is solved, the answer is then shared with other nodes in the system for validation. When it has been validated by enough of the other miners, the first record in the next block is the transfer of funds to the miner who solved the previous block.

The challenge is in the complexity of the mathematical equation that makes creating a new block difficult. No block can be submitted to the Blockchain unless the problem has been solved so it is nearly impossible for a single individual to verify a block on his own. Even with computers working on the problem, it can take as much as 10 minutes or more to find the solution (in the case of Bitcoin).

What is the Chain and Why is it Needed?

The chain or the Blockchain is all of the solved blocks linked together to make a single chain. It is the digital ledger that contains all of the transactions in a database that can be shared with all parties in the network.

While we have managed to get along quite well up until now without the Blockchain, it has some very important features that are already making significant differences in the way we do business today.

One of the reasons we need Blockchain Technology is to ensure that there is trust among strangers. With this type of technology, there are fewer people involved in a transaction; as a result, there are fewer hands in the pot, so to speak. As people create transactions, verify every one with a system of complex algorithms that will confirm the transfer of value and create a permanent register to be added to the chain there won't be a middleman to investigate the trustworthiness of the other party. This will all be done on the Blockchain giving the average person the power to work with whomever they please.

Blockchains are needed primary because they are more secure than storing data on a centralized ledger. Because transactions performed on the Blockchain can be completely anonymous, but at the same time transparent, it can be considered a more honest and trustworthy way to transfer anything with value.

While the Blockchain has very encouraging features, we have no idea in which direction it will take us in the future. Only time will tell its true value to the people and how it will change the world. However, even now signs are very clear that people are embracing this more secure means of transferring value in a world where few people can be trusted.

What Actually Happens in a Blockchain

So, let's now see how a block is created and added to the Blockchain.

1. When miners start to compute a block, they will choose all the transactions that they want to be included in their block. Since the Blockchain is global it is literally impossible to include every transaction made within a certain time frame in a single block. Usually, the miner looks at the pool of transactions and chooses which one he wants to work on. In addition, they must also add one Coinbase transaction to their address. This could include any transaction they choose to form the base of the tree (remember the Merkle Tree) of transactions.

2. For the miner's work on the block to be accepted as part of the chain, it can only include valid transactions. If even one of the transactions proves to be invalid or incorrect it will be rejected by the chain and the miner will have to start all over again.

3. The miner then begins to build the block header, which will be shown as a string of digits. This will include...

- a. The version (4 bytes)
- b. The hash of the block before (32 bytes)
- c. The Merkel root or the tree of transactions (32 bytes)
- d. A timestamp (4 bytes)
- e. Bits, which are the representations of the network difficulty (4 bytes)

f. Nonce (4 bytes)

Once the miner has completed the header he needs to start solving the problem. The solution always begins with the nonce at 0, hash (sha-256 2x) the block's header. Then he will check if the hash is under the right target. If it is not, he breaks up the parts and starts all over again. If the hash he comes up with is under the right target, the block has been solved. Then he transfers the header and all the associated transactions to the network where it is accepted or rejected by other nodes in the system.

Finding a block to add to the Blockchain is very rare and extremely complicated. This explanation is extremely simplified and difficult at best to decipher. However, if you're able to stick with it, the rewards are great. The payout for solving a single block on Bitcoin is 12.5 Bitcoins, which could amount to thousands of dollars so well worth the effort if you are patient enough to keep working on it.

Could the Blockchain Replace Our Institutions Altogether in the Future?

No one knows for sure where Blockchain Technology will go in the future. What we do know is that it has the potential to replace many of the systems that we have come to rely on every day. It is a means of giving power back to the people rather than having it dispersed amongst a host of disinterested persons who

are only interested in exploiting the value of the work that the majority of society will generate.

It is possible that we will see some very big changes in the near future but as for whether or not the Blockchain will replace our institutions altogether, that may be far off into the future. If history is to be repeated, the time it took for the world to adopt other technological advances could serve as an example and give us a clue as to what to expect.

When Thomas Edison first invented the light bulb, it was ridiculed and laughed at. Some mocked it as a fairy tale and others claimed it was not worthy of the attention of scientific men. Yet today, the light bulb has permeated society to such a degree that anyone without electricity or lights in their home are considered to be impoverished and living far below the normal standard of living.

Schoolchildren hear about how great the Wright Brothers were when they first invented the airplane in 1903 but others labeled it as a mere toy with no real scientific value. Today, the airplane is the single most important instrument that allows us to travel to distant regions of the earth in a matter of hours.

Even personal computers were rejected off hand when they were first introduced to the world. In fact, there was a certain fear associated with it and some claimed that even touching it could cause an affliction. It was even given a name,

“computerphobia.”

And automobiles were often considered the next evolutionary step from bicycles and were far out of the realm of reality for the masses.

Repeatedly, as new innovations were introduced, it took a while for the world to embrace these ideas. Eventually, all of that changed. In most cases, it took many years for the new technology to take root so if there is rejection today in regards to the Blockchain, don't worry about it. Just as with many other great new ideas, the people need time to warm up to it.

It may take another decade or it may take thirty or forty years. If Blockchain Technology does replace our institutions, it will be a gradual process, weaning the people off of the systems that they have become so accustomed to. It is a good guess, however, that once they do make the switch to the Blockchain, few will be willing to come back to the old system again.

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How Can It Be Used To Build Trust Between Strangers?

Trust is at the core of all types of transactions. Without it, there will always be a risk that one party or the other will default on their promises. However, this is a hard quality to develop when dealing with strangers. Without any background knowledge or

knowing their personal history, it is difficult to protect yourself in a transaction.

The Blockchain resolves this issue by means of hard-wiring the requirements and details of any transaction into the code and then automating it. By doing this, emotion is taken out of a transaction. The sad story as to why they couldn't make payment is removed, the deceptive talk of one individual in an attempt to persuade the other party to let go of his currency is eliminated.

By making the Blockchain the primary source, the users do not have to know all of these details of another party in a transaction because the stipulations of every agreement are already encoded into the system. In fact, the Blockchain is so good at managing trust that many corporations are already taking steps to embrace it.

IBM, Microsoft, and Intel are now using the Blockchain as another option for performing transactions mainly because of their public ledger giving transparency to every deal that is made.

Governments are also looking at it for managing other facets to prevent fraud. When the Blockchain was first introduced, it was used for the transfer of currency. Now, it can be used to transfer anything of value opening the door to voting situations, lotteries, identification cards, digital art, graphics, payments, jobs, and insurance.

Estimates show that the Blockchain will provide approximately \$176 billion in value to businesses within the next 6 years and by the year 2030 that number could amount to \$3.1 trillion.

The main reason for this growth is because of the security it provides and the trust that is built right into the system. Because there is no single point where the system can fail, it makes it less vulnerable to crime, and the complex encryption also makes it difficult to break.

The trust is so much a part of the Blockchain Technology that it is actually more secure than the traditional methods of exchange. Every day more and more people are beginning to realize the value of the Blockchain and how it can keep all of their assets safe when doing business with strangers.

CHAPTER 3: BLOCKCHAIN OUTSIDE OF FINANCE

Since the Blockchain was initially used with Bitcoin, most people are very familiar with how it works when exchanging currency from one place to another. The system is pretty straightforward. You start the transaction at one point and indicate where the funds will go and how much and the system pretty much does the rest.

The disconnection comes, however, when they try to grasp how the Blockchain can be used in other ways. These can sometimes be confusing and difficult to grasp simply because the technology generally moves in an entirely different direction.

By now everyone knows something about Bitcoin and every day more people are learning about Ethereum. These are the two largest and most recognizable uses of the Blockchain. But of the hundreds of other uses, you may not be so familiar.

Alternative uses for the Blockchain

With Bitcoin, we've already seen how the Blockchain works with finances. We can easily send money across borders from one place to the next without much hassle. However, even outside the area of finance, the Blockchain can do a lot more.

- Asset Management

With our traditional form of trade, the cost of sending money across borders can be extremely expensive. This is primarily because of the fees that must be paid to each party that participates in the process. When you send money across borders, you have a broker, a custodian, a settlement manager, and perhaps a few more. Most of these parties work in the background and neither the sender nor user know about their role in the movement of money. Still, they require a fee for their services that must be paid for.

With the Blockchain, however, the ledger reduces the number of middlemen involved in a transaction and by encrypting the records there is no specific chain of command that needs to be followed.

- Insurance

With the current system, it can be a frustrating and time-consuming experience to file a claim. Those whose job it is to process claims usually have to sift through many and weed out

those that aren't legitimate. Information received has to be verified, and they must ensure that they are paying out to true policyholders.

All of this is simplified on the Blockchain because of its transparency. Its system of encryption ensures that only legitimate claims can get through the process and the true owner of assets can be insured. It reduces the risk of human error by completely automating the process.

- Property Management

When it comes to transferring property, the Blockchain can be just as beneficial. Even with a tangible property like cars and houses or intangible property like patents and shares, each of these could have smart technology already embedded in them.

When a purchase is made the registration can be stored on the Blockchain in the form of a Smart Contract that will detail all of the contractual requirements for all parties involved in the sale. The keys used to gain access to the account can be passed on to the new owner every time the property changes hands.

This can be done in several ways.

- Hard Money Lending

With the traditional lending system, hard money lenders usually take advantage of those with poor or no credit by charging a large percentage of the loan amount for their fee. This starts a vicious cycle where the borrower fails to keep up with the high payments and eventually ends up defaulting on the loans. The lender then swoops in and takes the property as collateral and the borrower loses everything and has another mark on his credit as a result.

With the Blockchain, any stranger can lend money and record the details on the public ledger. Borrowers will have more freedom to choose which creditor they want to work with and the middleman that charges the higher fees can be completely cut out of the process.

- Managing Smart Property

We live in a very technological age and most of us now own some form of smart property. Keyless car entry, smartphones, home management systems are all part of the Internet of Things and connect us to the world around us. The problem with this existing system is that they can easily be tampered with.

The Blockchain will be able to reduce those risks. Your keyless car, for example, could be set up with an immobilizer, which would allow the car to only function when a certain protocol is met. The same could apply to the smartphone, which would only

work once certain codes encrypted on the Blockchain are met.

Ethereum

Just like Bitcoin took the world by surprise, Ethereum is doing the same but on a completely different scale. The second largest cryptocurrency in the world is a perfect example of how the Blockchain can move beyond the world of finance and be used in a wide number of situations.

Developers who use Ethereum can actually build their own software on top of the Ethereum Blockchain. They can then use the network's ledger to build trust right into their applications. Once the application is in the system there is no way to alter it or delete it. It is as if it has its own life that becomes part of the Ethereum platform.

This may not seem like much but in reality, it gives both parties to any agreement or transaction an incredible amount of security. Whenever an agreement or contract is programmed into the system, the conditions surrounding it become automated so neither party can pull back and claim fraud, deception, or otherwise manipulate the arrangement.

Ethereum works by using a scripting language for each application. The original idea was that they could use the Bitcoin Blockchain as a foundation but rewrite the base code so it would

be more compatible with additional programming languages. Later, it was decided that instead of using Bitcoin's Blockchain, they would start from scratch and create a whole new and separate Blockchain that would allow for these new changes.

Ethereum was launched in 2015 and in just a few short years has surpassed hundreds of other altcoins to become number two on the exchanges.

Smart Contracts

The major selling point for Ethereum is the Smart Contract. These are self-executing agreements that can be encoded into the language making each one unique from every other one. Because of the absence of the middleman, these contracts are far more accurate and secure than any traditional agreement.

If all parties involved in the contract maintain their agreement the contract will automatically execute without any third party involvement. If one party does not maintain their agreement, the funds, property, or whatever value was being transferred automatically reverts back to the original owner. In essence, smart contracts work in much the same way as the escrow company. They hold the assets until all parties meet their obligations and then close the deal.

Autonomous Organizations

Another feature that Ethereum uses is autonomous organizations. These utilize a series of smart contracts to implement and attain certain preset goals. This allows for the Blockchain to be used in voting, education, healthcare, and many other industries.

Through these organizations, Ethereum can be used to transfer files, maintain records, collect funds, monitor patients, and track the progress of ongoing projects without outside help. This could make huge differences in terms of delays in processing. Because the system is totally automated, actions can be taken as soon as certain goals are met, avoiding the wait for verification and approvals by human input.

There are many advantages to the use of a DAO. First, the system cannot be slanted to favor one party over the other; the guidelines and expectations are completely transparent so no one is left out of the loop. They are much more affordable options that can be implemented much faster than the current system most businesses are using.

ICOs

Ethereum also is used to create new coins tailored to meet the needs of certain markets or areas of interest. Because the Ethereum Blockchain allows for the creation of new applications,

and autonomous organizations, some users felt there was a need to have even more control over the token used to fuel their applications.

Now, every new application used on the Ethereum Blockchain has its own token that is used to power it. Basically, when a new idea is added to the Blockchain, funding is needed to get it implemented. The sale of the tokens released on Ethereum is their means of generating funds to implement the new program.

One needs only to have an idea and a white paper to get a new ICO set up and released on the Ethereum Blockchain. In 2017 alone, these new coins raised more than \$1.6 billion making it one of the most prolific kickstarter campaigns the world has ever seen.

Gaming

Blockchain Technology is also transforming the gaming industry as well. Probably the most obvious evidence of this is in the issuance of cryptocurrency as a reward for in-game achievements. Once you receive your coins, they can be used to purchase upgrades or to play other games.

Prior to this system, once you achieved a certain level in a game you were rewarded with tokens you could only use within that particular game platform. This gives players more freedom and

opportunity.

But that is only the beginning when we look closer at the Blockchain and gaming union because it is essentially a place to store data, the fundamentals of each game can actually be stored right on the Blockchain. The details of each game can be recorded and preserved down to every last move of each player. It can record records of high scores, and even monitor game tournaments.

With this technology, all of these things could be done automatically and the data would be easily retrieved for anyone who actually holds the keys. There are actually four key areas where players can benefit from Blockchain Technology.

Collaboration

It will be easier to collaborate with other people; you can form teams to meet certain game challenges or you could turn your time into a career as a game master. Prospective members can actually purchase a place on your team as payment for you taking the leadership role.

Through the ICO program, players could garner support in existing championships. You can send gifts to other members, and can freely interact with other players in more profitable ways.

Profit

Players can turn their game time into money-making ventures. Normally, serious players can accumulate a lot of rewards. The traditional system currently in use will generally disappear as soon as you quit the game. With the Blockchain, earnings can be stored and kept until you decide to use them however you want. With this system, a good player could actually earn enough to make a living playing the game he loves.

Growth

Players can also tap into their success as a player. By playing purchasing options, using upgrades, and getting help from other players in the game, they can accumulate rewards, study the moves of others players, and retain all that information on the Blockchain for future reference.

Create

Gamers will also be free to design their own games tailoring them to their own personal experience. One of the most frustrating experiences by gamers are the limitations that may be imposed by a game's developers. They feel as if they are playing someone else's game. However, for the serious gamer

who wants to take it to the next level, they are free to collaborate with whoever they choose to create their own games and their own set of rewards.

While gamers can definitely benefit from Blockchain technology, so can the developer. The ability to create games that provide meaningful rewards will mean that gamers will take the experience even more seriously. They won't be restricted by the parameters set by the old system but will have more flexibility in design, strategies, and collaborations.

The Blockchain and Cloud Storage

When it comes to storage, speed is everything. People do not want to wait for even a few extra seconds when they need access to their data. But with the traditional cloud storage situation that is exactly what happens.

In traditional cloud storage, you upload your data to the cloud with the idea that you can have access to it whenever you want to. When you are ready to retrieve it, you send your request to the data storage center and it sends it back to you. The idea is pretty straightforward but there have been some problems with this type of system.

1. If the data center is far from your location it could cause the delivery to be slow. You might end up waiting for a while for your

files to download.

2. Larger documents could also take a long time to retrieve.
3. They are expensive. Data centers need to be temperature controlled and the tech needs to be continuously updated.
4. Then there is the matter of security. These centers are always vulnerable to hacking and exposure to viruses and malware.
5. Finally, there is usually a risk of losing information due to human error.

With Blockchain Technology, data storage can eliminate many of those problems. There are already several companies that are using the Blockchain instead of the traditional system.

Storj and Sia

Storj creates a decentralized storage system without the need of server farms. With the Blockchain, data can be used and stored on the extra space people may hold on their hard drives.

For example, you have 250G of space on your hard drive that you don't expect to use. Through Storj, you can rent it out and get paid for the use of the space. The data still remains secure because it is encrypted and broken up before it is sent out into the system. No single computer will hold all your data and without the keys, no one can compile it again and retrieve it.

Users only pay for what they use and for the time they use it. There are no monthly contracts or time limits. This system is set up on the Ethereum Blockchain, which means it is all run by smart contracts, removing the risk of error.

The benefits are many. With easy access to your information, you can eliminate much of the wait times and more importantly, significantly reduce the costs associated with cloud storage. Right now, to store 5TB of data on the Blockchain will set you back around \$10 a month. This is a huge difference between that and regular data storage, which could easily run you several hundred dollars or more.

Digital Publishing

Many people are surprised to learn just how many problems the Blockchain has solved. Problems that they didn't even realize they had. But in the digital publishing industry, the Blockchain has addressed some issues that have been around for a very long time.

When it comes to the creative industry, the actual artists have historically struggled to get fair payments for their work. Licensing has been restricted, and few recognize the artist's rights. This is true in several aspects of the publishing world, especially in the area of digital publishing.

In our traditional system, we see a lot of free art being shared without payment. This can include music, books, articles, and images, much of which is not traceable back to its original source. This limits the amount of income a true artist can get for his work, usually only a tiny fraction of its true worth.

Another problem that presents itself is getting licenses to use this artistry. If you would like to find a picture to include in a book you are writing, there are many databases with free to use images that you can access. However, there is no guarantee that these are legally licensed to be used for free. This results in many cases of unauthorized use of artwork.

With Blockchain Technology, every piece of art, no matter what it is used for will be traceable back to the original owner. It will have its own digital fingerprint that will change with each license and ownership status change. If you choose to purchase a book, a song, a video, or any other type of digital art through the Blockchain, the creator will get immediate payment. There won't be any middlemen who are inclined to take a piece of the earnings.

But money is not the only way artists can benefit from the Blockchain. It will eliminate the need for third-party authorizations, which means there would no longer be a need to purchase ISBN numbers, copyright offices will no longer need to keep track of who owns what. With all its transparency, the Blockchain will make it clear who owns what work and who is authorized to use it.

The Blockchain was not designed to replace bookstores, publishers, or any of these agencies that have been involved in this type of industry but it was created to give transparency to these negotiations. Creators can receive micropayments for their work directly from the users without untold sums of their earnings going off to other people. Nothing can be more frustrating than an artist who sells their rights to work just to get enough money to meet their basic needs, only to find that it is being resold for thousands of times what they received. The Blockchain will eliminate all of that so that there is a more equalizing of events in the process.

Voting

One of the most surprising ways the Blockchain can be used is through voting. Rather than having lines of people standing outside in inclement weather to exercise their right to vote, with the Blockchain it can all be done in the privacy of their own homes.

Because of the tight security of the Blockchain, it could be an excellent way to secure votes without having to deal with the inconveniences of going to voting polls. As a matter of fact, the convenience of voting via the Blockchain may even get more people to vote than there are now.

By using the Blockchain, it would be easier to ensure that those who are voting are exactly who they claim to be and have the legal right. There would be much less risk of tampering and the chance of fraud will be drastically reduced. As each vote is uploaded and added to the Blockchain, there is no chance of it being altered in any way or form.

This feature has already been tested in several elections with huge success. It remains to be seen how long it will take for the rest of the world to embrace but, but it is really only a matter of time.

Many of the ideas mentioned above are only in the beginning stages of development. In time, it is expected that many more will be introduced. However, as of right now there are some that have already met with success and are already being rolled out to the masses very soon.

Bitcoin Cash

As an offshoot of the Big Daddy Bitcoin, Bitcoin Cash uses the same source code but in an entirely separate form of currency. Now, the third largest coin in the crypto world, the primary difference between Bitcoin and Bitcoin Cash is the block size. With Bitcoin, the blocks can only be 1MB in size but with Bitcoin Cash, it can be as much as 8MB per block.

This means faster transactions making it a more attractive option that may one day dethrone Bitcoin altogether.

Litecoin

Introduced in 2011, it is not as old as Bitcoin, which was released in 2009. They both have the same role in servicing the people but Litecoin also can process transactions much faster. With Bitcoin, it takes around 10 minutes but with Litecoin it can take as little as 2 minutes.

Because it is capable of creating new blocks more often, they can literally process transactions at a rapid speed. This means that new transactions can be added to the ledger and validated much faster.

ZCASH & Monero

Even now, people are beginning to improve on the existing Blockchain format. For example, while we continuously say that a transaction performed on the Blockchain is secure, it is only as secure as the user is in keeping his private address hidden. So, if someone can gain access to your private address, he has access to your wallet and to all of your assets.

Since your wallet is linked directly to your identity, it means that

the transaction history is not completely hidden from view. The result has been that some coins have already been blacklisted because of connections with suspicious or otherwise illegal activities. This was determined because they were able to trace the history of those coins back to their origins. With Monero, there is an additional level of security embedded directly into the Blockchain.

Monero has two separate technologies that can be used to conceal the identity of the parties involved in any transaction. It also conceals the transaction amount. Instead, the funds do not go directly to the recipient but go first to what is called a “stealth address” and from there will be sent to the waiting party.

By using this system, the personal information about the sender and the receiver is never uploaded into the Blockchain making it virtually impossible to follow the funds back to its original source.

ZCash has the same objective but works just a little differently. Instead of a stealth wallet as the go-between, its Blockchain uses cryptography to conceal the identity of the user. First, a key is created to initiate the transaction, the sender uses the details of the transaction along with a separate key (a prover key) and the initial generator key. The system will then study his responses to validate the transaction. If it is validated, then the transaction will go through.

When you use ZCash or Monero, you have options though. There is no need to hide your identity in every case but it is an added feature that you won't find on Bitcoin.

There are other methods for creating untraceable transactions using cryptocurrency. ZCash and Monero are just two of these types of options. With more than 1300 crypto coins on the market, with a little research, you'll find a coin for just about anything under the sun.

Regardless of what you expect from the Blockchain, this node-to-node network is taking off in many different directions. While it is still relatively new to the world, we can expect that it will be improved upon and expanded in the coming years so that it can be adopted in many other facets of life.

One thing we need to be conscientious of is its newness. There is still a great deal of regulation and ethical standards that have not yet been implemented into the program but that aside, this technology could prove to be a powerful game changer for all forms of business, government, healthcare, economics, and other areas of our daily lives.

Right now, it has slowly eased into the world's consciousness and there are still many who have yet to adopt it, this new innovation is not about to go away. With large corporations and major industries seeing its potential, even though it is expected to go through many changes in the future, it is definitely a

technology that will be embraced.

CHAPTER 4: BLOCKCHAIN MINING

After all the hype about the fortunes to be made in cryptocurrency, it is easy to see why people are so excited about it. There is no shortage of stories of how people have made fortunes in a very short period of time. No matter what country you're in, people are readily jumping on the bandwagon hoping to get a piece of the pie. It's really strange how when the big conglomerates, giant corporations, financial institutions, and governments are not controlling the money just how much of it is available to spread around.

The general consensus is that the Blockchain, if nothing else, will be a significant instrument in causing an equalizing of wealth in the world. So, it's no wonder that people are trying everything they can to make a little extra cash on the side.

There are two primary ways you can earn money with cryptocurrency. The first way, and by far the most popular, is by investing. You've probably already heard the incredible stories of great gains from just investing a few hundred dollars here and there.

However, if you're tapped out and don't have enough capital to invest there is another way that many have found successful. Mining cryptocurrency has the potential to produce the same results if not better than investing. Which avenue you should take is not for me to say but there are some pros and cons of investing and the same is true for mining.

Mining is an extremely important aspect of the cryptocurrency market. In fact, without miners, no cryptocurrency would exist. Therefore their role in the entire process is the most important of them all.

What Does a Miner Do?

The most fundamental job of a miner is to pick up transactions that have been uploaded into the system and then validate them. He must read the transactions and make sure that all the elements match.

A miner must also maintain and protect the privacy and the security of the Blockchain. As a new block is created, he must validate them and ensure that they are added to the chain and a copy is sent to all the other nodes in the system.

Miners are the people that create the blocks that make up the

Blockchain. They do this by picking up the latest unrecorded transactions, validating them and assembling them into a new block.

When a new block is added, the miner needs to solve the puzzle that is associated with the transactions contained inside. Each time this is done he must complete a series of complex algorithms to find the solution to the problem.

If all goes well, the miner will receive a substantial payment for his efforts. Because mining is such a complicated series of tasks, there is no guarantee that one will be successful in handling and working out all of the details. As a result, it could be weeks or even months before he can solve a single block. But the price for the work done can be so substantial that it is worth the effort dedicated to it.

Mining is very competitive, which also plays a significant part in their success. Those who can validate and solve the puzzles the fastest are those who will be able to make more money. Since they are only paid when they solve a block, the faster the task is done the better your chances of getting the reward you're working for.

Now, this is an oversimplified explanation of the miner's role. In this chapter, we'll look at a bit more closely at what really happens when you mine cryptocurrency.

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The Nonce

Before a block can be formed and validated it must have a variable included called a nonce. This is a number that can be used only one time but when used will change the results of the hash. The miner has one goal, to find the nonce with a specific number of zeros in the beginning.

Each block also contains a header, which is made up of the metadata that links it to the previous block, the nonce, and the hash output. It is all put together in a form called the Merkle Tree. It kind of looks like a family tree with a parent node on the top and then it is connected below by two nodes.

All of the data is placed inside the Merkle Tree structure. The top part of the tree is the root. All the other nodes in the tree are connected in some way to this top node but the bottom of the tree is where all the transactions contained within that particular block.

The miner's goal, once he has all of this information in order is to take all the new transactions and place them in the Merkle tree without exceeding the allowable limit for transactions for a particular block.

When the tree has been completed, the miner then adds a pointer to the block header and then searches for the nonce. This is mostly done by trial and error. You start with one iteration and continue on until you find the nonce number that matches perfectly. The nonce must have 32 bits in the header, so when doing the calculations you will have to count more than 32 possibilities to find your solution.

What makes it difficult is that you may end up exhausting every single permutation of the nonce you can think of and still not be able to find the solution. That, however, does not mean that you won't be able to mine your block. You may have to adjust your base value to find the solution. Whatever the case, all blocks are mineable, it is the miner's job to find the trick to figuring out the problem.

The first transaction found in a block also includes a special field that can use up to 100 bytes of arbitrary input from the miner. So, if you've already used up all your possibilities with the header nonce, you can shift your interactions to the Coinbase field to find the solution. However, if you change the value of anything on the tree, it must change at the root of the tree and in the block header as well.

Mining cryptocurrency is very hard and complicated. It is nearly impossible to do with just a simple calculator or by hand. This is why miners work with very special equipment that can shift through all of the iterations in a matter of seconds; something that the human brain could not possibly keep up with.

Getting Set Up With Mining Equipment

With more and more people embracing cryptocurrency, mining has become very popular and exciting. In fact, it has become such big business that just the sale of mining equipment could yield you a pretty tidy profit.

Just as the Blockchain has evolved over the years from the original Merkle Tree, mining hardware has also changed. It has become much faster and definitely more efficient. The type of hardware and software you use will depend on several different factors. The kind of coin you are mining and how deep you want to go into the mining venture.

The Hardware

Mining is primarily a volunteer job. While you do get into it for the money, the ability to mine successfully depends a great deal on the type of equipment you have, your location, and your state of determination. There is no guaranteed income and you could work for months before you receive any type of payment. This is why when payouts are made they are not referred to as income but as an award.

In the past, if you had the right equipment you could easily join a mining pool and start making money but given the number of people who are trying to mine now, you need to not only have the right equipment you need to have good equipment to get you on board. Depending on the kind of mining you want to do you have several options.

CPUs

When the Blockchain first made its appearance, miners simply used a central processing unit (CPU) chips from their home computers. To compute the hashes on their personal computers or their Macs was easy. At the time there were less than 10 lines of code and any type of home computer could solve the problem. However, it wasn't the best in terms of efficiency. It was slow and cumbersome. While you could still hook up to the system with a home CPU, these work so slowly that there isn't much of a chance of you being able to solve anything. Since mining works

by competition, the first one to solve the problem wins, you want something that will work much faster so you can have a good competitive edge.

GPUs

The next step up was the Graphics Processing Unit (GPU). These are actually graphics cards that work with the CPUs and can provide significant support for visually displaying all the elements on your monitor. When you use a GPU in mining, you can work on multiple hashes at a time, thus increasing your ability to solve the puzzle and find the nonce that much faster.

Another advantage of the GPU was that it wasn't very expensive and they were pretty easy to set up. Today, with the GPU you can calculate up to 200 million hashes per second. However, believe it or not, that is not fast enough to keep up with the fastest miners on the job.

There are some definite advantages to GPU mining.

- GPUs can overstock, which means that with the right adjustments you can actually get your device to run even faster than it was designed to do. When speed is more important than the quality of the visual, GPUs are the preferred choice.
- GPUs are also hardcoded to allow for faster computations.

- Scalability also works to your advantage. You can easily run several GPUs at a single time and control them all from one motherboard. Putting many of them together on a single mining rig, you can literally calculate millions of hashes at a single time.

The drawbacks to GPU mining

- They consume a great deal of energy, much of it has little to do with the mining process.
- They have a tendency to overheat very fast, especially when you stack them on top of each other or run them in a closed environment.
- This means you will get a pretty sizable electric bill, which could mean you'll have to spend a great deal of cash before you can solve your first nonce. This is why the majority of miners live in countries with lower power bills.

FPGA

Field Programmable Gate Array consists of specially integrated circuits you can program. These are the next step up from GPUs but they eliminate much of the drawbacks and problems. When your FPGA is set up properly, you can actually calculate billions of hashes per second. That may sound like it's very fast but, if you were mining Bitcoin, it would still take you years to find a single nonce even if you had a hundred of these.

While these may not be the ideal solution for Bitcoin mining there are other Blockchain currencies that you would do well with FPGA mining.

ASIC

Finally, the fastest of them all is the Application Specific Integrated Circuits. These are special chips that have only one use, to mine cryptocurrency. Right now, it is the only one fast enough to keep up with the fast-growing Bitcoin chain. Because Bitcoin's algorithms are constantly being adjusted and increased in complexity, it is difficult to determine the exact hash rate but estimates put it working at approximately 1000 GH/s, which is difficult to comprehend.

Mining Software

Once you have your hardware set up, you then need to determine which type of mining software you're going to need. There are several different options here as well. The best one for you will depend on the type of hardware you have and the kind of currency you plan to mine.

CG Miner

CG Miner is the favorite software for those who are just starting

to mine. It is compatible with practically all the operating systems so whether you're using Windows, Mac, or Linux you should be able to work well with it.

It uses a very basic Command Line Interface to receive and execute the commands. Once it is installed, just follow the prompts to get into your chosen mining pool and you're all set.

BitMinter

The BitMinter uses a Graphical User Interface and already has the mining pool settings installed when you receive it. However, its reach is limited because the settings are already set and cannot be changed. That means that you can only mine those currencies it is set up for and you cannot switch to a new pool or currency if you decide to change.

MultiMiner

This software works best with ASIC or FPGA hardware. It comes with additional features and is an easy option for those who are new to mining. Once installed, simply follow the prompts to add in the information for your chosen mining pool and you'll be ready to get started.

Mining Pools

In the past, mining was very easy to do. You didn't need special hardware or software to get started. You could stay at home and mine as many blocks as you needed. However, later as the field became more competitive, an individual mining alone would seriously struggle to compete. With faster equipment, you have a much better chance of solving a hash and recouping much of the money you've put out for the setup.

To beat this complication, mining pools were created where a number of miners pooled their resources to increase their chances of solving a puzzle. When a puzzle was solved, they would split the reward and everybody had a better chance of making money than trying to go it alone.

The Puzzles

We keep referring to hashes and nonces and other terms but the miner's job is really to solve a puzzle. Once the puzzle is figured out, a block can be added to the chain and an award is given.

The puzzles are extremely complex and are designed in a way that the amount of the reward justifies the amount of work that is being done. The mining puzzles are not like any puzzles that you may have worked on before so let's examine how they are made up now.

- They must be very difficult to solve but very easy to verify
- Their difficulty must be adaptable to varying circumstances
- It has to have a solution that anyone within the network is capable of solving.

There are different ways to create mining puzzles. Bitcoin uses the SHA256, which is a specific mathematical algorithm that takes something that is input into the system and turns it into an output. For example, an algorithm can be a code that takes all the numbers in a string and adds them together. So, the input would be 1234 and the output would be 10, which is the sum of $1+2+3+4 = 10$.

Now, this is a very simple explanation of how an algorithm works. The SHA256 algorithm is much more complex and extremely difficult to solve.

The Mining Rig

As you can see, the secret to successful mining has to be speed. While the CPUs are definitely capable of solving the complex algorithms to solve the puzzles for the blocks, they are incredibly slow in comparison to the ASICs or the GPUs. While they are still useful with some cryptocurrencies they are much too slow and cumbersome for others.

You could choose to purchase the more expensive ASICs if they are designed to work with the cryptocurrency you plan to mine but you could also create something called a mining rig. This is a number of devices (GPUs) connected so they could work together to increase your hash rate and mining power.

You may not be able to boost your speed on a single device but you can increase your chances of solving more hashes by having more than one device working at a time. With a mining rig, you can set up as many devices as you can handle and have them all working in sync to solve the puzzle.

Why is Mining Needed?

Considering the complexity of mining, one might think that solving these algorithms to solve a block could be a bit of overkill. The idea that it takes that much energy, effort, and speed to solve may not make much sense to the average person. But there is one other reason why mining is so important.

It guarantees a steady stream of crypto introduced to the market over a period of time. For example, when a block is solved with Bitcoin, the miner receives his award in Bitcoin. The current rate for a solved block is 12.5 Bitcoin. This is new currency entering the system. So rather than flooding the market at one time with all the currency, it ensures a steady release of Bitcoin until it reaches its maximum amount of 21 million.

Without mining, anyone could generate Bitcoin and the market could be overloaded and therefore killing to true value of the coin.

Mining also helps to reduce fraud. While the system is very crude, it is also expensive. Anyone bent on beating the system would be less likely to invest in costly equipment and go through the tedious and time-consuming work that miners do. The bigger payout is actually a stronger incentive, to be honest rather than try to take shortcuts to cheat or fake a block on the chain. In short, mining is the very thing that keeps the network functioning smoothly and all the users working honestly to everyone's benefit.

Alternative Methods of Mining a Blockchain

There is, however, another alternative to mining that is slowly beginning to emerge in the cryptocurrency world.

The Block Lattice

The block lattice is another peer-to-peer system that eliminates the need for miners. The concept is that each account in the system has its own Blockchain and only the holder of the account controls its use. With this type of system, no digital fingerprint is needed.

Rai Blocks, another popular cryptocurrency uses this system. So, for every transaction that is completed, there must be two blocks. The first block is created by the sender and the second block is created by the receiver. Once the funds have been received, there has to be a third block generated confirming that the transaction has been completed.

This system is done all with automation so there is no need for miners. The system does have a network of nodes that are constantly monitoring the system for fraud and will interrupt any transaction that is perceived as fraudulent and prevent it from going through.

While it has yet to be rolled out full scale, the thinking is that the block lattice will be much more scalable and easily adaptable to a wider range of Blockchain uses in the future.

Tangle

Another system that seems to be able to function without miners is Tangle. This system was built to take advantage of the IoT (Internet of Things), a host of common devices that are already connecting us together through the Internet.

Tangle allows you to send funds without having to pay any type of fee. One of the reasons for this is their ability to bypass the need for miners to verify transactions. Instead, they work

through a number of acyclic graphs to verify and process transactions. In this case, those who use the system are the ones that will verify the transactions of the other users. So, whenever you initiate any type of transaction that transaction triggers an algorithm that requires you to process two other transactions.

Once these transactions are solved, your own transaction is submitted to the tangle and is confirmed. As a result, everyone that uses Tangle also must contribute to the validation of the transactions in the system.

This is a very revolutionary idea that hopes to resolve many of the issues that come from mining the Blockchain. Right now, the only cryptocurrency that is using Tangle is IOTA with ParagonCoin anticipating adopting it in the future. While it has yet to be introduced full scale it will be worth it to monitor its progress to see what it holds for the future.

CHAPTER 5: THE STORY THAT STARTED IT ALL

No discussion of the Blockchain would be complete without a discussion of Bitcoin, the largest cryptocurrency in the world. In fact, it was Bitcoin that put the Blockchain on the map when it was first released. Even though elements of the Blockchain existed long before Bitcoin, few people knew about it. But in 2009 when Bitcoin was first released, it was just a matter of time before the technology that supported it was going to make its public appearance.

The idea behind Bitcoin was revolutionary, to say the least. It was both exciting and frightening as it allowed open access to anyone in the system without having any type of central authority to answer to. It gave the impression that anyone could do whatever they want on the Blockchain.

This, of course, was not the very first attempt to use Blockchain technology. As far back as 1998, there was B-Money. The idea was created by Wei Dai and suggested an “anonymous, distributed electronic cash system.” In this system, money could be transferred by means of broadcasting the details to all

participants in the transaction. In this system, while the transfer of funds could be made peer-to-peer, any contractual agreements between parties still had to be made with a third party serving as an arbitrator.

There was also Bit Gold, an idea that was based on calculating a string of “bits” to solve a puzzle. The puzzle would be extremely difficult to calculate making it less likely that people would want to create fraud.

Both of these ideas were formulated and even discussed at great length but neither of them was ever fully developed.

By 2008, ten years later, Bitcoin was introduced as a peer-to-peer electronic cash system. It was first mentioned in a paper written by Satoshi Nakamoto. The name was mysterious as no one knows his true identity. However, in his paper, he detailed exactly what is involved in this new system. Nakamoto’s ideas combined both the concept of B-Money and Bit Gold into one global idea with a few other changes to the program.

2009 – The software for Bitcoin was first released and the mining process was implemented for releasing new Bitcoins gradually into the system.

2010 – Bitcoin is valued for the very first time. At this point, it had not yet been traded but it had been mined so it was very

difficult to determine its true worth in terms of monetary value. But in 2010, someone decided to use 10,000 Bitcoin to purchase two pizzas. This was a historical landmark because the fact that a price was offered and accepted for something that people actually knew the value of, Bitcoin was given its worth for the first time. Interestingly enough, those 10,000 Bitcoins today would be valued at more than \$100 million.

2011 – As Bitcoin was just getting its footing, the competition was already beginning to pop up. The coin was already beginning to gain traction because of the concept of decentralization and encrypted currencies so the first altcoins began to pop up. They are called altcoins because they are alternatives to Bitcoin. Some of the first to appear was Namecoin and Litecoin. Today, there are more than 1,300 altcoins with more popping up every day.

2013 – Bitcoin crashes. Over the next two years, Bitcoin's price gradually moved up to pass the \$1000 mark. But then something happened and the price made a major dive to around \$300. At that point, it took Bitcoin more than two years to recover. Many people faced losses at that time but the ones who decided to stick with it and chose not to bail were rewarded once again as the coin recovered.

2014 – Bitcoin began to get a new reputation that was once again an attempt to undermine its potential. Rumors began to spread that Bitcoin was simply a tool that allowed criminals to conduct illegal activities using the anonymity behind the Blockchain to pay for it. This was immediately followed by the collapse of Mt.

Gox, the world's largest exchange of cryptocurrencies at the time. Owners of more than 850,000 Bitcoins lost everything – money that was never to be recovered.

Even today, investigations are still ongoing as they try to understand exactly what happened. The total loss amounted to more than \$450 million dollars worth of coin at the time. Today, that price would be somewhere in the vicinity of \$4.6 trillion. Still, Bitcoin was to recover from that devastating loss very quickly.

2016 – Alternative coins

Up until this point, Bitcoin was the major leader of all cryptocurrencies. However, a new coin was beginning to appear that was determined to steal its thunder. While there had been many altcoins that began to crop up after Bitcoin, none were going to come close to its success until Ethereum arrived.

Ethereum used an entirely new Blockchain and introduced the impressive feature of smart contracts and Dapps. Ethereum also introduced the concept of ICOs, small fundraising platforms that allowed investors to invest in start-up ventures for completely new coins and get in on the ground floor of promising new ventures. However, while many of these proved to be legitimate there were some that turned out to be completely fraudulent schemes that were designed to appear like legitimate investment tools. Some governments even went so far as to ban their sale

outright.

2017 – Bitcoin began its historic rise to the top. After hitting \$10,000/coin for the first time in November of the year, barely a month later, in December the price soared to nearly \$20,000.

Going well beyond anyone's expectations, many attributed this price increase to several factors. First, many concluded that the beginning of this sudden growth spurt had to do with a decision from Wall Street that was to allow regulated futures trading, as well as the implementation of the Lightning Network, a new system that would dramatically increase the speed of processing Bitcoin transactions.

Another factor that boosted the meteoric rise was the interest it was getting from major financial institutions. As more and more merchants were willing to accept Bitcoin, prominent banks like Barclays, Citi Bank, and the Deutsche Bank and BNP Paribas have all expressed interest in finding ways to incorporate Bitcoin into their systems. This meant that Bitcoin had finally made it to the mainstream and people were more willing to accept it.

2018 – What will the future of this coin hold? None of us knows for sure but after 2017, the question of whether or not Bitcoin is a fad that will soon fade away has finally been settled. Bitcoin and the Blockchain and all they can do is now an important part of our lives.

Regardless of your personal viewpoint on this single coin that made Blockchain Technology a household word, it is here to stay. Will it be the success and revolutionize the world as many predict? That remains to be seen, however, very few people will be able to ignore the potential possibilities that exist for it to happen.

There is little more that we can do at this point but sit back and watch the chips fall where they may. Whether you want to be a part of the Blockchain craze and invest, become a miner, or use the many altcoins already in place is up to you. Some people are still sitting on the sidelines waiting to see what happens. Either way, there will be an incredible show to watch in the coming months and years.

The Obstacles Bitcoin Faces

By now, Bitcoin has been on the scene for nearly a decade and it is slowly emerging into the mainstream with amazing success. Still, it is not without facing a few hurdles of its own. Some of those hurdles are definitely the result of mudslinging from outside doubters but there are quite a few obstacles the coin must overcome from within as well.

Volatility

When speaking to investors of all kinds, one of the first things it

points out is the coin's extreme volatility. From the moment of its inception, it has had major highs and lows with the price fluctuating more than \$1000 in a single day. Sometimes, it may plummet hundreds of dollars in a matter of minutes only to skyrocket to the stars in the next hour.

This kind of behavior makes investment decisions very tricky. It attracts an abnormally large group of speculators who buy the coin with the idea that the price will continue to rise only to sell at the top of the wave. When too many investors sell at the highs, it causes the price to plummet once again.

This is not an attractive option for the more stable investor who is looking at investing long. To leave your money in an investment tool like this can be very damaging in the long run. One must keep a constant watch on the price fluctuations so they know when their investment is taking a downward trend.

It is true that historically, Bitcoin's price has had a steady risen over the long term but the concerns for new investors or those with a lower risk factor cannot be ignored. Right now, the majority of people are wary of this type of volatility and therefore choose to stay away. To attract more people to this caliper, Bitcoin's prices will have to become much more stable than it is now.

Difficulty in Use

Another obstacle that is holding Bitcoin back is it is not very easy to use. While it has made some major improvements in simplifying the process of transferring funds from one location to the next, it is still not considered very user-friendly. This means that more people will still favor the more expensive approach of sending money because it isn't as confusing a process.

Today, if you want to buy Bitcoin, you must first open an account at an exchange, link a credit or debit card to the account and then wait for approval. Once approved you can purchase your Bitcoin. In many cases, this can be a real hassle and one big enough to discourage many people from taking that step. In addition, it is a major hurdle that will keep many from even considering using it as a major form of currency.

Investments in the stock market take only seconds to complete, purchasing precious metals, or many other types of investment instruments are much more appealing considering the complications associated with Bitcoin.

The introduction of Square (NYSE: SQ) into Bitcoin is possibly a means of fixing the issue. These would allow users to purchase it through an app with just a simple touch of a button. If this feature works, many more people who are not as astute technically will be more inclined to make the purchase.

It's Not Universally Accepted Yet

Every year, there are more merchants, retailers, and other types of companies that are willing to accept Bitcoin as payment for their products and services, but it is still a long way from universal acceptance. This could also be due to the slow processing times and the difficulty in use mentioned above.

While the Bitcoin development team is working to resolve these issues, it is still an obstacle that discourages many to accept it. What good is having a peer-to-peer transaction capability if you don't have a peer you can use it with? If their new relationship with Square works out or if another payment processing company chooses to work with Bitcoin, then it may become more accepted in the future. Once retailers and services start accepting it more the people will naturally follow.

Risk of Theft or Fraud

There are definitely security measures that make it difficult to steal Bitcoin but it is not impossible. The security measures are only as effective as you are in protecting your public and private keys. Those who are the best at protecting their assets are once again, those who are pretty tech savvy and understand the differences between types of wallets, how Bitcoin actually works, and the type of hardware and software needed to keep it secure. Those lacking in this basic knowledge worry that they won't be able to protect their assets once they have acquired them.

Yes, Bitcoin has some impressive security measures in place, but just like with any new instrument, it is just a matter of time before some criminals will be able to bypass the precautions and hack into their system. It has happened before and there is no reason to think it won't happen again. To protect your Bitcoin, it is important for you to know how to make sure your assets stay in your possession.

Unlike having a bank account with FDIC insurance to protect you Bitcoin has yet to reach that level. There are more regulations being introduced by the governments but Bitcoin's protection is nowhere near that level of protection that other financial instruments have.

Its Reputation

We all have to live with our own a bad reputation and Bitcoin is no different. In its early days, Bitcoin was often associated with criminal activities and while it has worked hard to be accepted in the mainstream, many people still recognize it for its use on the Dark Web, as a way to launder money, or as a tool to purchase illegal products.

It and several other cryptocurrencies are still the only means of making payments anonymously. Because of this feature, it naturally attracts those people who want to conduct illegal activities without being identified. Unfortunately, these were Bitcoin's first customers and it is a reputation that will probably

stay with it indefinitely. The hope is, however, that as more and more legitimate people begin to use Bitcoin, they will eventually crowd out that negative part of its history and it will fade into the background in much the same way as the fall of Mt. Gox did.

Uncle Sam

The government always wants to get its cut and as of right now, the IRS views any type of digital currency as “intangible property,” which automatically subjects them to capital gains taxes.

That means that if you buy Bitcoin and sell it for a profit, you are required by law to report that on your taxes. Also, when you use your Bitcoin to purchase a product or a service it becomes a “taxable event.” So, if you were to go to Starbucks and use your Bitcoin to buy a cup of coffee for \$5, but the coffee originally cost \$4 then you have made a profit, even if you never get to hold the dollar in your hand.

This means that even if you’re willing to make these types of purchases, it will require an excessive amount of record keeping where you will be expected to keep detailed records of every purchase using Bitcoin. If you fail to do this, then you could be liable to the IRS for penalties. In the face of those options, many have chosen to avoid using Bitcoin at all in their lives.

Scalability

There is also the issue of scalability that can be very limiting for Bitcoin users. Because the Bitcoin Blockchain limits just how much data can be contained in each block (1MB) it slows down the process of completing transactions. This means that as more people begin to use Bitcoin, the transaction speed will slow down and the miners will have a difficult time keeping up. Right now, Bitcoin miners can complete an average of 7 transactions per second while the centralized Visa network can complete approximately 2,000 transactions in the same amount of time.

This is the primary reason for the delays in getting transactions verified and added to the block. If Bitcoin will become a mainstream currency, it will have to take steps to adopt a better way of managing transaction speeds.

They are working on this right now, but what they decide to do in the long term still remains to be seen.

While Bitcoin has a great deal of promise in the long term, there are still a lot of bugs that need to be worked out for it to be universally accepted as a mainstream currency. Even though it is nearly a decade old, it is still a very young investment instrument in comparison to other more popular tools.

The good news is that the majority of the obstacles are being

worked on right now and a solution will eventually come down the pipeline. One needs to be conscious of the fact that it is a growing entity and like all new entities, it will have its growing pains, but one thing is for sure, Bitcoin and the Blockchain are now a permanent part of our lives and it won't be going away anytime soon.

CHAPTER 6: THE FUTURE OF THE BLOCKCHAIN

We all understand the power of the Blockchain and what we can reasonably expect to happen in the near future; but what about in the distant future? Surely, with the obstacles that the Blockchain already has, it is not possible to expect that it won't be changing in some major ways.

We need only to look at the past to see what we can anticipate in the future evolution of the Blockchain. Today, we have a system that allows us to perform the same functions without the use of a decentralized server. We can now send peer-to-peer transactions that allow us to completely eliminate the middleman from participating in our affairs.

Right now, we have the most powerful tool in our possession connecting us to the rest of the world. Your smartphone, tablet, or laptop can easily put you in touch with billions of people around the world at the flip of the switch. To show you how big of a deal that is, consider these facts.

In 1950 there were approximately 500 devices available to the mainstream population.

In 1975 that number had increased to 10,000 devices.

In 2003 the number had jumped to 500 million devices

In 2009 when Bitcoin was first released, it was more than 2.5 billion

In 2014 it had already reached 14 billion.

Where will it go from here? It is estimated that by 2020, a mere two years from now, the number of devices the world will depend on will be approximately 30 billion and by 2050, it could reach as high as 100 billion.

If history is to teach us anything, the future of the Blockchain will lie in those 100 billion devices being able to easily access it and perform even more functions than it has today.

The Future of Blockchain Technology in Business

When it comes to the Blockchain's impact on the future of

business, it looks very promising. As we stated before, there will, of course, be a period of growing pains where there will have to be a sifting out and solving a number of problems. Primarily the speed of transactions will have to be addressed.

As more devices connect in cyberspace complications and bottlenecks will naturally increase until each system finds a way to work around them. We saw this with the first introduction of the Internet. If you've been around during the AOL days, you remember the annoying whistle in the line when you tried to hook up to the Internet. That was because the Internet at the time was accessed mainly through phone lines, which weren't designed for that kind of connectivity. As time progressed though, it was easy to find alternatives for the Internet and gradually it began to use its own system. Now, that annoying whistle we get when we're online is barely even a memory.

We can expect similar situations coming up in the future. It is already evident that it will be the younger generation that will be the ones to make the Blockchain ready for everyone. According to one professor at New York University, digital currencies like the cryptocurrencies have already replaced gold as their primary investment choice. It is just a matter of time before these will begin to overtake the traditional fiat currencies we now depend on.

If that were to happen, Bitcoin could eventually replace the dollar, euros, pounds, rubles, yen, won, and a host of other currencies that are used today. While today, cryptocurrencies on

the Blockchain do meet the basic criteria for money, they just barely do. Their extreme volatility shows that there is still a lot of resistance to its overall acceptance but as they overcome the many obstacles they face, it is a strong possibility that it could happen.

No matter what type of business you're in, there are several needs that the Blockchain can supply in spades. First, is the element of trust; very few people still carry around cash for making large purchases, which means that a business must know that they can trust the person they're dealing with and that they will repay them for the credit they issue for any purchases they make over time.

Today, banks, lawyers, and accountants play that role but in the future, all of that can be done right on the Blockchain. This means that major institutions that we come to rely on could easily lose their control over the people. The banking industry, auditors, solicitors, and even some governmental offices will have to close as more people begin to bypass them in favor of Blockchain.

Cloud storage will also change as well. Right now, people generally have to pay a great deal of money for digital storage but as more people come to realize that the Blockchain can be used for more than making transactions online, we will begin to see a gradual shift to the Blockchain. Everything from personal wills to property deeds will be kept in permanent storage. And with the use of smart contracts, many of these documents can

become self-executing eliminating the need for expensive lawyers to validate them for us.

Better Privacy

Every year, people lose more and more of their right to privacy. We have been told repeatedly how our personal information is no longer our own. Businesses collect our information and actually sell it to other businesses to make a profit and there is nothing we can do or say about it. If someone has enough money to buy your phone number and address it is easily accessible to them. However, with the Blockchain, we can eliminate much of that.

With Blockchain technology, our online privacy will be more secure. We will be able to store our personal data on the Blockchain and have complete control over the people who should have access to it. Rather than some obscure organization deciding what we should see and hear, we will make decisions for ourselves on what kind of advertising we want, if any. So, the sale of your personal interests and information will no longer be an issue.

Businesses will, therefore, have to find other means of promoting their products or services to us.

On the other hand, the Blockchain opens the door to even more

business opportunities; especially in those less developed countries. Its ability to bypass their governmental bureaucracy and corrupt leaders, entrepreneurs can actually have a chance to get ahead without having to fight an endless fight to overcome the obstacles put in their way.

Yes, when it comes to business the Blockchain will definitely be disruptive and it will cause many problems for the existing infrastructure, but it will also open up many doors of opportunity that have previously been closed to many people. In short, the Blockchain can become the great equalizer preventing many large conglomerates from closing the door on struggling people and giving them a real chance to get ahead in life.

Other Blockchain Technologies You Should Know

You may not have noticed but already the Blockchain is changing the way you do business. But even without the future to look forward to, the Blockchain is already making huge inroads in the way we do things.

In Charity

Recently, the Opakeco Foundation started a new Blockchain venture that will have a direct impact on humanitarian causes. The ICO, which was recently launched in the fall of 2017 is already making a major impact. After a partnership with several

different charities, using the Blockchain to manage contributions will cut the fees in half, using smart contracts will regulate the slow release of founder's shares over a period of five years or more. This will ensure that founders of projects will remain with the project for a longer time period, giving each one a better chance of success.

More Start-Ups

With every new technique that the Blockchain offers, there will also be more new ideas coming down the pipeline. Many of them will push past our societal norms teaching us a better way to embrace our new technology and incorporate it into our lives. Even now, there are new companies appearing every day using the Blockchain as the foundation on which their business will function.

Abra: A digital wallet designed to work on a smartphone. From Agra, you will be able to send or receive funds from anywhere in the world completely peer-to-peer.

Augur: A betting system that allows you place bets on real-world events. Whether you want to bet on a game being played in China or you want to bet on the next lottery numbers in Australia, you'll be able to do it with Augur.

BlockCypher: A web services platform for Blockchain

applications.

Bluzelle: An application that supports all other Blockchain protocols making it possible to perform transactions between different cryptocurrencies without the need of repeated transfers into fiat currencies and then back into another cryptocurrency.

Credit Dream: a Blockchain designed to offer credit to people in areas where it is hard to get. Already this has been accepted in Brazil and other less privileged countries. Through credit dream, a lender can come from anywhere in the world and lend money to someone in a remote area to start a business or make an investment that will help them improve their present circumstances.

Enigma: A privacy Blockchain that takes all the privacy and security features, encrypts it, and protects your data from theft from someone else.

Slock.It: This one, built on the Ethereum Blockchain, helps start-up businesses to create smart contracts and use them in the Internet of Things. This way, anyone can rent, sell, or share their property without the use of an agent or any other type of middleman.

Plex: Also based on the Ethereum Blockchain, is designed to give

insurance companies a real-time estimate of cars and drivers.

ZCash: Implementing a program called zero-knowledge proof, they can make any transaction on the Internet completely anonymous. Even if it is mined on the public Blockchain, it can provide a cryptographic key that contains no private information. This means that if it is hacked, there are no identifiers available to link back to the person conducting the transaction.

CHAPTER 7: THE LEGAL SIDE OF BLOCKCHAIN

A look at the history of Bitcoin makes it painfully clear that Blockchain Technology is taking us into an entirely new age of development. While the old saying goes, history repeats itself, it bears mentioning that some things in history will never return. After the modern conveniences that we have all become accustomed to having been accepted by the masses, it would be difficult to imagine taking a step backward in an attempt to reclaim the past.

How many of us would be willing to live our lives permanently in the horse and buggy age? Or who among us would be willing to go back to the time of the hunters and gatherers in an attempt to preserve posterity? We don't even have to go back that far. Who would like to get rid of their many devices we have come to rely on? The iPad, PlayStation, MP3, MP4, smartphone, or just using a modern calculator.

While there may be many people who do not like the idea of change, change is a very real part of society and the Blockchain represents one of the biggest types of changes our world has ever

seen. With that in mind, looking at Blockchain Technology literally means that we will have to change the way we think about every aspect of our lives.

For example, there is an ongoing debate right now about the anonymity that the Blockchain offers to users. We already see that the IRS is challenging that privilege. We've already seen that the IRS views any money made on the Blockchain as capital gains, but how can they tax us if they don't know our identities.

Right now, there are two opposing views on the matter, one for making the Blockchain more open and transparent, revealing the identities of the users and the other against. This decision will have to be decided in the very near future. And right now, it looks as though those who are for revealing identities will win. The decision by the courts in November of 2017 ordered one exchange to reveal the identities of any investors who earned more than \$20,000 in a single year.

While the exchange lost this case, it is not completely settled. Many feel it sets a bad precedent for financial privacy for individuals. So, we need to all be watching carefully to see where this debate will take us.

The Blockchain also brings up another issue of concern and that is the definition of a digital identity. Since all transactions done on the Blockchain are executed and completed in cyberspace, determining exactly what makes up a digital identity is

something everyone should be concerned about.

The identification that is associated with your key will have to be maintained someplace in cyberspace and the security measures to protect it must also be addressed. From a legal standpoint, this could present major legal issues in the future should a dispute arise.

From the Blockchain perspective, the digital identity belongs to the person who holds the keys to the wallet holding the funds. That means that no matter who opens the account or who deposits money into it, the person who has the key is the one who owns its contents. So, if someone were to steal your key or you were to hand it over to someone, you would be literally handing over access to all of your cryptocurrency.

This is very different from a bank where you can deposit money into an account and only you have access to it. You have to present proper identification and give adequate proof that you are authorized to access the funds in the account. With any type of Blockchain access, there is only the key that determines who gets what's inside. You can see how this could lead to a great deal of legal debate in the coming months and years.

Laws That Can Impact Blockchain Technology

When it comes to predictions about the future, the vast majority

of claims are related to business and finance. Few people are looking at the impact that the Blockchain will have on the legal system. There is no question that this new technology will be disruptive across the board in all sorts of industries and because of that, the impact on the legal system these industries rely on will eventually have to make some adjustments.

This doesn't mean that the disruption will be bad or that it will impact society in a negative way. We have heard all too many positive reports about the Blockchain to predict gloom and doom but we will have to readjust our thinking about how laws will be enforced and the role that lawyers will be playing in the future.

The very fact of eliminating the middleman means that the Blockchain will be absorbing many of the jobs that lawyers now do today. Now the fact that much of this will not happen until the Blockchain is accepted and adopted by the masses means that these changes may be years down the line but already we see many law firms preparing for what some may feel is inevitable. These law firms want to be in a position to direct their clients in a completely new way of doing business with them.

Many different roles that are now common in the legal industry will have to change.

Client Advisors

The interesting thing about law is that it has its fingers in every sort of industry there is. Clients from every region of the business world refer to their lawyers for guidance and direction in all sorts of matters. From the moment they develop an idea for a business until it finally closes its doors, a lawyer's primary role is to give expert advice on the regulations, laws, and policies they need to be concerned about.

With that in mind, it stands to reason that there will eventually be a division of lawyers who specialize in all things Blockchain and many will refer to them for advice. Already there are questions being compiled that need to be answered on the legalities of many different activities related to this new technology.

Contracts/Smart Contracts

With the introduction of Ethereum and the new smart contracts, the role of lawyers who practice contract law will begin to change. In the future, this will probably become the job of a computer specialist who will know how to create these contracts and build computer codes that can dictate exact variables unique to each agreement. Once these become widely accepted there won't need to be very many court related cases that focus on defaults of an agreement.

We can fully expect that contract law will change to adapt this new aspect to the legal system. Those who play the role of the

middleman in these conditions will definitely see some change. Since the smart contracts are all self-executing there will no longer be a need to hire a legal intermediary to give notices, file appeals, or otherwise manage the myriad of problems we see today in contract law.

Property Title and Deed Management

Keeping track of who owns what property will also be managed on the Blockchain so lawyers who are responsible for that aspect of **property** management will find they have a lot less to do. This could represent a major change in property law not just in one country but globally. When you think of all the property owned by governments, businesses, religious organizations, and private enterprises, once these things are uploaded to the Blockchain there won't be any need for lawyers to keep massive amounts of paperwork to keep track of them or how to transfer them from one party to the next.

Intellectual Property Rights

Another area where lawyers will see a huge change is in the area of intellectual property. Similar to how the Blockchain can record home ownership and land rights, it can also create an unbreakable record of intellectual property that could span the entire globe. Because every transaction performed on the Blockchain is not only permanent, it is time stamped and irreversible. Disputes over who was the creator of any type of

intellectual property can be resolved quickly and easily. This can apply to copyright laws, trademarks, patents, and any other type of artistic or intellectual creation someone might create.

Maintaining Public Records

Governments have the responsibility of maintaining public records for everyone in the country. They do this by periodically collecting census data; they keep birth records, criminal records, health records, and employment records. Other records that will probably shift to the Blockchain could be business licenses, marriage licenses, driver's licenses, death records, and a host more.

Historically, getting a copy of these past documents was always a time-consuming process that sometimes would take weeks if not months to obtain. As it gradually shifts to the Blockchain it could become instantly accessible making it possible for everyone to have immediate access to their own data, significantly reducing the role of lawyers in helping clients to obtain this information.

Dispute Mediation

It stands to reason that the field of dispute mediation in law will see a significant change. As more and more people begin to rely on smart contracts on the Blockchain there will inevitably be

some problems. Lawyers will have to know how to manage these issues and still maintain the anonymity of the parties involved.

At this point, no one knows how one could resolve an issue in regards to a dispute over a smart contract. Since they are self-executing, if party A decides that party B misled him when he coded the contract there will be some challenges that legally must be figured out, if it is at all possible.

Since transactions verified on the Blockchain are permanent and unchangeable and the fact that the parties are for the most part anonymous, what recourse would a party have to dispute a transaction and who would they file a claim against.

Practicing law in the age of the Blockchain doesn't mean that they will be out of a job but it does mean that the dynamics of their job will change in a very big way. Many of these issues have yet to be resolved and there are still many things that must be figured out before the Blockchain is fully absorbed as a major part of society.

Proposed Laws on the Blockchain Itself

There is no doubt that as the Blockchain continues to grow and be accepted by the people, the government will force its hand and try to control and regulate it. Already we see signs of this happening and in areas where the government cannot gain

control, they have made it outright illegal to use.

As the price of Bitcoin has hit its all-time peak of \$20,000 at the end of 2017, it has suddenly got the attention of governments all over the world. Now, as more and more people are seen joining the party, they are thinking seriously about regulations and how they can implement them. While other aspects of the legal issues relating to the Blockchain may be years away, these government concerns are not. We can fully expect Uncle Sam to put the hammer down very soon.

US Regulations

We should not be surprised at the range of reactions among government members across the country. Some have shown real surprise that the technology has been embraced by so many and others have become extremely concerned as to what this will mean, how it will impact the government fluidity as a whole, and what are the implications for the future.

As it stands now, the Federal government has the constitutional power to preempt and regulate the Blockchain whenever it feels it should. So far, that has not happened nor has the government expressed any intention of doing so. So for now, we are all free to follow whatever guidelines our local states have imposed in regulating Blockchain Technology.

As far as individual states, we are already beginning to see some of these happening. For example, in June of 2015, New York became the first state to impose regulations on digital currency. By 2017, at least eight more states had followed suit. In the future, we can reasonably expect that more states will begin to pass laws that will regulate the Blockchain and if there are too many inconsistencies across state borders, the Federal government is already poised to step in and create universal regulations that will apply all across the nation.

There have also been many legal precedents that have caused governmental recognition of the Blockchain. Some of them settled in landmark cases.

- Arizona legally recognized the use of smart contracts
- Vermont courts recognized the Blockchain as evidence
- Chicago used the Blockchain for real estate records
- Delaware recognized the authorization of registration of shares in several companies in the Blockchain form.

Bitcoin has already received the same financial protection as any other traditional asset, and the US Commodity Futures Trading Commission has granted one cryptocurrency, LedgerX, the approval to become the first federally regulated digital currency options exchange and clearinghouse in the entire United States.

The reality is that this is just the beginning and those who are already involved in the cryptocurrency craze can fully expect that they will see a lot more coming. But this is just in the United States, Blockchain Technology in other nations is just as exciting.

Blockchain on a Global Scale

Everywhere you look, in just about every country, world governments, businesses, and private people are embracing this new technology. There are very strong feelings for and against the Blockchain depending on how vulnerable people feel.

Some governments feel that if they allow cryptocurrencies to flow freely in their country, they would lose their economic hold over the people. Others may be allowing it for now but are reserving judgment until they know more, all the while giving careful scrutiny over every aspect of the new phenomenon.

Europe

In Europe, the overall acceptance of the Blockchain has been very encouraging. The EU follows the philosophy of innovation first, which in the end supports the development of digital currencies. They see it as a tool to help entrepreneurs to start new businesses based on the Blockchain and it gives them an opportunity to observe and test out its impact on the governance

of laws in these areas.

The European Union has recently announced that it is already working on a Blockchain project of its own to support the distributed ledger concept, and the European Commission is also working on exploring the benefits and challenges for applying this technology in their own financial services.

Switzerland specifically has made itself the main hub for all of Europe when it comes to the development of the Blockchain. Their Crypto Valley Association, a non-profit Blockchain ecosystem has even started to create its own ICO Code of Conduct to counteract the ban imposed by China on token crowdsales.

Blockchain in Other Countries

No matter where you go in the world, the Blockchain is becoming the preferred form of technology for existing businesses but especially for new startups. These are not just popping up in the more affluent countries but even in the most distant regions of the world.

Many countries are now at least partially supporting the introduction and use of digital currencies, but they are also encouraging these new businesses to try new ways of transacting business and see it as a welcome boost to their economy. There

are still some countries that are rejecting it outright, labeling it as illegal and a negative influence on their respective economies but as one Congressional resolution proposed in the US explained...Blockchain technology along with the right protections can actually have the power to change the way we trust and secure transactions in the areas of financial services, payments, health care, energy, property management, and even intellectual property.

Those who reject that idea as false only need to look at the thousands of already proven cases that are popping up all around the globe. As the rest of the world slowly embraces this new technology it is just a matter of time before those other nations who are holding out realize that the train is leaving the station and they had better get on it.

Blockchain Across Borders

In those areas where Blockchain Technology has already been accepted and has been in use, we see there is a global acceptance of the reality that going digital is the way to go. Everywhere you look businesses are already gearing up to prepare for the new changes that they are sure are coming down the pipeline.

By now, most people have heard of Bitcoin but few really understand the true nature of the Blockchain goes far beyond transferring funds across borders. In the future, they will come to see how the Blockchain can impact every industry in the world

in some form or another.

Finance

No doubt, this is the area that people are most familiar with. Evidence has already been shown in various results in different countries. For example, in China, the OMX Group along with the NASDAQ joined Chain, a new Blockchain startup and together they tested the trading of shares in the NASDAQ private market. The result? There was an increase in the accuracy of the trading process while there was a significant reduction in the settlement process. Payments were faster, even when trading across borders and the costs in fees were significantly lower. In short, the Blockchain increased the level of efficiency over the older system of things and reduce costs at the same time.

Data Sharing

Evidence has also been shown how data is shared across borders. Some even predict that in the future, rather than having companies to transmit data from one country to the next there should instead be a data transfer protocol. Because of the efficiency of the Blockchain in its use of a decentralized system, the movement of data can be done safely and securely with minimal influence by anyone that is not an active participant in the transfer.

Blockchain is expected to reinvent that data sharing industry through several cryptocurrencies already in place. The Lunyr Platform, built on the Ethereum Blockchain has already introduced a censorship-resistant and autonomous way to transfer data from one place to the next regardless of where it is in the world. Those who have the proper access can simply log into the system and retrieve data on anything they are authorized to see, confident that it will be accurate and verifiable.

Supply Chain Management

We often ship products all over the globe and it is expected that supply chain management will be heavily impacted by the Blockchain Technology. It will improve transparency and traceability as a product travels from one hand to the next. Those who handle any product in the supply chain will be able to record the date of the transfer, the condition the product is in, its location throughout the process, and even certify the price of the goods all on the Blockchain. And when it is all completed, even if they are shipping from one nation to the next, payments can be instantly shipped right on the same Blockchain. This will make for faster business and fewer downtimes when it comes to shipping and handling.

Cybersecurity

There will be a reduction in fraud in the digital world, and

businesses will be conducted much more efficiently. This will also impact the accounting and auditing field as the Blockchain will ensure the ability to keep better and more accurate records of the transactions and business they conduct. Even some banks in different countries are already preparing to adopt extra security measures based on what they have learned from the Blockchain. The Commonwealth Bank of Australia is already using the Blockchain to help them to increase their efficiency when offering services to their customers.

Healthcare

Even in the healthcare industry, the Blockchain is making an impact. Already we see the improvement in their data storage records and the ease of accessibility of patient information when it is needed. This goes beyond just maintaining a patient's medical records. When all the data is compiled and easily accessible, researchers can collect specific data that can be applied to several studies to help with determining common causes for diseases, accidents, and injuries and come up with a better treatment or even a possible cure faster. Data details like age, gender, and medical similarities with other patients can easily be grouped together to provide information that would otherwise take weeks if not months to acquire.

They can also connect to the Internet with numerous mobile devices that have been dispensed to monitor patient care. Patients can immediately get in touch with their health care providers, monitor the activity on their devices, and increase

efficiency and the quality of care for all patients. Imagine what would happen when your pacemaker goes off track and it can instantly notify your doctor even before you know it. Once information can be shared seamlessly from one location to another regardless of the distance, delays in proper health care services will be a thing of the past.

These are not the only ways the Blockchain will work across borders or within your own country. You can fully expect to see evidence of the Blockchain appear in real estate, the media, social services, charity work, and even in child care. While every business differs in the kind of goals they want to achieve, they all have one thing in common, they need to work faster and more efficiently and the Blockchain is the common denominator that can help them all to achieve it.

Blockchain and Taxes

Of all the things we can look forward to in the future of the Blockchain, taxes are probably going to receive the biggest impact. Up until now, citizens have had very little they could do about paying Uncle Sam his due. But now there may be another way for everyone to address the tax issues, with a new solution offered by the Blockchain.

Because it can produce more accurate records of transaction history, that is completely unalterable, the information contained inside is literally an indisputable record of

authenticity. On the one hand, it will be more difficult for someone to “cheat” on their taxes because there is no wiggle room to fudge the numbers and there are thousands of copies spread throughout the Blockchain so even if one node in the system would permit a change the others would not. Therefore, the actual records kept on the Blockchain will create a form of trust between the taxpayer and the government.

So, how does this work when you’re talking about taxes? Since it has already been established that the tech is extremely efficient and secure the Blockchain can handle a wide variety of different transactions. It could conceivably allow each taxpayer the ability to choose how much of their tax dollars will go to what areas of the national budget. People would be much more inclined to pay their taxes if they had a say in how the money would be spent.

Before we can understand how this could actually work, let’s first take a look at the current system that we are using now. Presently, the budget is broken up into two primary areas, the mandatory spending programs, and the discretionary spending programs. Under these programs are several areas that usually take the lion’s share of the tax money we pay.

Mandatory Spending Programs

Social Security makes up 24% of the tax budget

Healthcare subsidies usually take up approximately 26%

Defense and International Security uses about 16%

Programs to help hardship families takes up about 9%

Interest on the National Debt is about 6%

Benefits for federal retirees and veterans 8%

Discretionary Spending Programs

Infrastructure 2%

Education 2%

Science and medical research 2%

Non-security related international programs 1%

And the remaining 3% is for all other types of federal spending

As a taxpayer, there are no choices. You can't tell Uncle Sam that you want your tax money to go to healthcare and not to international defense. You can't say that you want your tax dollars to go to education rather than to hardship families. However, if the time comes when the Blockchain is used for filing taxes you could have the option to dedicate a percentage of your taxes to mandatory spending and another percentage of discretionary spending.

While the option to pay taxes won't be eliminated and the option to not pay into the mandatory spending will most likely be off

the table. This includes the top expenses on the list. Social security, healthcare, veterans' benefits, transportation, food, and agriculture. You would be able to direct your discretionary spending dollars where you want them to go.

This could be possible with Blockchain Technology because it has a built-in way to both authenticate and authorize all of these allocations. The IRS may stipulate that you must contribute to every program on the list but you would have the freedom to dedicate a certain percentage to the areas that are of more interest to you.

This idea would be very effective in motivating more people to want to pay their taxes rather than trying to avoid them or hide their assets from the government.

Tax Fraud

We may not have seen a full-blown tax rebellion in the past years but we have seen plenty of people trying to avoid paying taxes. We've already discussed how the Blockchain can work with voting and recording property ownership and now we've seen how it can serve as a better means for dispensing our tax dollars giving us more control over how we want that money to be spent.

Tax authorities not just here but around the world are beginning

to see the benefits of using this technology in preventing cases of tax evasion. In the UAE (United Arab Emirates) for example, they are implementing a revised tax system that will make it much easier for its citizens to comply with the requirements. In the UK, they are already working on a plan to create a digital tax system.

And even in Europe, there is a plan to use smart contracts that will be prepared well in advance of tax day for the remittance of the proper tax amount. While learning that you won't be able to cheat on your taxes anymore may not be good news for the taxpayer there are quite a few advantages to the new system. It could eliminate the need to file taxes altogether. As long as the government Blockchain is connected to your employer, creating a smart contract with the taxpayer will allow them to collect the taxes from your pay as you go. It could basically be filling out your returns in advance. Then with each paycheck, the government can just pull out the money as you earn it.

But it is not all slanted in favor of the government. Because of the transparency of the Blockchain, taxpayers can easily see what the government is doing with their money any time they want. It will make for more accountability and leave an easy to follow the trail of where the money is spent making it easy for tax representatives to figure out what taxes need to be collected.

While these are just speculation and Blockchain Technology is still in its infant stage, the possibilities for the future are very real. Every day more people are beginning to understand the

power of a built-in trust system, one that is secure enough to protect your investment but cautious enough to guarantee anonymity with each transaction made. The reality is that whatever will happen, it is certainly going to happen now because the Blockchain is here to stay.

CONCLUSION

Thank you for making it through to the end of Blockchain: Understanding the Technology of Bitcoin and Cryptocurrency, let's hope it was informative and able to provide you with all of the tools you need to achieve your goals whatever it may be.

No matter where you look there is evidence that the Blockchain has permeated our society in countless ways. It has already taken root and has inserted itself in a vast number of industries. If it has not already, it is just a matter of time before the Blockchain becomes a part of everyone's life whether they choose to embrace it or not.

The question is – what side of the coin do you want to fall on? Some have seen it as a threat and something to be wary of. China feels that way and has unceremoniously banned many new coins from even being traded in their country. The fact that they have done so attests to the power of this new currency and could actually give it more leverage than if they had just ignored it.

However, as long as The Blockchain Technology can be utilized, no matter how many currencies they decide to ban, there will always be a use for it. Yes, there will be bumps in the road as we go along. This is only to be expected. It is the common knee-jerk reaction of many people when faced with something new. And there are still plenty of obstacles that need to be addressed. The price is still very volatile and predictions are all over the place but the chances of there being a ban on all things Blockchain is not likely and maybe not even possible. It only takes one to push its way through and be accepted for the tide to turn in its favor.

There is no question that there is huge potential to revolutionize many aspects of our old system of things and only time will tell what new things the Blockchain will come up with in the future. The impact it could have on our economy, legal system, governmental infrastructure, and healthcare could be immense. There would be some battles to be won for sure, but the results could be improved in ways many of us never thought imaginable.

Where do we go from here? That's entirely up to each of us to decide. One thing for sure, Blockchain Technology will take us all on an exciting and adventurous ride to parts unknown so make sure you fasten your seatbelt. It's going to be a bumpy ride.

Finally, if you found this book useful in any way, a review on Amazon is always appreciated!